1 Section II.A in detail: anyon permutation, defects, and gauging

Target text: Section II.A, "Non-Abelian Anyons from Gauging," of arXiv-2602.12228v1.
Level: advanced undergraduate quantum mechanics, with prior exposure to the toric code and stabilizer codes.

1.1 What this subsection is trying to establish

Section II.A is not yet the lattice construction. Its purpose is to explain *why* gauging a particular symmetry of two toric codes produces non-Abelian topological order. The key chain of ideas is

$$\begin{aligned} \text{controlled-Z symmetry} &\implies \text{permutation of anyon types} \\ &\implies \text{symmetry defect} \implies \text{dynamical gauge flux with non-Abelian braiding.} \end{aligned}$$

The result is a model with $\mathcal{D}(D_4)$ topological order. Here $\mathcal{D}(G)$ denotes the quantum double of a finite group G, and the paper uses D_4 for the eight-element dihedral group. The derivation of the actual gauged stabilizers is postponed to Section II.B. This note explains the physics and algebra that motivate that derivation.

1.2 Starting point: two decoupled toric codes

Put one qubit on each edge of a square lattice. For one toric code, introduce

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \in \partial p} Z_e.$$

A_v is the vertex (or star) stabilizer and B_p is the plaquette stabilizer. The Hamiltonian can be written

$$H_{\mathrm{TC}} = -\sum_v A_v - \sum_p B_p.$$

The ground space is the simultaneous $+1$ eigenspace of every A_v and B_p. In this note, copy B and copy C have independent Pauli operators:

$$X_e^B, Z_e^B, \qquad X_e^C, Z_e^C.$$

Their decoupled Hamiltonian is $H_{\mathrm{TC}}^B + H_{\mathrm{TC}}^C$. The superscripts are labels of the two codes, not powers.

1.2.1 Excitations and string operators

An e anyon is a vertex where $A_v = -1$. If γ is a path on the direct lattice, then

$$W_e^B(\gamma) = \prod_{e \in \gamma} Z_e^B$$

commutes with all B_p^B and anticommutes only with the two star stabilizers at the endpoints of γ. Acting on a ground state, it therefore creates a pair of e_B anyons at those endpoints.

Likewise, if $\gamma^\star$ is a path on the dual lattice, then

$$W_m^B(\gamma^\star) = \prod_{e \perp \gamma^\star} X_e^B$$

creates a pair of m_B anyons at the two endpoint plaquettes. The same statements hold for code C.

We use the usual $\mathbb{Z}_2$ fusion rules:

$$e_B^2 = m_B^2 = e_C^2 = m_C^2 = 1.$$

For example, $m_B e_C$ is a composite excitation carrying an m flux of code B and an e charge of code C. Before gauging, every excitation of the two-copy theory has a unique fusion outcome: this is an Abelian anyon theory.

1.2.2 Logical operators and the bar

On a torus, a closed string that cannot be shrunk to a point is not a product of stabilizers. It acts nontrivially within the ground-state subspace and is a logical operator. A bar distinguishes such an encoded operation from a single-qubit Pauli:

$$\bar{X}_B,\ \bar{Z}_B,\ \bar{X}_C,\ \bar{Z}_C.$$

For instance, $\bar{X}_B$ is a noncontractible X^B-string (with the precise direct/dual choice depending on convention). It acts as Pauli X on one encoded qubit of code B. The bar does **not** mean complex conjugation. It means "logical operator."

1.3 The controlled-Z symmetry

The two codes are placed with a small relative displacement. The circuit U_{CZ} in the paper is a product of physical controlled-Z gates: each horizontal edge qubit of one copy is coupled to a nearby vertical edge qubit of the other copy, and conversely. The geometry is chosen so that the extra Z factors produced by these gates form plaquette stabilizers of the other code.

Everything follows from the one-pair identities

$$\mathsf{CZ}_{12}\,(X_1)\,\mathsf{CZ}_{12} = X_1 Z_2, \qquad \mathsf{CZ}_{12}\,(X_2)\,\mathsf{CZ}_{12} = Z_1 X_2,$$

$$\mathsf{CZ}_{12}\,(Z_1)\,\mathsf{CZ}_{12} = Z_1, \qquad \mathsf{CZ}_{12}\,(Z_2)\,\mathsf{CZ}_{12} = Z_2.$$

There is no dagger on the right in these equations because $\mathsf{CZ}^\dagger = \mathsf{CZ} = \mathsf{CZ}^{-1}$.

1.3.1 Why it preserves the ground space

Consider a B-code star operator. Conjugating each of its X^B factors by the nearby controlled-Z gates produces a pattern of Z^C factors. For the pairing shown in the paper, that pattern is precisely a neighboring C-code plaquette operator. Schematically,

$$U_{\mathsf{CZ}} A_v^B U_{\mathsf{CZ}}^\dagger = A_v^B B_{p(v)}^C.$$

Similarly, a C-code star is sent to itself times a B-code plaquette. Plaquette stabilizers themselves are unchanged because controlled-Z commutes with every Z.

Thus U_{CZ} does *not* leave each term of the microscopic Hamiltonian fixed. Nevertheless, if $|\psi\rangle$ is a ground state, then $A_v^B|\psi\rangle = B_p^C|\psi\rangle = |\psi\rangle$, so

$$(A_v^B B_{p(v)}^C)|\psi\rangle = |\psi\rangle.$$

All transformed stabilizers still have eigenvalue $+1$. Therefore U_{CZ} maps the ground-state subspace to itself. This is what the paper means by an **emergent symmetry of the ground-state subspace**.

1.3.2 Its logical action

The same local conjugation rule applied along noncontractible strings gives

$$\begin{aligned} U_{\mathsf{CZ}}\bar{X}_B U_{\mathsf{CZ}}^\dagger &= \bar{X}_B\bar{Z}_C, \quad U_{\mathsf{CZ}}\bar{Z}_B U_{\mathsf{CZ}}^\dagger = \bar{Z}_B, \\ U_{\mathsf{CZ}}\bar{X}_C U_{\mathsf{CZ}}^\dagger &= \bar{Z}_B\bar{X}_C, \quad U_{\mathsf{CZ}}\bar{Z}_C U_{\mathsf{CZ}}^\dagger = \bar{Z}_C. \end{aligned}$$

These are exactly the Heisenberg-picture conjugation rules of a controlled-Z gate acting on the two encoded qubits. Since every physical CZ squares to identity,

$$U_{\mathsf{CZ}}^2 = 1.$$

The symmetry group is consequently $\mathbb{Z}_2$.

1.4 From the logical gate to an anyon permutation

The logical equation already suggests the local anyon action. More directly, take an m_B string. It is made from X^B operators. Under conjugation by U_{CZ}, every X^B in the string acquires a nearby Z^C. Symbolically,

$$U_{\mathsf{CZ}} W_m^B(\gamma^\star) U_{\mathsf{CZ}}^\dagger = W_m^B(\gamma^\star)\, W_e^C(\gamma).$$

The exact path γ is set by the shifted edge pairing, but its endpoints are at the same physical locations as the endpoints of the m_B string. Thus each endpoint now carries both an m_B and an e_C excitation:

$$g(m_B) = m_B e_C.$$

Repeating the argument with B and C exchanged gives

$$g(m_C) = m_C e_B, \qquad g(e_B) = e_B, \qquad g(e_C) = e_C.$$

The map g is an **anyon automorphism**. It is not merely a phase assignment: it changes the superselection label of an excitation.

It is useful to write the complete action once. Every two-copy toric-code anyon can be labeled as

$$x = e_B^a m_B^b e_C^c m_C^d, \qquad a, b, c, d \in \{0, 1\}.$$

All exponents are understood modulo two. The symmetry acts as

$$g(x) = e_B^{a+d} m_B^b e_C^{c+b} m_C^d.$$

This formula makes two facts transparent:

1. g respects fusion. Since fusion adds the four exponents modulo two, $g(xy) = g(x)g(y)$.
2. $g^2 = 1$. Applying the map twice adds $2b$ and $2d$, both zero modulo two.

For example, the two-element orbit of m_B is

$$\mathcal{O}_{m_B} = \{m_B,\ m_B e_C\}.$$

1.5 A symmetry patch is a defect line

Until now, U_{CZ} was applied everywhere. A **symmetry patch** means applying the same circuit only in a finite region R. Denote this partial operation by $U_{\mathsf{CZ}}(R)$.

Inside R, an anyon label is related to the outside label by g. Thus an m_B anyon crossing the boundary has the following description:

$$m_B \quad \xrightarrow{\text{cross } \partial R} \quad m_B e_C.$$

The boundary ∂R is therefore not an ordinary boundary between two materials. It is a line across which anyon labels are *twisted* by the permutation g. Such a line is called an **extrinsic symmetry defect** or **branch cut**.

"Extrinsic" matters. At this stage we chose R by hand and applied an external operation. The defect line is not yet a fluctuating degree of freedom of the Hamiltonian. This distinction is exactly what gauging changes.

1.6 What gauging does

Gauging a global $\mathbb{Z}_2$ symmetry means two related things:

1. Add new gauge degrees of freedom, called the A sublattice in the paper.
2. Replace the global symmetry by a local redundancy, so that a symmetry transformation can be applied independently in different regions.

After gauging, the choice of region R is no longer externally specified. The information formerly carried by the imposed branch cut ∂R is carried by a configuration of the A gauge field. In the language of the paper, a closed A-gauge-field loop realizes the boundary of a controlled-Z symmetry patch.

Consequently, crossing the gauge-field loop has the same effect as crossing the old defect line:

$$\text{A-gauge loop}: \qquad m_B \longmapsto m_B e_C, \qquad m_C \longmapsto m_C e_B.$$

The loop may be opened into a string. Its endpoints are now genuine point-like excitations of the gauged theory; the paper calls such a gauge-flux excitation m_A. The string is the dynamical descendant of the old extrinsic defect line.

Section II.B will show how the A qubits, a cluster state, a symmetry-enrichment circuit, and projection implement this statement in a commuting-projector lattice model. Section II.A only needs the physical consequence: an A flux is an endpoint of a line that performs g.

1.7 Why the resulting braiding is non-Abelian

Take an m_B anyon around an m_A flux once. Topologically, its path crosses the m_A defect string once. Therefore

$$m_B \quad \xrightarrow{\text{braid once around } m_A} \quad m_B e_C.$$

For comparison, in an Abelian anyon theory a full braid of two fixed anyon types changes a state only by a complex number:

$$|\psi\rangle \longmapsto e^{i\theta}|\psi\rangle.$$

Here the braid changes the anyon label. It acts nontrivially on the orbit $\mathcal{O}_{m_B}$, rather than multiplying one fixed label by a scalar. After gauging, a label such as m_B is not separately gauge invariant: local gauge transformations can exchange the two labels in its orbit. The gauged excitation must therefore retain an internal two-component space associated with that orbit.

This is the physical mechanism behind non-Abelian statistics. More precisely, a non-Abelian anyon theory has fusion spaces of dimension larger than one and braiding acts by noncommuting matrices on those spaces. The type-changing braid above is the observable sign that this structure has appeared. The complete $\mathcal{D}(D_4)$ theory obtained after the lattice construction supplies the corresponding fusion and braiding data.

1.8 The torus diagnostic mentioned in the paper

The final paragraph of Section II.A refers to two pairs of nontrivially braiding anyons wound around the two independent cycles of a torus. The important point is not a failure of charge conservation. Rather, the two windings cannot be treated as two independent Abelian processes: their ordering and fusion channel affect the final state.

When the second pair is fused, a nontrivial excitation can remain at the fusion point. The figure single_anyon.pdf is a diagnostic of the non-Abelian fusion space and of the nontrivial topology of the torus. The paper will later look for an analogous diagnostic in qLDPC codes, where no ordinary geometric torus is available.

1.9 Where D_4 enters

Each input toric code is a $\mathbb{Z}_2$ gauge theory, so the pair has Abelian gauge structure $\mathbb{Z}_2 \times \mathbb{Z}_2$. Gauging the extra $\mathbb{Z}_2$ symmetry whose action is g does not give three independent commuting $\mathbb{Z}_2$ factors. The new symmetry acts nontrivially on the old topological data.

At the group level, this is captured by

$$D_4 \cong (\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_2.$$

The symbol $\rtimes$ means semidirect product: the final $\mathbb{Z}_2$ acts nontrivially on the first factor. The classification of gauged anyon-permuting symmetries identifies the resulting topological order as $\mathcal{D}(D_4)$. Section II.A invokes this known correspondence; the next subsection constructs a specific lattice realization.

1.10 Check your understanding

1. Using $\mathsf{CZ} X_1 \mathsf{CZ} = X_1 Z_2$, explain why an X^B-string can acquire a Z^C-string.
2. Use the four-exponent formula for g to verify explicitly that $g^2(x) = x$.
3. Explain the difference between a symmetry defect before gauging and a gauge flux after gauging.
4. Why is a label-changing braid stronger than the Abelian statement "braiding gives a phase"?

1.11 Worked solutions

1.11.1 1. Conjugating an X^B-string

Let Γ be the set of B-code edges in an m_B string:

$$W_m^B(\Gamma) = \prod_{e \in \Gamma} X_e^B.$$

For the circuit in the paper, each relevant B edge e is paired with a nearby C edge, which we call $\phi(e)$. On that pair,

$$\mathsf{CZ}_{e,\phi(e)}\, X_e^B\, \mathsf{CZ}_{e,\phi(e)} = X_e^B Z_{\phi(e)}^C.$$

All controlled-Z gates commute with one another because they are diagonal in the computational basis. Pauli operators on distinct qubits also commute. We can consequently conjugate the whole string one factor at a time:

$$\begin{aligned} U_{\mathsf{CZ}} W_m^B(\Gamma) U_{\mathsf{CZ}}^\dagger &= \prod_{e \in \Gamma} \left(X_e^B Z_{\phi(e)}^C \right) \\ &= \left(\prod_{e \in \Gamma} X_e^B \right) \left(\prod_{e \in \Gamma} Z_{\phi(e)}^C \right) \\ &= W_m^B(\Gamma)\, W_e^C(\phi(\Gamma)). \end{aligned}$$

The first factor is the original m_B string. The second is a Z^C string on the partner path $\phi(\Gamma)$. A Z^C string anticommutes with the C-code star stabilizers at its endpoints, so it creates an e_C anyon at each of those endpoints. The shifted geometry chosen in the paper ensures that these endpoints coincide with the endpoints of the m_B string. Thus each endpoint changes from m_B to $m_B e_C$.

The essential point is local: controlled-Z turns every X in the original string into XZ. The global anyon transmutation follows because the extra Z factors join into a second string.

1.11.2 2. Verifying $g^2 = 1$

Write a general anyon as

$$x = e_B^a m_B^b e_C^c m_C^d, \qquad a, b, c, d \in \mathbb{Z}_2.$$

The note derived

$$g(x) = e_B^{a+d} m_B^b e_C^{c+b} m_C^d.$$

It is helpful to regard this as an update rule for the four exponents:

$$(a, b, c, d) \xrightarrow{g} (a + d,\ b,\ c + b,\ d).$$

Apply the same rule again:

$$\begin{aligned}(a + d,\ b,\ c + b,\ d) &\xrightarrow{g} ((a + d) + d,\ b,\ (c + b) + b,\ d) \\ &= (a + 2d,\ b,\ c + 2b,\ d) \\ &= (a, b, c, d).\end{aligned}$$

The last equality holds because the exponents lie in $\mathbb{Z}_2$, where $2b = 2d = 0$. Hence $g^2(x) = x$ for every anyon x. This agrees with the operator statement $U_{\mathsf{CZ}}^2 = 1$.

1.11.3 3. Extrinsic defect versus dynamical gauge flux

Before gauging, choose a region R and actively apply $U_{\mathsf{CZ}}(R)$. Its boundary ∂R is a symmetry defect: crossing it changes an anyon label by g. It is called **extrinsic** because the Hamiltonian did not create it. We introduced it from outside by choosing R and applying a circuit. We can also move or remove it by changing that externally imposed circuit.

After gauging, the choice of whether the symmetry is applied locally is stored in new A-sublattice gauge variables. A branch cut is then a configuration of the gauge field, rather than an externally prescribed boundary. Its endpoint is a gauge-invariant, point-like excitation of the gauged Hamiltonian: the gauge flux m_A.

There is an important subtlety. The *path* of the defect string is not itself an observable after gauging; local gauge transformations can deform it. Its endpoint is observable, for example by braiding an m_B anyon around it. Thus the same label-changing line has two different statuses:

- before gauging: an externally imposed defect, not an intrinsic excitation;
- after gauging: a dynamical gauge string whose endpoint is an intrinsic quasiparticle.

1.11.4 4. Why label-changing braiding is stronger than a phase

For Abelian anyons, the state space for a fixed set of anyon labels and a fixed fusion channel is one dimensional. A full braid must therefore act as

$$|\psi\rangle \longmapsto e^{i\theta}|\psi\rangle.$$

The output has the same anyon labels as the input; only its phase changes.

In the present example, one circuit around m_A sends

$$m_B \longmapsto m_B e_C.$$

The two labels span an orbit. If we use the basis $\{|m_B\rangle, |m_B e_C\rangle\}$, then the label part of this braid is represented, up to possible phase conventions, by

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

P is not a scalar multiple of the identity: it mixes two components. Consequently a single Abelian phase cannot describe this braid. After gauging, the two components must be regarded as internal states of one gauge-invariant excitation.

The swap matrix by itself does not prove every mathematical property of a non-Abelian anyon theory; it squares to the identity and hence commutes with itself. The full conclusion comes from the fusion and braiding data of the gauged $\mathcal{D}(D_4)$ theory, where suitable braids act by noncommuting matrices on multidimensional fusion spaces. The label-changing process is the physically transparent signal of why such a non-Abelian description is needed.

2 Section II.B in detail: gauging on the lattice

Section II.A explained what gauging is meant to achieve. Section II.B is the constructive part: it gives a local circuit and measurements whose remaining degrees of freedom realize that gauging. The construction has three ingredients:

1. an auxiliary cluster state, built from new qubits (a) and (A);
2. a controlled-controlled-Z circuit that couples this cluster state to the two toric codes; and
3. X-measurements of all the a qubits.

The notation is deliberately suggestive: lower-case a denotes qubits which will be measured away, while upper-case A denotes the gauge qubits which remain. The original two toric codes continue to be called B and C.

2.1 The auxiliary lattice and its cluster state

Put an a qubit at every vertex of the square lattice and an A qubit at the midpoint of every edge. This vertex–edge lattice is often called a Lieb lattice. An A edge e has two endpoint vertices; write them as v and w.

Initially put every new qubit in $|+\rangle$. The stabilizer of an individual $|+\rangle$ is $X = +1$. Apply a controlled-Z gate between every incident a–A pair, and then apply a Hadamard to every A qubit. The resulting state is a slightly rotated form of the usual two-dimensional cluster state. Its stabilizers are

$$K_v^a = X_v^a \prod_{e \ni v} X_e^A, \qquad K_e^A = Z_v^a Z_e^A Z_w^a. \tag{1}$$

Here and below, $\prod_{e \ni v}$ means the product over the four square lattice edges touching vertex v.

2.1.1 Deriving Eq. (1), rather than memorising it

Two elementary conjugation rules are enough:

$$\mathrm{CZ}_{ij}\, X_i\, \mathrm{CZ}_{ij} = X_i Z_j, \qquad HXH = Z, \quad HZH = X. \tag{2}$$

Before the final Hadamards, the stabilizer which began as X_v^a has passed through one controlled-Z for each neighbouring edge. It is therefore

$$X_v^a \prod_{e \ni v} Z_e^A.$$

The Hadamards on the A qubits exchange their Z's for X's. This gives the first operator in Eq. (1). Similarly, the X_e^A stabilizer acquires a Z^a at each endpoint under the two controlled-Z gates:

$$X_e^A Z_v^a Z_w^a.$$

The final Hadamard changes its X_e^A into Z_e^A, giving the second operator in Eq. (1). This short calculation is useful: all later stabilizer bookkeeping is just this same idea applied to a larger circuit.

2.2 Why this cluster state is the correct resource for gauging

The cluster state has two symmetries. The first is an ordinary, or "0-form'', global symmetry,

$$G_a = \prod_v X_v^a. \tag{3}$$

The second is a loop, or "1-form'', symmetry. For every closed collection of A edges γ, it is

$$G_A(\gamma) = \prod_{e \in \gamma} Z_e^A. \tag{4}$$

For example, G_a commutes with K_e^A because that stabilizer has two Z^a factors, each of which anticommutes with its corresponding X^a. The two minus signs cancel. A closed A loop meets each vertex in zero or two edges, so the same even-number-of-minus-signs argument shows that $G_A(\gamma)$ commutes with every K_v^a.

The key identity is obtained by multiplying the vertex stabilizers in any connected region R:

$$\prod_{v \in R} K_v^a = \left(\prod_{v \in R} X_v^a \right) \left(\prod_{e \in \partial R} X_e^A \right). \tag{5}$$

Every A-edge strictly inside R occurs in two vertex stabilizers and therefore contributes $(X_e^A)^2 = 1$. Exactly the edges crossing the boundary occur once, which proves Eq. (5).

What is true is the state-dependent identity

$$G_a(R)\, X_A(\partial R) |\text{cluster}\rangle = |\text{cluster}\rangle.$$

This follows because the left-hand operator is precisely the product of the vertex stabilizers in R:

$$\prod_{v \in R} K_v^a = G_a(R) X_A(\partial R).$$

Since $X_A(\partial R)^2 = 1$, we may rearrange this to obtain

$$G_a(R)|\text{cluster}\rangle = X_A(\partial R)|\text{cluster}\rangle.$$

Equation (5) is the elementary lattice form of the SPT "patch-to-boundary'' principle. Acting with the a-symmetry only inside R is inseparable, on this state, from an operator living on the boundary of R. Notice that the boundary operator is X^A, whereas the 1-form symmetry is a loop of Z^A. They anticommute when they cross once. That is the seed from which a gauge field and its flux–charge relation will emerge.

2.3 The general gauging move

Let U be a global symmetry of a matter state $|\psi\rangle$, so

$$U|\psi\rangle = |\psi\rangle. \tag{6}$$

The role of the additional circuit Ω is to *attach* that matter symmetry to the cluster-state symmetry. The required algebraic condition is

$$\Omega G_a \Omega^\dagger = G_a U. \tag{7}$$

This equation deserves a literal reading. Before Ω, a global flip of the disposable a qubits is G_a. Afterwards, exactly the same operation is accompanied by the matter transformation U. Combining Eq. (7) with the patch identity (5) says that a partial U action is tied to an A-edge boundary. That is precisely the relation we wanted from the discussion of symmetry-defect lines in Section II.A.

Write the symmetry operation applied only inside a region R as

$$G_a(R) = \prod_{v \in R} X_v^a.$$

Equation (5) says that, for the auxiliary cluster state,

$$G_a(R) \quad \text{has exactly the same action as} \quad X_A(\partial R) = \prod_{e \in \partial R} X_e^A.$$

Why? Every A-edge inside R is touched twice and cancels; only the edges on the boundary remain. So an a-symmetry *patch* leaves an A-edge operator on its boundary.

Now add the CCZ coupling circuit Ω. Equation (7),

$$\Omega G_a \Omega^\dagger = G_a U,$$

means: after coupling to the matter, performing the a-symmetry also performs the matter symmetry U.

Because Ω is local, the same statement works region by region:

$$\Omega G_a(R) \Omega^\dagger = G_a(R)\, U(R),$$

where $U(R)$ is the controlled-Z circuit applied only in region R.

Putting this into the patch identity gives, schematically,

$$G_a(R)\,U(R) \quad \longleftrightarrow \quad X_A(\partial R).$$

Finally, measuring all a qubits in the $X = +1$ state makes $G_a(R) = 1$. Therefore, in the remaining A, B, C system,

$$U(R) \quad \longleftrightarrow \quad X_A(\partial R).$$

This is the quoted sentence's content: applying the matter symmetry only in R is inseparable from creating an A-field string on the boundary of R.

That matches Section II.A because a partial U operation has a symmetry-defect line on its boundary: crossing that line changes an anyon's type. After gauging, that boundary is no longer an externally chosen defect; it is represented by physical A-gauge degrees of freedom.

Finally measure every a qubit in the X basis and retain the $+1$ outcomes for the moment. In this postselected description,

$$X_v^a = +1 \quad \text{for every } v. \tag{8}$$

The a variables disappear, but the A variables retain the record of where the formerly global operation was applied. Thus local changes of that record are gauge transformations, and the unmeasured A qubits are the gauge field. Section II.C of the paper explains how non-$+1$ measurement outcomes are corrected by feedforward; for the derivation here, Eq. (8) is the cleanest starting point.

A useful sanity check is the familiar special case $U = X$ on a collection of matter qubits. Choosing Ω to be a product of CNOT gates from matter to the neighbouring a qubits satisfies Eq. (7). Measuring the a's then produces the usual toric-code gauging construction. The paper uses the same architecture for a less elementary symmetry: logical controlled-Z on the two toric codes.

2.4 Attaching the controlled-Z symmetry with CCZ gates

For the present problem $U = U_{\mathrm{CZ}}$, the finite-depth circuit that implements the anyon relabelling described in Section II.A. The appropriate three-qubit gate is

$$\mathrm{CCZ}_{abc}|x_a x_b x_c\rangle = (-1)^{x_a x_b x_c}|x_a x_b x_c\rangle. \tag{9}$$

It is diagonal, and it has the following especially important conjugation identities:

$$\begin{aligned} \mathrm{CCZ}_{abc}\, X_a\, \mathrm{CCZ}_{abc} &= X_a\, \mathrm{CZ}_{bc}, \\ \mathrm{CCZ}_{abc}\, X_b\, \mathrm{CCZ}_{abc} &= X_b\, \mathrm{CZ}_{ac}\,. \end{aligned} \tag{10}$$

The first follows directly by comparing the phase before and after flipping x_a: their ratio is $(-1)^{x_b x_c}$, which is exactly the phase of a CZ_{bc}. The second identity is the same statement with the labels exchanged. Every Z operator is unchanged because CCZ is diagonal.

The paper chooses a local pattern of CCZ gates. Each gate has one a vertex as its control and targets one nearby B edge and one nearby C edge. Denote the set of B–C pairs coupled to vertex v by $\mathcal{P}(v)$, and write

$$\Omega_{aBC} = \prod_{v} \prod_{(b,c)\in\mathcal{P}(v)} \mathrm{CCZ}_{vbc}\,. \tag{11}$$

The diagrams in the paper specify $\mathcal{P}(v)$. What matters for the derivation is that, across all vertices, each controlled-Z gate in U_{CZ} occurs exactly once. Equation (10) then gives

$$\Omega_{aBC} X_v^a \Omega_{aBC}^\dagger = X_v^a \prod_{(b,c)\in\mathcal{P}(v)} \mathrm{CZ}_{bc}\,. \tag{12}$$

Multiplying Eq. (12) over all vertices yields the promised symmetry enrichment condition,

$$\Omega_{aBC} G_a \Omega_{aBC}^\dagger = G_a U_{\mathrm{CZ}}. \tag{13}$$

This is the central derivation of Section II.B. A CCZ gate is not chosen by analogy: it is chosen because conjugating its X on the a leg literally creates the desired CZ on the B, C legs.

2.5 What survives the X^a measurements

We can now track the stabilizers. This is a good place to distinguish two simple rules.

- A stabilizer with no a-support commutes with the measurement and is retained immediately.
- A stabilizer containing X_v^a also commutes with the X_v^a measurement; after projection its X_v^a can be replaced by $+1$.

An operator containing a lone Z_v^a anticommutes with that measurement, so it cannot be retained by itself. We must multiply such relations together, or rewrite them using other stabilizers, until their a dependence cancels.

2.5.1 Diagonal plaquette relations

We first make the notation completely explicit. A **plaquette** p is one elementary square. Its boundary, written ∂p, is the closed loop made from its four sides. Hence $e \in \partial p$ means that e is one of the four lattice edges lying along the perimeter of p, and $\prod_{e\in\partial p}$ means to multiply one operator for each of those four edges.

For corners v_1, v_2, v_3, v_4, name the four sides $\partial p = \{e_{12}, e_{23}, e_{34}, e_{41}\}$. Then the compact notation expands to

$$\prod_{e\in\partial p} Z_e^B = Z_{e_{12}}^B Z_{e_{23}}^B Z_{e_{34}}^B Z_{e_{41}}^B. \tag{14a}$$

The superscript specifies the layer: the same geometrical square has a B-plaquette operator, a C-plaquette operator, and, after gauging, an A-plaquette operator.

We also need a precise meaning of "survives the measurements." Let $|\Psi\rangle$ be the state immediately before measuring the a qubits. Conditioning on a $+1$ result for every X_v^a applies

$$P_+ = \prod_v \frac{1 + X_v^a}{2}. \tag{14b}$$

Thus the unnormalised post-measurement state is $P_+|\Psi\rangle$. If a pre-measurement stabilizer S obeys $S|\Psi\rangle = |\Psi\rangle$ and commutes with P_+, then

$$SP_+|\Psi\rangle = P_+ S|\Psi\rangle = P_+|\Psi\rangle. \tag{14c}$$

So S remains a $+1$ constraint on the projected state. This, rather than simply being "left over," is what we mean by *survives*.

CCZ is diagonal in the computational basis and therefore commutes with every Z operator. The enrichment circuit Ω_{aBC} leaves the old B- and C-plaquette stabilizers unchanged. They act only on B or C qubits, while P_+ acts only on a qubits, so they commute with P_+. Equation (14c) therefore proves that the projected state obeys

$$B_p^B = \prod_{e \in \partial p} Z_e^B, \qquad B_p^C = \prod_{e \in \partial p} Z_e^C. \tag{14}$$

The A-layer relation needs one more step. The pre-measurement cluster stabilizer on an A edge $e = (v, w)$ is

$$K_e^A = Z_v^a Z_e^A Z_w^a. \tag{14d}$$

It too is unchanged by Ω_{aBC}, because CCZ is diagonal. But K_e^A does **not** survive by itself: it contains Z_v^a and Z_w^a, which anticommute with the X^a measurements at their respective vertices. Equivalently, it does not commute with P_+.

Now multiply these edge relations around the closed boundary of a single plaquette. With the labels introduced above,

$$\begin{aligned} \prod_{e \in \partial p} K_e^A &= (Z_{v_1}^a Z_{e_{12}}^A Z_{v_2}^a)(Z_{v_2}^a Z_{e_{23}}^A Z_{v_3}^a)(Z_{v_3}^a Z_{e_{34}}^A Z_{v_4}^a)(Z_{v_4}^a Z_{e_{41}}^A Z_{v_1}^a) \\ &= (Z_{v_1}^a)^2 (Z_{v_2}^a)^2 (Z_{v_3}^a)^2 (Z_{v_4}^a)^2 \, Z_{e_{12}}^A Z_{e_{23}}^A Z_{e_{34}}^A Z_{e_{41}}^A \\ &= \prod_{e \in \partial p} Z_e^A. \end{aligned} \tag{14e}$$

Every corner lies on exactly two sides of the square. Hence every Z_v^a appears twice and cancels because $(Z_v^a)^2 = 1$. The final operator has no a-support, hence it commutes with P_+. The criterion in Eq. (14c) therefore proves that the projected state satisfies

$$B_p^A = \prod_{e \in \partial p} Z_e^A. \tag{15}$$

This is the magnetic-flux, or plaquette, constraint for the newly created A gauge field. The contrast is essential: an open edge relation K_e^A is incompatible with the X^a measurement, but the closed product of four such relations is compatible because all dependence on the discarded a qubits cancels exactly.

2.5.2 The dressed A-star constraint

The enriched version of the first cluster stabilizer in Eq. (1) is immediate from Eq. (12):

$$\widetilde{K}_v^a = X_v^a \left(\prod_{e \ni v} X_e^A \right) \left(\prod_{(b,c) \in \mathcal{P}(v)} \mathrm{CZ}_{bc} \right). \tag{16}$$

It commutes with the measurement, so applying Eq. (8) gives the surviving constraint

$$\widetilde{A}_v^A = \left(\prod_{e \ni v} X_e^A \right) \left(\prod_{(b,c) \in \mathcal{P}(v)} \mathrm{CZ}_{bc} \right). \tag{17}$$

Without the second factor, this is the usual toric-code star: it flips the four gauge bits adjacent to v. The product of CZ gates dresses that flip by a local transformation of the B, C matter. This is the microscopic form of the slogan that an A-gauge transformation carries the old controlled-Z symmetry along with it.

2.5.3 Why the B- and C-star constraints take more work

This is the only genuinely delicate measurement update in the construction, so we now do it completely for one representative B-star. Every other B-star is a translate or rotation of this calculation. The C-star calculation is the same with B and C interchanged.

Let A_v^B be the ordinary B-code star centered at v. Near it, label the four C-edge qubits on the relevant neighbouring C plaquette by c_1, c_2, c_3, c_4, in cyclic order. The CCZ incidence pattern in the paper has three nearby disposable vertex qubits:

vertex ancilla	controlled-Z factors attached after CCZ conjugation
a_{12}	$\mathrm{CZ}_{a_{12},c_1}\,\mathrm{CZ}_{a_{12},c_2}$
a_3	CZ_{a_3,c_3}
a_4	CZ_{a_4,c_4}.

(18)

The subscripts are labels, not powers. They describe the local pattern: a_{12} is attached to the two C edges c_1, c_2, while a_3 and a_4 are attached to c_3 and c_4, respectively. Let A_3 and A_4 be the two A-edge qubits connecting a_{12} to a_3 and to a_4. This is exactly the part of the incidence diagram used in the paper's rewrite.

For any qubit q, define its computational-basis occupation number by

$$n_q = \frac{1 - Z_q}{2}. \tag{19}$$

It is 0 on a $Z_q = +1$ state and 1 on a $Z_q = -1$ state. Therefore

$$\mathrm{CZ}_{r,s} = Z_r^{n_s}. \tag{20}$$

The enriched B-star is consequently

$$
\begin{aligned}
S_v^B &= A_v^B \,\mathrm{CZ}_{a_{12},c_1}\,\mathrm{CZ}_{a_{12},c_2}\,\mathrm{CZ}_{a_3,c_3}\,\mathrm{CZ}_{a_4,c_4} \\
&= A_v^B Z_{a_{12}}^{n_{c_1}+n_{c_2}} Z_{a_3}^{n_{c_3}} Z_{a_4}^{n_{c_4}}.
\end{aligned}
\tag{21}
$$

The first line is the post-CCZ stabilizer; the second line is only the identity (20) applied four times. At this stage the operator cannot be kept after an X^a measurement: it contains Z's on the measured qubits a_{12}, a_3, a_4.

The cluster state supplies the way to remove them. Its edge stabilizers on A_3 and A_4 say

$$
Z_{a_{12}}^a Z_{A_3}^A Z_{a_3}^a = 1, \qquad Z_{a_{12}}^a Z_{A_4}^A Z_{a_4}^a = 1. \tag{22}
$$

Equivalently,

$$
Z_{a_3}^a = Z_{a_{12}}^a Z_{A_3}^A, \qquad Z_{a_4}^a = Z_{a_{12}}^a Z_{A_4}^A. \tag{23}
$$

Substitute Eq. (23) into Eq. (21). The two endpoint a-variables are moved onto the two A edges, and all remaining a-dependence collects at the single vertex a_{12}:

$$
S_v^B = A_v^B \left(Z_{a_{12}}^a\right)^{n_{c_1}+n_{c_2}+n_{c_3}+n_{c_4}} \left(Z_{A_3}^A\right)^{n_{c_3}} \left(Z_{A_4}^A\right)^{n_{c_4}}. \tag{24}
$$

The first factor in Eq. (24) looks troublesome, but it is actually the existing C-plaquette constraint in disguise. Write $n_{a_{12}} = (1 - Z_{a_{12}}^a)/2$. On a computational-basis state,

$$
\begin{aligned}
\left(Z_{a_{12}}^a\right)^{\sum_{i=1}^4 n_{c_i}} &= (-1)^{n_{a_{12}} \sum_{i=1}^4 n_{c_i}} \\
&= \prod_{i=1}^4 \left(Z_{c_i}^C\right)^{n_{a_{12}}} \\
&= \left(\prod_{i=1}^4 Z_{c_i}^C\right)^{n_{a_{12}}} = \left(B_p^C\right)^{n_{a_{12}}} = 1.
\end{aligned}
\tag{25}
$$

The penultimate equality uses the fact that $c_1, \ldots, c_4$ form the boundary of that C plaquette p. The final equality is legitimate on the code state because $B_p^C = +1$, and Section 2.5.1 proved that this plaquette stabilizer is still valid after the a-measurements.

Removing the identity factor in Eq. (25) leaves an operator with no a-support:

$$
\begin{aligned}
\widetilde{A}_v^B &= A_v^B \left(Z_{A_3}^A\right)^{n_{c_3}} \left(Z_{A_4}^A\right)^{n_{c_4}} \\
&= A_v^B \,\mathrm{CZ}_{A_3,c_3}\,\mathrm{CZ}_{A_4,c_4}\,.
\end{aligned}
\tag{26}
$$

Equation (26) is the explicit final dressed B-star represented by the paper's diagram. It commutes with every X^a measurement because it contains no a qubit at all. The old B-star is therefore not discarded; it is converted into a gauge-invariant local constraint whose two control qubits are now physical A-gauge-field edges.

For a different B-star, the names A_3, A_4, c_3, c_4 change according to translation or rotation. A notation that covers every location is

$$\widetilde{A}_v^B = A_v^B \prod_{(A_e,c)\in\mathcal{Q}_B(v)} \mathrm{CZ}_{A_e,c}, \tag{27}$$

where $\mathcal{Q}_B(v)$ is the local pair of A- and C-edge qubits read from the translated incidence pattern. In the representative above, $\mathcal{Q}_B(v) = \{(A_3, c_3), (A_4, c_4)\}$.

Repeating the same four steps for a C-star gives

$$\widetilde{A}_w^C = A_w^C \prod_{(A_e,b)\in\mathcal{Q}_C(w)} \mathrm{CZ}_{A_e,b}\,. \tag{28}$$

Here the opposite-code edges b belong to the B layer, and the residual a-factor is removed by the neighbouring B-plaquette constraint $B_p^B = +1$. Thus the two derivations are exact layer-swapped copies of each other.

2.6 The final stabilizer structure and what is unusual about it

After the projection, the state is characterized locally by three diagonal plaquette families,

$$B_p^A, \qquad B_p^B, \qquad B_p^C, \tag{22}$$

and three dressed star families,

$$\widetilde{A}_v^A, \qquad \widetilde{A}_v^B, \qquad \widetilde{A}_v^C. \tag{23}$$

The tildes matter. These are no longer Pauli stabilizers in the elementary toric-code sense, because they contain controlled-Z factors. In particular, two bare-looking X-type constraints need not commute as literal operators. Instead, as the paper notes, interchanging two of them can produce a product Q of the diagonal plaquette constraints:

$$S_i S_j = S_j S_i Q. \tag{24}$$

On a physical ground state, every such Q has eigenvalue $+1$, so the two orders act identically there. This is why the construction is consistent even though its local terms are not simply a mutually commuting Pauli stabilizer group. Put differently, the diagonal flux constraints carry exactly the bookkeeping needed to make the dressed gauge constraints compatible.

The output is the lattice realization of the gauged phase discussed in Section II.A: the A field supplies dynamical symmetry-defect strings, and its flux endpoints enact the controlled-Z-induced relabelling of B, C anyons. The construction is therefore a preparation of the $\mathcal{D}(D_4)$ topological order, not merely three independent toric codes.

2.7 The construction in one line per stage

cluster resource	encodes "symmetry patch = A boundary",
Ω_{aBC}	changes that symmetry into $G_a U_{\mathrm{CZ}}$,
X^a measurement	removes a and leaves A as the gauge field.

The first stage supplies the geometry, the second supplies the desired anyon automorphism, and the third turns the formerly imposed symmetry patch into a dynamical gauge configuration. That is the entire logical skeleton of Section II.B. We now replace its ideal postselection by the actual, random-outcome measurement procedure of Section II.C.

2.8 Section II.C: preparation by measurement and feedforward

2.8.1 What has to be fixed

Up to now we pretended that every measurement of X_v^a gave the outcome $+1$. That is useful for deriving the code, but it is not a preparation protocol: quantum measurements do not allow us to choose their outcomes. Section II.C makes the construction deterministic:

1. Measure all disposable a qubits in the X basis.
2. Treat the pattern of negative outcomes as a syndrome.
3. Apply a Pauli correction on the unmeasured A qubits.

The correction removes *only Abelian electric charges* of the new A gauge field. It does not have to undo, or even identify, any non-Abelian excitation. This is why ordinary decoder-style feedforward is enough.

2.8.2 Keep the measurement record instead of postselecting it

Write the outcome at a-vertex v as a bit $s_v \in \{0,1\}$:

$$X_v^a = (-1)^{s_v}. \tag{32}$$

Thus $s_v = 0$ denotes the old $+1$ branch, whereas $s_v = 1$ denotes a -1 outcome. The projector onto the complete outcome record $s = \{s_v\}$ is

$$P_s = \prod_v \frac{1 + (-1)^{s_v} X_v^a}{2}. \tag{33}$$

If $|\Psi\rangle$ is the state immediately before these measurements, write the normalised branch state as $|\psi_s\rangle \propto P_s|\Psi\rangle$. The outcome bits are classical information; the a qubits do not remain part of the code.

2.8.3 A negative outcome is exactly an e_A excitation

Before measuring, Eq. (16) gave the enriched vertex stabilizer

$$\widetilde{K}_v^a = X_v^a \widetilde{A}_v^A, \qquad \widetilde{K}_v^a |\Psi\rangle = |\Psi\rangle. \tag{34}$$

Here $\widetilde{A}_v^A$ is the dressed A-star from Eq. (17). It acts only on retained qubits, so it commutes with P_s. Use this and Eq. (34):

$$\begin{aligned} \widetilde{A}_v^A P_s|\Psi\rangle &= X_v^a \widetilde{K}_v^a P_s|\Psi\rangle \\ &= X_v^a P_s|\Psi\rangle \\ &= (-1)^{s_v} P_s|\Psi\rangle. \end{aligned} \tag{35}$$

After normalisation,

$$\widetilde{A}_v^A|\psi_s\rangle = (-1)^{s_v}|\psi_s\rangle. \tag{36}$$

This is the complete syndrome calculation. In a toric code, an electric anyon e at a vertex is defined by a -1 eigenvalue of the star constraint at that vertex. Equation (36) is the identical definition for the new, dressed gauge field: every $s_v = 1$ marks an e_A charge at v. No A-plaquette, B-plaquette, or C-plaquette eigenvalue is flipped. In particular, the random measurements do not create an m_A flux, the excitation responsible for the type-changing braiding in Section II.A.

Although $\widetilde{A}_v^A$ contains CZ gates on B, C, its eigenvalue is still a single $\mathbb{Z}_2$ sign. The charge e_A is therefore Abelian, exactly as stated in the paper.

2.8.4 The correction string, derived operator by operator

Choose a collection γ of A-edges. Its boundary $\partial\gamma$ is the set of vertices incident on an odd number of chosen edges:

$$(\partial\gamma)_v = \sum_{e\ni v} \gamma_e \pmod 2, \qquad \gamma_e \in \{0,1\}. \tag{37}$$

If γ is a path, every interior vertex touches two chosen edges and drops out modulo two; only its endpoints lie in $\partial\gamma$. Apply the Pauli string

$$R_\gamma = \prod_{e:\,\gamma_e=1} Z_e^A. \tag{38}$$

Each Z_e^A anticommutes with the X_e^A in the dressed star at the endpoints of e, and commutes with all CZ dressing factors because they are diagonal. Hence

$$\widetilde{A}_v^A R_\gamma = (-1)^{(\partial\gamma)_v} R_\gamma \widetilde{A}_v^A. \tag{39}$$

Read Eq. (39) literally: R_γ flips the star eigenvalue precisely at the vertices in $\partial\gamma$, and nowhere else. It commutes with the plaquette constraints because they are products of Z's (or do not act on A). It also commutes with the dressed B- and C-stars: their A-dependent parts are diagonal CZ gates, while their X's act on B or C, not on A.

Let a classical decoder choose γ so that

$$\partial\gamma = s. \tag{40}$$

In words: pair the vertices with negative measurement outcomes and draw an A-edge path for each pair. Then

$$\begin{aligned} \widetilde{A}_v^A R_\gamma|\psi_s\rangle &= (-1)^{(\partial\gamma)_v} R_\gamma \widetilde{A}_v^A|\psi_s\rangle \\ &= (-1)^{s_v}(-1)^{s_v} R_\gamma|\psi_s\rangle \\ &= R_\gamma|\psi_s\rangle. \end{aligned} \tag{41}$$

Thus $R_\gamma|\psi_s\rangle$ has the desired $+1$ A-star eigenvalue at every vertex. Since the other final constraints were unaffected, it is the desired $\mathcal{D}(D_4)$ ground state in the chosen global sector.

2.8.5 Why the charges can be paired

On a closed lattice, local e_A charges cannot occur singly: the allowed syndrome has even total parity in the sector prepared here, so its vertices can be partitioned into pairs. This is why Eq. (40) has a solution made from open paths. It is the familiar toric-code fact that a Z string has two endpoints, never one.

For a finite lattice with boundaries, a path may instead end at a boundary that condenses e_A. On a torus, two pairing choices can differ by a closed noncontractible Z_A loop. This is a logical operator, so a decoder must use a convention (or track a logical frame) to prepare a specified ground-state sector.

The paper calls this a finite-depth adaptive circuit. Once a correction is chosen, every factor in Eq. (38) is a single-qubit Z gate, so all of them can be applied simultaneously in one quantum layer, irrespective of path length. Finding a good pairing from a noisy syndrome is the separate classical decoding problem.

2.8.6 The deterministic adaptive circuit

Section II.C is therefore the following procedure:

$$\begin{aligned} \text{prepare cluster state and the two toric-code states} &\longrightarrow \Omega_{aBC} \\ &\longrightarrow \text{measure all } X_v^a \text{ and record } s \\ &\longrightarrow \text{classically choose } \gamma \text{ with } \partial\gamma = s \\ &\longrightarrow R_\gamma = \textstyle\prod_{e\in\gamma} Z_e^A. \end{aligned} \tag{42}$$

The first two stages are fixed finite-depth circuits, and all X^a measurements are parallel. The final operation depends on the recorded outcomes: that dependence is the *feedforward*. It replaces the unphysical instruction "keep only the all-+1 branch" with a procedure that succeeds for every allowed record.

This completes Section II.C. Its role is narrow but crucial: Section II.B constructs the desired phase in an ideal measurement branch; Section II.C shows that branch randomness only creates correctable Abelian e_A excitations, so the preparation is deterministic.

2.9 What exactly is the $\mathcal{D}(D_4)$ quantum double?

This section fills in the name that has appeared throughout the discussion. The short answer is that $\mathcal{D}(D_4)$ is a particular non-Abelian topological order: it is the quantum-double gauge theory whose finite gauge group is the eight-element symmetry group of a square. It has 22 anyon types on the plane and consequently 22 ground states on a torus.

There are two closely related uses of the phrase *quantum double*, which are worth separating.

- Kitaev's **quantum-double lattice model** is a solvable gauge-theory Hamiltonian built from a finite group G.
- $\mathcal{D}(G)$, also called the Drinfeld double of G, denotes the topological order — equivalently, the complete fusion and braiding data — of that model.

Thus the statement "the construction gives $\mathcal{D}(D_4)$" does *not* mean that every displayed local term has to look like the textbook group-element Hamiltonian. It means that, at long distances, the superselection sectors and their fusion and braiding are those of the D_4 quantum double.

2.9.1 First reminder: the ordinary finite-group quantum double

For a finite group G, put a G-valued degree of freedom $|h_e\rangle$, $h_e \in G$, on each oriented lattice edge. Reversing the orientation replaces h_e by h_e^{-1}. The standard quantum-double Hamiltonian has two kinds of local constraints:

$$H_{D(G)} = -\sum_v A_v - \sum_p B_p. \tag{43}$$

The plaquette term B_p checks **flatness**. If the oriented boundary edges of p carry $h_1, \ldots, h_n$, it keeps precisely the states with

$$h_1 h_2 \cdots h_n = 1_G. \tag{44}$$

The vertex term A_v imposes **gauge invariance**. More precisely, A_v is the average over all local gauge transformations:

$$A_v = \frac{1}{|G|} \sum_{g \in G} A_v^g. \tag{45}$$

Here A_v^g multiplies every edge variable leaving v by g, and every edge variable entering v by g^{-1}. The two requirements, flatness and gauge invariance, are the group-valued analogues of the plaquette and star constraints of the toric code.

For $G = \mathbb{Z}_2 = \{1, -1\}$, Eq. (44) becomes the usual Z-plaquette constraint and Eq. (45) becomes the usual X-star constraint. Thus

$$\text{toric code} = \mathcal{D}(\mathbb{Z}_2). \tag{46}$$

This is the reason it was natural to describe the two B, C toric codes as an Abelian $\mathbb{Z}_2 \times \mathbb{Z}_2$ starting point.

2.9.2 Flux and charge labels in $\mathcal{D}(G)$

An anyon in $\mathcal{D}(G)$ carries two related pieces of information:

$$\boxed{\text{anyon type} = ([h], \pi).} \tag{47}$$

The first label $[h]$ is the **conjugacy class** of a group element h:

$$[h] = \{ghg^{-1} : g \in G\}. \tag{48}$$

It is the gauge-invariant version of magnetic flux. A local gauge transformation changes a particular representative h by conjugation, so only its whole conjugacy class is observable.

The second label π is an irreducible representation of the **centralizer**

$$C_h = \{g \in G : gh = hg\}. \tag{49}$$

This is the allowed electric charge attached to that flux. A particle with no flux has $h = 1$, hence $C_1 = G$, so its pure electric charges are the ordinary irreducible representations of G.

The quantum dimension is

$$d_{([h],\pi)} = |[h]| \dim \pi. \tag{50}$$

This one formula shows where non-Abelian particles come from. A dimension larger than one can arise because a flux has several gauge-related representatives ($|[h]| > 1$), because it carries a multidimensional charge representation ($\dim \pi > 1$), or both.

For an Abelian group all conjugacy classes have one element, every centralizer is the whole group, and every irreducible representation is one-dimensional. Equation (50) then gives $d = 1$ for every anyon. This recovers the fact that all toric-code anyons are Abelian.

2.9.3 The group D_4

The convention in both papers is

$$D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle. \tag{51}$$

Geometrically, r is a quarter-turn of a square and s is a reflection. The elements are

$$\{1, r, r^2, r^3, s, rs, r^2 s, r^3 s\}. \tag{52}$$

The notation varies in the literature: some authors call this eight-element group D_8, because it has eight elements. This note follows the notation of the two local papers: D_4 means the symmetry group of the four-sided square, hence $|D_4| = 8$.

The relevant conjugacy classes and centralizers are:

flux class $[h]$	$\lvert[h]\rvert$	C_h	number of irreps of C_h	anyon dimensions in this row
$\{1\}$	1	D_4	5	$1, 1, 1, 1, 2$
$\{r^2\}$	1	D_4	5	$1, 1, 1, 1, 2$
$\{r, r^3\}$	2	$\mathbb{Z}_4$	4	$2, 2, 2, 2$
$\{s, r^2 s\}$	2	$\mathbb{Z}_2^2$	4	$2, 2, 2, 2$
$\{rs, r^3 s\}$	2	$\mathbb{Z}_2^2$	4	$2, 2, 2, 2$

(53)

There are $5 + 5 + 4 + 4 + 4 = 22$ rows-with-charge-labels, so $\mathcal{D}(D_4)$ has **22 anyon types**. As a check, their dimensions satisfy

$$\sum_a d_a^2 = |D_4|^2 = 8^2 = 64. \tag{54}$$

The total quantum dimension is therefore $\mathcal{D}_{\text{total}} = 8$. On a torus, the ground-state degeneracy of a quantum double equals the number of anyon types, explaining the number 22 that appears in the paper and in the trapped-ion experiment.

The two-dimensional entries in Eq. (53) are already enough to establish non-Abelian order. For example, the pure charge associated with the unique two-dimensional irrep of D_4 has $d = 2$, even though its flux

is trivial. The word "flux" should therefore not be used as a synonym for "non-Abelian anyon": $\mathcal{D}(D_4)$ contains non-Abelian charges, non-Abelian fluxes, and flux-charge composites.

2.9.4 How the Section II.A–II.C derivation reaches this phase

The derivation in this note does not begin by putting eight-level D_4 variables on edges. Instead it starts from qubits and proceeds through

$$\underbrace{\mathcal{D}(\mathbb{Z}_2)\times\mathcal{D}(\mathbb{Z}_2)}_{\text{two Abelian toric codes}} \xrightarrow[\text{anyon-permuting } \mathbb{Z}_2]{\text{gauge}} \mathcal{D}(D_4). \tag{55}$$

There are three concrete stages behind this arrow.

1. **The starting Abelian order.** B and C each contribute a $\mathbb{Z}_2$ gauge sector. Before gauging, every excitation has dimension one.
2. **A nontrivial action on the anyons.** The controlled-Z circuit fixes e_B, e_C but acts as

$$m_B \mapsto m_B e_C, \qquad m_C \mapsto m_C e_B. \tag{56}$$

   This is not merely a phase symmetry. It changes the superselection label. Its defect line is therefore a type-changing line.
3. **Gauging the symmetry.** The A qubits make that defect line dynamical. An A-flux endpoint is now a genuine excitation, and transporting an old B, C excitation around it enacts Eq. (56). Because m_B and $m_B e_C$ belong to one gauge orbit, the resulting gauge-invariant object carries a two-component internal space. This is the physical counterpart of a dimension-two entry in Eq. (53).

The semidirect-product identity

$$D_4 \cong (\mathbb{Z}_2\times\mathbb{Z}_2)\rtimes\mathbb{Z}_2 \tag{57}$$

is a compact group-theory mnemonic for the outcome: the added $\mathbb{Z}_2$ does not remain an independent commuting factor. A useful explicit realization takes two commuting involutions x, y and an involution t which swaps them,

$$x^2 = y^2 = t^2 = 1, \qquad xy = yx, \qquad txt = y, \quad tyt = x. \tag{58}$$

Then xt has order four because $(xt)^2 = xy$, so the group is non-Abelian and isomorphic to D_4.

Equation (57) should not be read too literally as a statement about an individual B- or C-edge variable: Eq. (56) is an automorphism of the *anyon theory* and mixes a magnetic label with an electric label. The precise conclusion of the lattice derivation is that, after gauging, the resulting topological order has the $\mathcal{D}(D_4)$ anyon data. The semidirect product is the group-theoretic presentation that records the nontrivial extension, not a replacement for the stabilizer calculation in Section II.B.

Section II.B provides the microscopic constraints that realize this arrow: the diagonal plaquettes are the flatness-type constraints, while the dressed stars are the gauge-invariance-type constraints. Their CZ decorations are the local memory of Eq. (56). Section II.C then makes the preparation deterministic: a -1 X^a result gives only an Abelian e_A charge, and a Z_A string removes it. In particular, the feedforward corrects preparation byproducts; it does not remove the intended non-Abelian A-flux sectors.

2.9.5 Relationship to arXiv:2305.03766

The local paper arXiv-2305.03766v2, *Non-Abelian Topological Order and Anyons on a Trapped-Ion Processor*, is an experimental realization of the same $\mathcal{D}(D_4)$ topological order. It prepared a 27-qubit kagome-lattice state on Quantinuum's H2 trapped-ion processor and used braiding and torus-sector tests to probe its non-Abelian properties. The connection is direct at the level of the phase, but the two works use different microscopic presentations:

	this note / arXiv:2602.12228	arXiv:2305.03766
starting point	two toric codes plus an A gauge field	a type-III $\mathbb{Z}_2^3$ SPT construction
geometrical lattice	three square-lattice sublattices	kagome lattice
nontrivial local factor	CCZ enrichment, then CZ-dressed stars	CZ-dressed 12-body stars
measurement byproduct	e_A charges corrected by Z_A strings	Abelian star excitations corrected by Z strings
long-distance order	$\mathcal{D}(D_4)$	$\mathcal{D}(D_4)$

(59)

The apparently different "gauge group" language is not a contradiction. The experimental paper writes its kagome Hamiltonian as a **type-III twisted quantum double**

$$\mathcal{D}^{\alpha}(\mathbb{Z}_2^3), \qquad \alpha(a,b,c) = (-1)^{\rho_a \gamma_b \beta_c}, \tag{60}$$

where $a = R^{\rho_a} G^{\gamma_a} B^{\beta_a}$. Without the CZ factors in its star operators, its three colours would be three decoupled toric codes, namely an untwisted $\mathbb{Z}_2^3$ gauge theory. The type-III twist makes the fluxes carry projective, sometimes two-dimensional, internal representations. Thus an Abelian-looking input group can nevertheless produce non-Abelian anyons.

Crucially, the experimental paper gives an explicit anyon table identifying its $\mathcal{D}^{\alpha}(\mathbb{Z}_2^3)$ anyons with the $\mathcal{D}(D_4)$ labels of Eq. (47). This is a **braided equivalence**: the two descriptions have the same anyon types, quantum dimensions, fusion rules, and braiding data. It is not a claim that $\mathbb{Z}_2^3 \cong D_4$; those groups are plainly different, since the former is Abelian and the latter is not.

For instance, the experiment uses three dimension-two non-Abelian bosons m_R, m_G, m_B. One of its characteristic fusion rules is

$$m_B \times m_B = 1 + e_R + e_G + e_R e_G. \tag{61}$$

The four possible outcomes on the right have total dimension four, matching $d_{m_B}^2 = 2^2$. Braiding m_G around m_B changes their fusion channel; after the pairs are fused, it can leave Abelian e_R charges. This is the same qualitative phenomenon that motivated the "single anyon on a torus" discussion in Section II.A.

Finally, the measurement-and-feedback logic of arXiv:2305.03766 is almost the same as Section II.C. There, -1 outcomes of measured SPT ancillas signal **Abelian** star excitations, which are paired by conditional Z-strings. Here, Eq. (36) showed that -1 outcomes signal Abelian e_A charges, again paired by Z_A-strings. Both protocols exploit the same crucial separation: random measurement outcomes create only correctable Abelian byproducts, whereas the desired non-Abelian topological order remains in the unmeasured system.

2.9.6 What to retain

1. $\mathcal{D}(D_4)$ is the non-Abelian quantum double of the eight-element dihedral group; it has 22 anyons and torus ground states.
2. Anyons of $\mathcal{D}(G)$ are labelled by a flux conjugacy class and a charge representation of its centralizer.
3. Dimension-two anyons are the sharp algebraic signature of the non-Abelian internal spaces anticipated by the defect-gauging argument.
4. The present derivation and arXiv:2305.03766 realize the same phase by different local circuits and lattices.
5. The experimental paper's twisted $\mathcal{D}^{\alpha}(\mathbb{Z}_2^3)$ description is braided-equivalent to $\mathcal{D}(D_4)$, not an assertion that the two underlying groups are isomorphic.

2.10 Section III: from 2D gauging to non-Abelian qLDPC codes

The paper now leaves the square lattice. This is a real change of setting, not just a change of notation. Sections II.A–II.C started with two toric codes, where vertices, edges, plaquettes, strings, and anyon motion were visible in ordinary two-dimensional space. A general qLDPC code still has local *checks* in the coding-theory sense, but it normally has no embedding in a fixed-dimensional Euclidean lattice. One cannot assume that it has literal plaquettes, geometric paths, or mobile point particles.

The question guiding the rest of the paper is therefore:

Can the measurement-based gauging idea survive after ordinary spatial geometry is removed?

(62)

The authors' answer is yes, provided that "geometry" is replaced by the algebraic incidence structure of the code and by a carefully chosen ancillary gauge system. This opening section announces the two ways they do so.

2.10.1 What a qLDPC code supplies, and what it does not

For the purposes of this paper, an Abelian CSS qLDPC code supplies:

- physical qubits;
- a sparse family of X-type checks and a sparse family of Z-type checks;
- commutation between every X-check and every Z-check;
- logical operators and, sometimes, transversal logical Clifford gates.

"Low density" means that each check acts on only $O(1)$ qubits and each qubit occurs in only $O(1)$ checks as the code grows. This is the coding-theory analogue of locality. It does **not** mean that the qubits must lie near one another in a plane or in three-dimensional space.

The 2D toric-code picture had a convenient dictionary:

square-lattice toric code	general CSS qLDPC code
edge	physical qubit
vertex/star	X-check
plaquette	Z-check
which qubits touch a vertex or plaquette	the check–qubit incidence pattern
boundary of a region	a relation among checks / an incidence map

(63)

The final row is the subtle one. On a square lattice, "the boundary of a boundary is zero" is visibly why stars and plaquettes commute and why closed strings have no endpoints. In a qLDPC code, the same statements are encoded algebraically. The next subsection of the paper introduces chain complexes precisely to record this incidence data without drawing a lattice.

For now, it is enough to keep this distinction in mind:

$$\text{qLDPC locality} \neq \text{ordinary spatial locality}. \tag{64}$$

That distinction is the source of both the difficulty and the novelty of the paper. The authors cannot simply copy the red A-sublattice of the square-lattice construction. They must decide what the new gauge degrees of freedom should be connected to.

2.10.2 The input is an emergent logical symmetry, not necessarily a Hamiltonian symmetry

The 2D prototype gauged a controlled-Z circuit U_{CZ}. As stressed earlier, this circuit need not leave each microscopic Hamiltonian term unchanged. Rather, it maps stabilizers to products of stabilizers and hence preserves the code subspace. Such a circuit is an **emergent symmetry of the encoded information**.

Exactly the same is true here. Start with two copies, called B and C, of an Abelian CSS qLDPC code. The desired input is a transversal Clifford gate U between them:

$$U = \bigotimes_j U_j, \tag{65}$$

where each U_j has bounded support, often on the corresponding qubits of the two copies. "Transversal" is important operationally: even though the logical action of U can be global, the physical circuit has constant depth and preserves the LDPC character of the construction.

The paper distinguishes these gates from the code's usual logical Pauli operators. Logical X- and Z-operators are often described algebraically as cycles and cocycles. They are important symmetries, but they are **not** the symmetries being gauged here. The construction instead gauges a finite-depth transversal *Clifford* operation which acts on the encoded degrees of freedom, such as:

$$\mathsf{CZ}, \qquad \mathsf{SWAP}, \qquad \text{or, in a different qudit example, charge conjugation.} \tag{66}$$

This is the central generalization of Section II:

$$\boxed{\text{gauge an encoded transversal gate, rather than a geometrically onsite microscopic symmetry.}} \tag{67}$$

The distinction matters because it is what makes the proposal relevant to qLDPC codes. A good qLDPC code may have no useful Euclidean geometry at all, yet it can possess a sparse physical implementation of a meaningful logical Clifford gate.

2.10.3 The same three-stage mechanism survives

Although the geometry changes, the logical skeleton from Section II remains unchanged:

$$\begin{aligned} \text{ancilla resource state} &\longrightarrow \text{a system with an ancilla symmetry,} \\ \text{symmetry enrichment } \Omega &\longrightarrow \Omega G_a \Omega^\dagger = G_a U, \\ X^a \text{ measurement} &\longrightarrow \text{a gauged non-Abelian qLDPC code.} \end{aligned} \tag{68}$$

Here G_a denotes a symmetry of the ancilla resource state, and U is the transversal gate of the B, C code pair that we intend to gauge. The second line is the same symmetry-enrichment equation encountered in Eq. (13), now written without assuming a square lattice.

The new ancillary qubits play the role previously held by the a-vertex and A-edge qubits:

- the measured a-type qubits carry the ancilla symmetry;
- the unmeasured A-type qubits retain the gauge information after measurement;
- Ω ties local changes of that ancilla symmetry record to the logical gate U on the code.

The decisive design problem is therefore not "how do we draw the old square lattice again?" It is:

$$\boxed{\text{Which ancillary incidence structure has the right symmetry to gauge the chosen } U?} \tag{69}$$

This is why the paper calls the ancillary degrees of freedom an **ancilla chain complex** later on. In the most general language, it is an algebraic object describing which ancilla checks touch which ancilla qubits. In a special case it is literally a graph; in another it can mirror the chain-complex structure of the input code.

2.10.4 Why there are two constructions

In two dimensions, the obvious ancillary lattice is a copy of the original toric-code lattice. In the qLDPC setting, that choice is sometimes natural but not always possible or useful. The authors identify two regimes:

Homological gauging requires the input code to have a Poincare-duality-like algebraic structure. The ancilla complex is then a copy of, or close analogue of, the input code's complex. The payoff is that the construction can exploit a structured family of transversal gates, naturally including CZ gates, and can make some of them addressable. The price is that a generic qLDPC code need not have the necessary duality.

Graph gauging makes no Poincare-duality assumption. The ancillas instead form a separately chosen graph. Its single global symmetry is matched to one selected transversal gate; transversal SWAP is always available between two identical code copies. The payoff is broad applicability. The price is that the graph must be tailored to the chosen gate, and it does not automatically provide the structured family of gates available in the homological case.

Poincare duality will be defined carefully later. At this stage, its role can be stated without technical machinery: it is the extra algebraic structure that lets the authors identify stabilizer redundancies with a family of compatible transversal logical gates. That match allows the ancilla system to inherit the code's structure and gauge more than one related symmetry.

Graph gauging makes the opposite choice. It sacrifices that structured family in exchange for generality. A connected graph has one global $\mathbb{Z}_2$ ancilla symmetry, so it is suited to gauging one selected global $\mathbb{Z}_2$

transversal gate. The paper later shows that this is enough to gauge transversal SWAP for two copies of essentially any suitable input code.

The two methods are not competing answers to the same constrained problem. They answer two different needs:

$$\begin{gathered}\text{Use homological gauging when the code has the needed duality and you want to exploit it.}\\ \text{Use graph gauging when broad applicability to a chosen gate is the priority.}\end{gathered} \tag{71}$$

2.10.5 What "non-Abelian qLDPC code" means here

The phrase should be read with care. For the toric code, the non-Abelian claim could be visualized directly through particles moving in a plane and braiding. For a generic qLDPC code, there may be no ambient plane in which to define a conventional braid. The paper does not claim that every such code literally realizes a 2D gas of mobile D_4 anyons.

Instead, it constructs codes with non-Pauli, controlled-Clifford dressed constraints whose ground-state and logical-operator algebra has a non-Abelian character inherited from gauging U. Later, in Section IV, the authors provide an operational diagnostic: suitable logical operations can create a state with a single violated gauge-charge constraint in a subsystem-symmetry sector. This is the qLDPC analogue of the "single anyon on a torus" phenomenon from Section II.A.

So the logical order of the paper is:

$$\begin{array}{ll}\text{Section III} & \text{construct the gauged qLDPC codes,}\\ \text{Section IV} & \text{show their non-Abelian-like physical signatures,}\\ \text{Section V} & \text{use gauging and ungauging as logical-gate measurement.}\end{array} \tag{72}$$

The present introduction promises the construction; it does not yet prove all those later physical claims.

2.10.6 Road map for the next subsections

The remaining parts of Section III can initially be read with the following questions in mind:

1. **Toy model:** How can the same stack of toric codes admit two different ancilla choices, and what symmetry does each choice gauge?
2. **Chain-complex dictionary:** How are qubits and checks encoded without a Euclidean lattice?
3. **General gauging rule:** What condition must Ω and the ancilla complex satisfy so that an ancilla symmetry becomes $G_a U$?
4. **Two constructions:** What additional structure permits homological gauging, and how does graph gauging avoid needing it?

For a first pass, the most important result of this introduction is not a formula. It is the design principle:

$$\boxed{\text{the choice of ancillary gauge structure must match the transversal logical symmetry one wants to gauge.}} \tag{73}$$

Everything that follows is an increasingly explicit realization of that principle.

2.11 Section III.A: toy model — the ancilla chooses which symmetry is gauged

The paper's first example is deliberately familiar: take N independent two-dimensional toric-code layers, and take two identical stacks of them. Call the stacks B and C. Thus each layer $\ell = 1, \dots, N$ contains a pair

$$(\text{toric code } B_\ell) \quad \text{and} \quad (\text{toric code } C_\ell). \tag{74}$$

Nothing initially couples distinct layers. The full Hamiltonian is simply

$$H_{\text{in}} = \sum_{\ell=1}^{N} \left(H_{\text{TC}}^{B_\ell} + H_{\text{TC}}^{C_\ell} \right). \tag{75}$$

This is not yet a non-Abelian model; it is just $2N$ independent Abelian toric codes. The advantage of this example is that we already know how to gauge a controlled-Z symmetry for one pair of layers. We can now ask the new question in its simplest form:

Should we gauge each layer's symmetry separately, or only one symmetry acting on the entire stack? (76)

The authors show that both choices are legitimate, but they require different ancillary gauge structures and lead to different physics.

2.11.1 The symmetry content of the input stack

For each layer, let

$$U_\ell = U_{\text{CZ}}^{(\ell)} \tag{77}$$

denote the controlled-Z circuit between B_ℓ and C_ℓ. It is the direct layer-by-layer analogue of the circuit in Section II: within that layer it permutes the appropriate anyon labels and is an emergent symmetry of the encoded subspace.

Because the layers are decoupled, the N operations U_ℓ commute:

$$[U_\ell, U_{\ell'}] = 0 \qquad (\ell \neq \ell'). \tag{78}$$

Each has order two, $U_\ell^2 = 1$. Consequently the stack has a whole family of possible $\mathbb{Z}_2$ symmetries:

$$\left\{ \prod_{\ell=1}^{N} U_\ell^{q_\ell} \;\middle|\; q_\ell \in \{0,1\} \right\} \cong \mathbb{Z}_2^N. \tag{79}$$

Two choices within this family will matter:

$$\begin{aligned} &\text{layer-resolved generators:} && U_1, U_2, \dots, U_N, \\ &\text{one stack-wide product:} && U_{\text{all}} = \prod_{\ell=1}^{N} U_\ell. \end{aligned} \tag{80}$$

The first list allows us to change the topological sector of one selected layer without touching the others. The second operation instead changes every layer simultaneously. They are not interchangeable pieces of information, even though U_{all} is the product of the first list.

This distinction is the whole toy model in one sentence:

$$\boxed{\begin{array}{c}\text{independent ancilla symmetry generators}\\ \Longleftrightarrow \\ \text{independent input symmetries that can be gauged}\end{array}} \tag{81}$$

2.11.2 Choice 1: one ancillary cluster layer per toric-code layer

First give every physical toric-code layer its own copy of the two-dimensional ancilla resource used in Section II. In each ancillary layer there are a-qubits at vertices and A-qubits on edges; together they form a Lieb-lattice cluster state. The picture is therefore

$$\begin{array}{ccccc} (B_1, C_1) & \longleftrightarrow & \text{ancilla layer } 1 & \longleftrightarrow & U_1, \\ (B_2, C_2) & \longleftrightarrow & \text{ancilla layer } 2 & \longleftrightarrow & U_2, \\ \vdots & & \vdots & & \vdots \\ (B_N, C_N) & \longleftrightarrow & \text{ancilla layer } N & \longleftrightarrow & U_N. \end{array} \tag{82}$$

For a connected two-dimensional cluster layer, the relevant 0-form symmetry is the simultaneous X flip of all its a-vertex qubits. Write it as

$$G^a_\ell = \prod_{v \text{ in ancilla layer } \ell} X^a_{v,\ell}. \tag{83}$$

Because the ancillary layers are disconnected from one another, these are N independent symmetries:

$$\mathcal{G}^{\text{layer}}_{\text{anc}} = \langle G^a_1, \dots, G^a_N \rangle \cong \mathbb{Z}_2^N. \tag{84}$$

This independence is physical. Applying G^a_3, for example, changes only the a-qubits in layer 3. It does not have to be accompanied by an operation in layers 1, 2, 4, and so on.

The enrichment circuit is chosen independently in every layer:

$$\Omega_{\text{layer}} = \prod_{\ell=1}^{N} \Omega_\ell, \qquad \Omega_\ell G^a_\ell \Omega^\dagger_\ell = G^a_\ell U_\ell. \tag{85}$$

Equation (85) is just the Section II relation $\Omega G_a \Omega^\dagger = G_a U_{\text{CZ}}$, repeated N times. After measuring all a-qubits, the A-qubits in each layer become that layer's gauge field. Thus the procedure gauges every U_ℓ independently.

For this decoupled toy input, the output is correspondingly a stack of independently gauged two-dimensional systems: each layer has the same local $\mathcal{D}(D_4)$-type construction discussed in Section II. The important word is *independently*. A defect or gauge transformation can be specified separately in each layer, because the ancilla resource had one symmetry per layer.

The paper calls this the prototype of **homological gauging** because the ancilla inherits the same layer-by-layer structure as the input code. In the later qLDPC generalization, "same structure" becomes "the ancilla and input are described by closely related chain complexes."

2.11.3 Choice 2: one connected three-dimensional ancillary graph

Now keep exactly the same B, C stack, but choose a completely different ancilla resource: a connected three-dimensional cubic cluster state spanning all layers. The a-qubits live at cubic-lattice vertices and the A-qubits live on cubic-lattice edges. The crucial structural difference is that the ancillas are now connected *between* layers.

A connected cluster resource of this kind has one relevant global 0-form symmetry,

$$G^a_{\mathrm{3D}} = \prod_{v \text{ in the full 3D ancilla}} X^a_v. \tag{86}$$

There is no independent factor analogous to G^a_ℓ for an individual horizontal slice. Flipping only one slice is not a symmetry of this three-dimensional cluster state, because the vertical ancilla-edge constraints link that slice to neighbouring slices.

Hence the symmetry matching condition can tie this sole ancilla symmetry only to one input operation:

$$\Omega_{\mathrm{3D}} G^a_{\mathrm{3D}} \Omega^\dagger_{\mathrm{3D}} = G^a_{\mathrm{3D}} U_{\mathrm{all}} = G^a_{\mathrm{3D}} \prod_{\ell=1}^{N} U_\ell. \tag{87}$$

The procedure now gauges the single stack-wide controlled-Z, not the N layer-resolved controlled-Z gates. Measuring the a-qubits leaves a three-dimensional A gauge system, because the unmeasured A-qubits occupy the edges of the connected cubic ancillary lattice.

It is tempting to say that Eq. (87) is merely a simultaneous version of Eq. (85), but that would miss the point. The two outputs differ because their gauge fields have different connectivity:

	layer-resolved choice	connected-3D choice
ancilla 0-form symmetry	$\mathbb{Z}_2^N$	$\mathbb{Z}_2$
gauged input operation	$U_1, \ldots, U_N$ separately	U_{all} only
unmeasured A field	N independent 2D gauge fields	one connected 3D gauge field
resulting interpretation	stack of 2D non-Abelian layers	hybrid 3D/2D non-Abelian system

(88)

The source paper summarizes the second output by saying that its three-dimensional gauge field has non-Abelian mutual statistics with two-dimensional gauge fields of restricted mobility. The toy subsection does not derive those statistics, and neither do we need to derive them yet. The safe conclusion at this point is simply that this is **not** a stack of independent $\mathcal{D}(D_4)$ layers: the connected ancillary field correlates what happens in different layers.

2.11.4 Why the second choice is more general

The layer-resolved construction required a strong property of the input: each individual layer had to possess its own useful emergent controlled-Z symmetry U_ℓ. In the toy model this is automatic because the input was deliberately chosen as N decoupled toric-code pairs.

A more general qLDPC code may have no meaningful decomposition into layers, and it may not admit a family of separately addressable transversal gates. It can nevertheless have one transversal gate U on the entire code. A connected ancillary graph with one matching global symmetry is then enough to gauge that U.

This is the conceptual trade:

$$\begin{array}{rcl} \text{more independent input symmetries} & \Longleftrightarrow & \text{more independent ancilla symmetry generators,} \\ \text{one available global input symmetry} & \Longleftrightarrow & \text{one connected ancillary graph is sufficient.} \end{array} \tag{89}$$

It is not that the graph method is an approximation to the homological method. It is a different design choice: it gives up layer-by-layer addressability in order to work when the input code supplies only one global symmetry.

2.11.5 The lesson for the qLDPC construction

The toy model is a controlled experiment. The matter code is held fixed; only the ancilla architecture changes. That change alters

$$\text{ancilla symmetry} \longrightarrow \text{which logical gate is gauged} \longrightarrow \text{connectivity and character of the emergent gauge field.} \tag{90}$$

This is exactly the chain of reasoning that will survive in the general qLDPC setting. The rest of Section III replaces the visible layer geometry by chain complexes and graphs, but the selection rule remains Eq. (81).

2.11.6 Check your understanding

1. Why is $\prod_\ell U_\ell$ not equivalent, for gauging purposes, to having the N generators U_ℓ available separately?
2. In Eq. (85), what would go wrong if one tried to enrich G_1^a with U_1U_2 while leaving G_2^a untouched?
3. Why does connecting ancillary layers by vertical edges reduce the number of independent 0-form symmetries?
4. Which feature makes graph gauging suitable for a code that has one transversal gate but no layer decomposition?

2.12 Interlude: binary linear algebra over $\mathbb{F}_2$

The next subsection replaces the visible edges, vertices, and plaquettes of the toric code by matrices. The required mathematics is ordinary linear algebra, with one change: all numbers belong to the two-element field

$$\mathbb{F}_2 = \{0, 1\}. \tag{B1}$$

Its addition and multiplication are performed modulo two:

$$\begin{array}{c|cc} + & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array} \qquad \begin{array}{c|cc} \cdot & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}. \tag{B2}$$

Thus $1+1=0$. Subtraction is exactly the same operation as addition, because $-1=1$ in $\mathbb{F}_2$. The word *field* means that the usual rules of linear algebra still apply: addition and multiplication distribute, and every nonzero number has an inverse (here, the inverse of 1 is 1).

2.12.1 Vectors are subsets, and addition is XOR

The vector space $\mathbb{F}_2^n$ consists of binary strings $v=(v_1,\dots,v_n)$. Choose labels $e_1,\dots,e_n$ for some objects (for example, qubits). The vector

$$v=\sum_{i=1}^{n} v_i e_i \tag{B3}$$

can be read as the subset of labels for which $v_i=1$. Adding two vectors is bitwise XOR, so it takes the **symmetric difference** of their subsets: an element included twice disappears. For instance,

$$(1,0,1,1)+(0,1,1,0)=(1,1,0,1). \tag{B4}$$

This cancellation is the algebraic form of the Pauli facts $X_i^2=Z_i^2=I$. If a binary vector records the support of a product of Z's, then adding support vectors records multiplication of those Pauli products, with repeated factors cancelling.

2.12.2 Linear maps are binary matrices

A linear map $M:\mathbb{F}_2^n\to\mathbb{F}_2^m$ is an $m\times n$ matrix with entries 0 or 1. It acts by the usual matrix multiplication, but every sum is reduced modulo two. In particular,

$$(Mv)_a=\sum_{i=1}^{n} M_{ai}v_i \pmod 2. \tag{B5}$$

The entry $M_{ai}=1$ says that input label i is incident on, or is selected by, output label a. Hence such a matrix is often called an **incidence matrix**. The value $(Mv)_a=1$ means that output a sees an odd number of selected input labels; value 0 means that it sees an even number.

Three familiar linear-algebra notions will recur:

- The **image** $\operatorname{im} M$ is the set of outputs Mv that can be produced.
- The **kernel** $\ker M=\{v:Mv=0\}$ is the set of input patterns invisible to the map.
- The **rank** is $\dim\operatorname{im} M$. Rank–nullity still holds: $\dim\ker M+\operatorname{rank} M=n$.

As over the real numbers, these quantities can be found by Gaussian elimination. The only nontrivial row operation is "add one row to another," which is again XOR; there is no distinction between adding and subtracting a row.

2.12.3 The mod-two dot product and transpose

For $u,v\in\mathbb{F}_2^n$, define

$$u\cdot v=\sum_{i=1}^{n} u_i v_i \pmod 2. \tag{B6}$$

It equals 1 precisely when the two corresponding subsets overlap in an odd number of labels. We call u and v **orthogonal** when $u \cdot v = 0$, i.e. when their overlap is even. This is the CSS commutation test: the X-product with support u and the Z-product with support v commute exactly when $u \cdot v = 0$.

The transpose $M^{\mathsf{T}} : \mathbb{F}_2^m \to \mathbb{F}_2^n$ reverses the direction of an incidence matrix and obeys

$$u \cdot (Mv) = (M^{\mathsf{T}} u) \cdot v. \tag{B7}$$

This identity lets us translate freely between "which checks contain this qubit?" and "which qubits are selected by this collection of checks?"

2.12.4 The one equation to watch for

If two maps can be composed,

$$\mathbb{F}_2^r \xrightarrow{N} \mathbb{F}_2^n \xrightarrow{M} \mathbb{F}_2^m, \tag{B8}$$

then $MN = 0$ means: every column of N, after being tested by every row of M, has even overlap. Equivalently, $\operatorname{im} N \subseteq \ker M$. In the next section this becomes $\partial_1 \partial_2 = 0$. It is simultaneously the abstract statement "a boundary has no boundary" and the concrete CSS statement that every X-check and Z-check overlap an even number of times.

Nothing deeper than this binary bookkeeping is being assumed. The new names (*chains*, *cochains*, *boundary maps*) organize these same vectors and matrices so that the toric-code construction can be stated without relying on a lattice picture.

2.13 Section III.B: generalizing gauging to a qLDPC code

The toy model made the central design choice visible by drawing layers: independent ancilla layers gave independent symmetries, whereas one connected ancilla gave one global symmetry. A general qLDPC code need not come with layers, surfaces, or any other useful picture. This subsection replaces the picture with an algebraic description that works for any CSS code.

The paper's strategy is:

encode the incidence pattern of qubits and checks in a chain complex;
use that description to specify both the input code and the ancillary gauge system;
then impose the same symmetry-matching condition as in the toy model.

(91)

This is not a new physical mechanism. The measurement-based gauging mechanism remains Eq. (68). What changes is the language used to answer the question "which qubits and checks are connected to which?"

2.13.1 From labelled qubits to a CSS chain complex

We now make the word "sum" fully precise. Let Q, C_X, and C_Z be finite sets labelling, respectively, the physical qubits, the X-check generators, and the Z-check generators. For a finite set S, write $\mathbb{F}_2[S]$ for the free $\mathbb{F}_2$-vector space on S: it has one **canonical basis vector** $[s]$ for every $s \in S$. Thus the three spaces in this paper are, precisely,

$$D_2 = \mathbb{F}_2[C_Z], \qquad D_1 = \mathbb{F}_2[Q], \qquad D_0 = \mathbb{F}_2[C_X]. \tag{92}$$

For example, an element $b \in D_1$ is a formal binary sum

$$b = \sum_{q \in Q} b_q \, [q], \qquad b_q \in \mathbb{F}_2. \tag{93}$$

It is not a sum of physical qubits: it is a vector whose coefficient b_q records whether the label q is selected. In particular, $\sum_{q \in S}[q]$ means the vector with coefficient one on the subset $S \subseteq Q$ and zero elsewhere. This is exactly the XOR/symmetric- difference addition introduced in the interlude.

Choose two binary incidence matrices H_Z and H_X, or equivalently two linear maps,

$$0 \longrightarrow D_2 \xrightarrow{\partial_2} D_1 \xrightarrow{\partial_1} D_0 \longrightarrow 0, \qquad \begin{aligned} \partial_2[z] &= \sum_{q \in Q} (H_Z)_{qz} \, [q], \\ \partial_1[q] &= \sum_{x \in C_X} (H_X)_{xq} \, [x]. \end{aligned} \tag{94}$$

Here $[z]$ and $[x]$ are canonical basis vectors labelled by $z \in C_Z$ and $x \in C_X$. Therefore $(H_Z)_{qz} = 1$ means that qubit q belongs to Z-check z, while $(H_X)_{xq} = 1$ means that qubit q belongs to X-check x. The arrows in Eq. (94) are simply the two incidence matrices written as linear maps.

The notation becomes physical through two support-to-Pauli maps. As in the paper, we write $D^1 = \mathrm{Hom}_{\mathbb{F}_2}(D_1, \mathbb{F}_2)$ for the cochain space. Here $\mathrm{Hom}_{\mathbb{F}_2}(V, W)$ is the vector space of $\mathbb{F}_2$-linear maps from V to W; this will be unpacked carefully below. Let $\mathbf{q} \in D^1$ be the coordinate function defined by $\mathbf{q}([q']) = \delta_{q,q'}$. Then, for $b = \sum_q b_q[q] \in D_1$ and $\mathbf{a} = \sum_q a_q \mathbf{q} \in D^1$, define

$$\mathsf{Z}(b) := \prod_{q \in Q} Z_q^{b_q}, \qquad \mathsf{X}(\mathbf{a}) := \prod_{q \in Q} X_q^{a_q}. \tag{95}$$

The exponents are bits: exponent 0 means the identity and exponent 1 means the indicated Pauli. Consequently vector addition in D_1 corresponds to multiplication of Z-Pauli products, and cochain addition in D^1 corresponds to multiplication of X-Pauli products.

2.13.2 Why restrict to CSS qLDPC codes?

Using the paper's coboundary notation $d_0 : D^0 \to D^1$, defined carefully below, the actual check operators are therefore

$$S_z^Z = \mathsf{Z}(\partial_2[z]), \qquad S_x^X = \mathsf{X}(d_0 \mathbf{x}), \tag{96}$$

and a CSS stabilizer Hamiltonian is

$$H_{\mathrm{CSS}} = -\sum_{x \in C_X} S_x^X - \sum_{z \in C_Z} S_z^Z. \tag{97}$$

The paper writes the natural pairing of a cochain with a chain as evaluation: $\mathbf{a}(b) = \sum_q a_q b_q \pmod 2$. The Pauli commutation rule is exactly

$$\mathsf{X}(\mathbf{a})\mathsf{Z}(b) = (-1)^{\mathbf{a}(b)}\mathsf{Z}(b)\mathsf{X}(\mathbf{a}). \tag{98}$$

Thus S_x^X and S_z^Z commute precisely when $(d_0\mathbf{x})(\partial_2[z]) = 0$. By the transpose identity from Eq. (B7), this pairing is the (x, z) entry of the product $H_X H_Z$, equivalently the coefficient of $[x]$ in $\partial_1\partial_2[z]$. All CSS checks commute if and only if

$$\partial_1\partial_2 = 0 \qquad \text{(equivalently, } H_X H_Z = 0 \text{ over } \mathbb{F}_2\text{)}. \tag{99}$$

Read Eq. (99) from right to left. For one Z-check z, $\partial_2[z]$ is its binary support vector. Applying ∂_1 then counts, modulo two, how many qubits of that support belong to each X-check. The zero vector says that all those numbers are even.

The toric code is the standard geometric realization:

	basis label	support vector and operator
C_X	v (vertex)	$d_0\mathbf{v} = \sum_{e\ni v}\mathbf{e}, \quad S_v^X = \prod_{e\ni v} X_e$
C_Z	p (plaquette)	$\partial_2[p] = \sum_{e\in\partial p}[e], \quad S_p^Z = \prod_{e\in\partial p} Z_e$

(100)

The equality $\partial_1\partial_2 = 0$ says that a vertex and a square plaquette share zero or two edges. For a general CSS qLDPC code there may be no geometric vertex or plaquette, but the same matrix equation remains the complete commutation condition.

2.13.3 A tiny non-geometrical example

Let $Q = \{e_1, \ldots, e_5\}$, $C_Z = \{\beta\}$, and $C_X = \{\alpha_1, \alpha_2, \alpha_3\}$. Bracketed symbols below are the canonical basis vectors introduced above, whereas unbracketed symbols are labels. Choose the Z-check support vector to be

$$\partial_2[\beta] = [e_1] + [e_2] + [e_4] + [e_5]. \tag{101}$$

The corresponding physical stabilizer is not the vector on the left; it is the Pauli operator obtained from it using Eq. (95):

$$S_\beta^Z = \mathsf{Z}(\partial_2[\beta]) = Z_{e_1} Z_{e_2} Z_{e_4} Z_{e_5}. \tag{102}$$

Assume the other incidence map has the values

$$\begin{aligned} &\partial_1[e_1] = [\alpha_1] + [\alpha_2], \quad \partial_1[e_2] = [\alpha_2] + [\alpha_3], \\ &\partial_1[e_4] = [\alpha_1] + [\alpha_3], \quad \partial_1[e_5] = 0. \end{aligned} \tag{103}$$

Linearity over $\mathbb{F}_2$ now gives

$$\begin{aligned} \partial_1\partial_2[\beta] &= ([\alpha_1] + [\alpha_2]) + ([\alpha_2] + [\alpha_3]) + ([\alpha_1] + [\alpha_3]) + 0 \\ &= 2[\alpha_1] + 2[\alpha_2] + 2[\alpha_3] = 0 \qquad \text{in } D_0 \text{ over } \mathbb{F}_2. \end{aligned} \tag{104}$$

This says that every X-check overlaps the support of S_β^Z an even number of times. For example, α_1 sees e_1, e_4, and α_2 sees e_1, e_2. Let $\alpha_i \in D^0$ denote the indicator cochain for the label α_i, as defined generally

below. Consequently the physical X-check operators $S^X_{\alpha_i} = \mathsf{X}(d_0\alpha_i)$ commute with S^Z_β. The example has no drawing, but it has all the algebraic information the construction needs.

2.13.4 The dual language: why "cochains" will appear

The paper denotes the cochain spaces by upper indices:

$$D^0, \qquad D^1, \qquad D^2, \tag{105}$$

called **cochains**, and defines them by

$$D^n = \mathrm{Hom}_{\mathbb{F}_2}(D_n, \mathbb{F}_2). \tag{106}$$

Read Hom as "the set of homomorphisms," which in this vector-space setting means **linear maps**. Thus $\mathbf{a} \in D^n$ is a function $\mathbf{a} : D_n \to \mathbb{F}_2$ satisfying

$$\mathbf{a}(u+v) = \mathbf{a}(u) + \mathbf{a}(v), \qquad \mathbf{a}(cu) = c\,\mathbf{a}(u) \quad (u, v \in D_n,\ c \in \mathbb{F}_2). \tag{107}$$

Such functions form a vector space, with addition defined pointwise. For example, $(\mathbf{a}+\mathbf{b})(u) = \mathbf{a}(u) + \mathbf{b}(u)$. The paper uses boldface for cochains: for every qubit label q, the 1-cochain $\mathbf{q} \in D^1$ is the indicator function $\mathbf{q}([q']) = \delta_{q,q'}$. Hence $\mathbf{a} = \sum_q a_q \mathbf{q}$ is a row-vector way to select qubits, whereas $b = \sum_q b_q [q]$ is the corresponding column-vector chain.

In precisely the paper's notation, the associated cochain complex is

$$0 \longleftarrow D^2 \xleftarrow{d_1} D^1 \xleftarrow{d_0} D^0 \longleftarrow 0, \qquad d_n = \partial^{\mathsf{T}}_{n+1}. \tag{108}$$

The superscript on D^n reminds us that this is a cochain space, not the chain space D_n. The left-pointing arrows make the reversal of direction visible. In the bases introduced above, the co-boundary maps act as

$$d_0\mathbf{x} = \sum_{q \in Q} (H_X)_{xq}\mathbf{q}, \qquad d_1\mathbf{q} = \sum_{z \in C_Z} (H_Z)_{qz}\mathbf{z}, \tag{109}$$

where $\mathbf{x} \in D^0$ and $\mathbf{z} \in D^2$ are the analogous indicator cochains for check labels. Thus d_0 turns the cochain for one X-check label into the cochain selecting precisely the qubits in that check. While ∂_1 asks "which X-checks touch this qubit?", d_0 asks "which qubits are selected by this collection of X-checks?" Likewise, $d_1 : D^1 \to D^2$ takes a 1-cochain selecting qubits and returns a 2-cochain on Z-check labels. For a Z-check z,

$$(d_1\mathbf{a})([z]) = \mathbf{a}(\partial_2[z]). \tag{109a}$$

It equals 1 precisely when the selected qubits have **odd overlap** with the support of z. Thus d_1 asks "which Z-checks have odd overlap with this selected collection of qubits?" The shorter phrase "which Z-checks use these qubits?" is only an intuition: a Z-check containing two selected qubits gives 0 because all counting is modulo two. In the toric code, applying d_0 to one vertex cochain selects its incident edges—the support of the star operator.

Accordingly, the paper represents:

chain language	Pauli operator
$b = \sum_{e\in Q} b_e[e] \in D_1$	$\mathsf{Z}(b) = \prod_{e\in Q}(Z_e)^{b_e}$
$\mathbf{a} = \sum_{e\in Q} a_e\mathbf{e} \in D^1$	$\mathsf{X}(\mathbf{a}) = \prod_{e\in Q}(X_e)^{a_e}$

(110)

and Eq. (99) becomes $d_1 d_0 = 0$ after transposition. In the paper's function notation this is immediate: for $\mathbf{x} \in D^0$ and $p \in D_2$, $(d_1 d_0 \mathbf{x})(p) = \mathbf{x}(\partial_1\partial_2 p) = 0$.

2.13.4.1 Cocycles, coboundaries, and the quotient H^1 The word **cocycle** has a precise and very practical meaning here. A 1-cochain $\mathbf{a} \in D^1$ is a **1-cocycle** exactly when

$$d_1\mathbf{a} = 0. \tag{110g}$$

Equivalently, it belongs to the kernel of d_1,

$$\ker d_1 := \{\mathbf{a} \in D^1 \mid d_1\mathbf{a} = 0\}. \tag{110h}$$

This is not just a name inherited from topology. By Eq. (109a), each coefficient of $d_1\mathbf{a}$ records the parity of the overlap between the support of $\mathbf{a}$ and one Z-check. Therefore

$$\mathbf{a} \in \ker d_1 \quad \Longleftrightarrow \quad X_{\mathbf{a}} \text{ commutes with every (Z)-check.} \tag{110i}$$

So a 1-cocycle is precisely a candidate X-type logical operator: it has zero Z-syndrome. It need not look like a loop. "Cycle" in this word means only that the co-boundary map gives zero.

There is one further distinction. A **1-coboundary** is a 1-cochain of the form $d_0\mathbf{x}$ for some $\mathbf{x} \in D^0$. The set of all of them is the image of d_0:

$$\operatorname{im} d_0 := \{d_0\mathbf{x} \mid \mathbf{x} \in D^0\} \subseteq D^1. \tag{110j}$$

Physically, $\mathbf{x}$ selects a collection of X-check labels, and $X_{d_0\mathbf{x}}$ is the product of exactly those X-check stabilizers. It consequently acts as the identity on every code state. The identity $d_1 d_0 = 0$ says that every coboundary is automatically a cocycle:

$$\operatorname{im} d_0 \subseteq \ker d_1. \tag{110k}$$

This inclusion is what makes the quotient well-defined. The slash in

$$H^1(D_\bullet) := \ker d_1 / \operatorname{im} d_0 \tag{110l}$$

does **not** mean ordinary numerical division. Here is the precise definition. Abbreviate

$$V := \ker d_1, \qquad W := \operatorname{im} d_0. \tag{110l-a}$$

We have already proved $W \subseteq V$. For every cocycle $\mathbf{a} \in V$, its **coset** modulo W is the set

$$\mathbf{a} + W := \{\mathbf{a} + \mathbf{w} \mid \mathbf{w} \in W\}. \tag{110l-b}$$

In words: begin with $\mathbf{a}$, add every possible coboundary, and put all resulting cocycles into one set. The quotient vector space is the set of all these cosets:

$$V/W := \{\mathbf{a} + W \mid \mathbf{a} \in V\}. \tag{110l-c}$$

Thus an element of $H^1 = V/W$ is not a single cochain. It is one entire coset, i.e. a whole collection of cochains. Two cocycles represent the same quotient element precisely when their cosets are equal. Equivalently, they differ by an element of W. In the present notation,

$$\mathbf{a} \sim \mathbf{a}' \quad \Longleftrightarrow \quad \mathbf{a}' = \mathbf{a} + d_0\mathbf{x} \quad \text{for some } \mathbf{x} \in D^0. \tag{110m}$$

The notation $[\mathbf{a}]$ denotes the coset $\mathbf{a} + W = \mathbf{a} + \operatorname{im} d_0$, not an arbitrarily chosen representative $\mathbf{a}$. All cochains in this coset give Pauli operators that differ only by an X-stabilizer, and so have exactly the same action on the encoded subspace. A nonzero class in H^1 is therefore an independent logical X operator.

Mathematically, $H^1(D_\bullet)$ is a **quotient vector space over** $\mathbb{F}_2$. It is not a subset of $\ker d_1$: its elements are the cosets $[\mathbf{a}] = \mathbf{a} + W$. Since every vector space is an abelian group under addition, H^1 is also an abelian group. Its zero element is the coset

$$[0] = W = \operatorname{im} d_0, \tag{110l-c-i}$$

which contains *all* X-stabilizer products, not only the empty operator. The corresponding statement for the chain quotient $H_1 = \ker \partial_1 / \operatorname{im} \partial_2$ is identical: it is a vector space over $\mathbb{F}_2$, hence an abelian group, whose elements are cosets of Z-cycles modulo Z-stabilizer boundaries.

Addition in the quotient is defined on cosets by

$$[\mathbf{a}] + [\mathbf{b}] := [\mathbf{a} + \mathbf{b}]. \tag{110l-d}$$

This does not depend on which representatives are chosen. Indeed, replacing $\mathbf{a}$ by $\mathbf{a} + \mathbf{w}$ and $\mathbf{b}$ by $\mathbf{b} + \mathbf{w}'$, with $\mathbf{w}, \mathbf{w}' \in W$, changes the sum by $\mathbf{w} + \mathbf{w}' \in W$, which remains in the same coset. In short:

object	algebraic condition	physical meaning
1-cocycle $\mathbf{a}$	$d_1\mathbf{a} = 0$	$X_{\mathbf{a}}$ commutes with every (Z)-check
1-coboundary $d_0\mathbf{x}$	$\mathbf{x} \in D^0$	$X_{d_0\mathbf{x}}$ is a product of (X)-stabilizers
class $[\mathbf{a}] \in H^1$	$\mathbf{a} \in \ker d_1$ modulo $\operatorname{im} d_0$	a logical (X) action

(110n)

The later gauging construction also uses **0-cocycles**. These are not 1-cocycles placed at vertices; they are elements $\mu \in D^0$ with

$$d_0\mu = 0, \qquad \ker d_0 = \{\mu \in D^0 \mid d_0\mu = 0\}. \tag{110o}$$

Because this level-$(2+1)$ complex begins at D^0, there is no preceding co-boundary map whose image must be quotiented out at degree zero. Thus $H^0 = \ker d_0$ here. A 0-cocycle is a binary choice of X-check labels for which every qubit is selected an even number of times. Equivalently, the corresponding product of X-check stabilizers is the identity. In Section III.C these 0-cocycles become the ancilla subsystem symmetries and the relations among its final X-checks.

The notation is more general than the lattice picture, but its job is still only to track which qubits a Pauli product acts on.

2.13.5 A tiny non-geometrical example: cochains

Return to the five-qubit example above. Its single Z-check has support

$$\partial_2[\beta] = [e_1] + [e_2] + [e_4] + [e_5]. \tag{110a}$$

To make one cochain calculation fully definite, complete the previously unspecified incidence data by taking $\partial_1[e_3] = 0$. This choice does not change the earlier calculation: e_3 does not occur in the support of $\partial_2[\beta]$. Let $\alpha_2 \in D^0$ be the indicator cochain of the X-check label α_2. The displayed incidences in Eq. (103) say that this check contains e_1 and e_2, and no other qubit. Therefore its co-boundary is

$$d_0\alpha_2 = \mathbf{e}_1 + \mathbf{e}_2 =: \mathbf{a} \in D^1. \tag{110b}$$

The right-hand side is a 1-cochain, not a chain. It is the function whose value is one on $[e_1]$ and $[e_2]$, and zero on $[e_3]$, $[e_4]$, and $[e_5]$. Its associated physical Pauli operator is consequently

$$X_{\mathbf{a}} = X_{e_1} X_{e_2}. \tag{110c}$$

Now test whether $\mathbf{a}$ is a cocycle. There is only one Z-check label, so it is enough to evaluate $d_1\mathbf{a}$ on $[\beta]$:

$$\begin{aligned} (d_1\mathbf{a})([\beta]) &= \mathbf{a}(\partial_2[\beta]) \\ &= \mathbf{a}([e_1] + [e_2] + [e_4] + [e_5]) \\ &= 1 + 1 + 0 + 0 = 0 \qquad \text{in } \mathbb{F}_2. \end{aligned} \tag{110d}$$

Since D^2 has just the one basis label β, this proves $d_1\mathbf{a} = 0$. Equivalently, the X-operator in Eq. (110c) overlaps the Z-stabilizer $S^Z_\beta = Z_{e_1} Z_{e_2} Z_{e_4} Z_{e_5}$ twice, and therefore commutes with it.

This cocycle is not a new logical X operator, because Eq. (110b) shows that it is a co-boundary: $\mathbf{a} \in \operatorname{im} d_0$. The operator $X_{\mathbf{a}} = X_{e_1} X_{e_2}$ is exactly the X-stabilizer associated with α_2. This is the concrete reason for the quotient

$$H^1(D_\bullet) = \frac{\ker d_1}{\operatorname{im} d_0} : \tag{110e}$$

$\ker d_1$ first selects all X-operators that commute with every Z-check, and then quotienting by $\operatorname{im} d_0$ removes the ones that are already X-stabilizers. A cocycle not in $\operatorname{im} d_0$ would represent a logical X operator.

For contrast, consider the cochain $\mathbf{a}' = \mathbf{e}_1 + \mathbf{e}_3$. It gives

$$(d_1\mathbf{a}')([\beta]) = 1 + 0 = 1, \tag{110f}$$

so it is not a cocycle. Correspondingly, $X_{e_1}X_{e_3}$ has odd overlap with S^Z_β and anticommutes with it. The two examples show the full logic of the cochain complex: d_1 computes an X-operator's mod-two commutation syndrome against the Z-checks, while $\operatorname{im} d_0$ identifies the zero-syndrome operators that are already stabilizers.

2.13.6 What this formalism must accomplish for gauging

The authors now use the same kind of chain-complex data twice:

$$\begin{array}{rcl} D_\bullet & : & \text{the chain complex of the input } B, C \text{ qLDPC codes,} \\ A_\bullet & : & \text{the chain complex of the ancillary } a, A \text{ gauge system.} \end{array} \tag{111}$$

This notation does **not** say that $A_\bullet = D_\bullet$. The toy model already taught us why that equality can either hold or fail:

- for homological gauging, the ancilla tends to mirror the input structure, so $A_\bullet$ is chosen to be a copy or close relative of $D_\bullet$;
- for graph gauging, $A_\bullet$ is instead the chain complex of a chosen graph and can be very different from $D_\bullet$.

The rest of this subsection will formulate two questions in this language:

Question 1: Which transversal logical symmetry U of the two input codes is to be gauged?

Question 2: Which ancillary complex $A_\bullet$ has enough appropriate symmetry to gauge that U? (112)

These are exactly the qLDPC versions of the questions answered by the toy model. The chain-complex formalism does not choose the answer automatically; it gives a precise way to test whether a proposed ancilla structure and symmetry-enrichment circuit match.

2.13.7 What to retain before the next subsection

1. A CSS qLDPC code is sparse check–qubit incidence data satisfying an even-overlap rule.
2. The chain complex $D_2 \to D_1 \to D_0$ packages Z-checks, qubits, and X-checks, respectively.
3. $\partial_1\partial_2 = 0$ is the algebraic version of "a boundary has no boundary"; physically, it proves X- and Z-checks commute.
4. d_1 maps the support of an X-operator to its mod-two syndrome against the Z-checks. A 1-cocycle is an element of $\ker d_1$, so its X-operator has zero such syndrome.
5. $\operatorname{im} d_0$ consists of X-stabilizer products. Hence $H^1 = \ker d_1 / \operatorname{im} d_0$ labels zero-syndrome X-actions modulo stabilizers: the logical X operators.
6. Gauging requires both an input complex $D_\bullet$ and an ancilla complex $A_\bullet$. Their relation determines whether the homological or graph construction is appropriate.

2.14 Section III.C: symmetries and gauging in qLDPC codes

We can now state the problem without referring to a square lattice. Take two identical CSS qLDPC codes, called B and C, with common chain complex

$$0 \longrightarrow D_2 \xrightarrow{\partial_2} D_1 \xrightarrow{\partial_1} D_0 \longrightarrow 0. \tag{113}$$

The purpose of this subsection is not to invent a new measurement protocol. The protocol is still the one encountered in Section II: prepare an ancilla cluster state, entangle it with the two input codes in a symmetry-respecting way, and measure the temporary ancillas in the X basis. What changes is the bookkeeping. The paper asks two questions:

1. Which transversal Clifford symmetry of the pair B, C should be gauged?
2. Which ancilla chain complex has the right subsystem symmetries to gauge that symmetry?

(114)

The answers are linked: choosing the symmetry constrains the possible ancilla, and choosing the ancilla determines how many independently addressable symmetries can be gauged.

2.14.1 First distinction: logical-string symmetries are not the target

Every CSS code described by Eq. (113) has two familiar kinds of logical Pauli operator. In the paper's chain notation, their labels are

$$H_1(D_\bullet) = \frac{\ker \partial_1}{\operatorname{im} \partial_2}, \qquad H^1(D_\bullet) = \frac{\ker d_1}{\operatorname{im} d_0}. \tag{115}$$

Here is a direct reading of the first quotient. A 1-chain $\eta \in D_1$ satisfying $\partial_1 \eta = 0$ is a **cycle**. Its operator Z_η commutes with all X-checks. But if $\eta = \partial_2 p$ for some $p \in D_2$, then Z_η is a product of Z-stabilizers and acts trivially on the encoded space. The quotient therefore identifies cycles which differ by a stabilizer boundary. Its **nonzero** classes label nontrivial logical Z operators. The second quotient says the dual thing: a 1-cocycle $\gamma \in \ker d_1$, modulo the co-boundaries $\operatorname{im} d_0$, labels a logical X operator X_γ.

2.14.1.1 Where do the stabilizer boundaries go? They are not absent from $H_1(D_\bullet)$. Instead, all of them become the *same* element of the quotient: its zero class. Indeed, the chain condition $\partial_1 \partial_2 = 0$ implies

$$\operatorname{im} \partial_2 \subseteq \ker \partial_1. \tag{115a}$$

Thus a boundary $b = \partial_2 p$ is certainly an allowed cycle in the numerator. But its class in the quotient is

$$[b] = b + \operatorname{im} \partial_2 = \partial_2 p + \operatorname{im} \partial_2 = \operatorname{im} \partial_2 = [0]. \tag{115b}$$

The notation $[b]$ means the coset containing b, not the original chain by itself. Equation (115b) says that a Z-stabilizer product and the empty operator represent the same *logical* action: both act as the identity on every encoded state. Consequently, it would be incorrect to say that quotienting simply throws stabilizers away. It groups every stabilizer boundary into the identity logical operation.

For a minimal abstract example, suppose

$$\ker \partial_1 = \{0, b, \ell, \ell + b\}, \qquad \operatorname{im} \partial_2 = \{0, b\}. \tag{115c}$$

Here b is a stabilizer boundary and ℓ is a cycle which is not a boundary. There are only two quotient classes:

$$H_1(D_\bullet) = \underbrace{\{0, b\}}_{[0]\text{: identity logical action}} \cup \underbrace{\{\ell, \ell + b\}}_{[\ell]\text{: one nontrivial logical } Z}. \tag{115d}$$

Multiplying Z_ℓ by the stabilizer Z_b changes its representative to $Z_{\ell+b}$, but cannot change what it does to the encoded qubits. The quotient records exactly this physical equivalence. In summary,

$$\begin{array}{c|l} \ker \partial_1 & \text{all } Z\text{-strings with no } X\text{-syndrome} \\ \operatorname{im} \partial_2 & \text{the syndrome-free } Z\text{-strings that are stabilizers} \\ H_1(D_\bullet) & \text{logical } Z \text{ actions modulo } Z\text{-stabilizers.} \end{array} \tag{115e}$$

So the zero class contains all the stabilizers; the nonzero classes are the genuinely distinct logical Z operators. For the toric code, a contractible closed loop belongs to the zero class, while a loop winding around the torus represents a nonzero class.

For a toric code these are the usual noncontractible loop operators. The paper calls them 1-cycle and 1-cocycle symmetries, generalizing the toric code's 1-form symmetries. In a qLDPC code they need not look like geometric loops or surfaces; η and γ are simply sparse or non-sparse binary vectors satisfying the displayed kernel conditions.

These logical Pauli symmetries are **not** what this construction gauges. The target is a transversal Clifford gate U acting on the two copies B, C. "Transversal" here means that it factors into gates of bounded size acting on corresponding or otherwise locally prescribed physical qubits. It can act on every qubit, or only on a specified subsystem. The important condition is that it preserves the code space: under conjugation, it maps each stabilizer to a product of stabilizers. It need not fix each Hamiltonian term individually. Thus it is an **emergent symmetry** of the encoded subspace—the same sense in which the controlled-Z in Section II.A was an emergent symmetry.

This distinction matters. A logical Pauli Z_η is an operator *inside* one code's protected logical algebra. A transversal Clifford U is a physical circuit that *acts on* that logical algebra, for example by permuting its logical Pauli operators. It is the latter action that the gauging construction promotes to a dynamical gauge action.

2.14.2 The ancilla is another chain complex

Replace the square-lattice ancilla by an arbitrary level-$(2+1)$ chain complex

$$0 \longrightarrow A_2 \xrightarrow{\partial_2^a} A_1 \xrightarrow{\partial_1^a} A_0 \longrightarrow 0. \tag{116}$$

Let V_A label a basis of A_0 and E_A a basis of A_1. The construction places one temporary ancilla qubit a_v at each $v \in V_A$, and one surviving gauge-field qubit A_e at each $e \in E_A$. The labels "vertex" and "edge" are only mnemonic: for a general qLDPC complex, they mean a 0-chain and a 1-chain basis label, not a location in ordinary space.

For a vertex label v, let $\mathbf{v} \in A^0$ be its indicator cochain. For an edge label e, let $[e] \in A_1$ be its chain basis vector. The ancilla cluster state is the simultaneous $+1$ eigenspace of the following two kinds of stabilizer:

$$K_v^a = X_{a_v} X^A_{d_0^a \mathbf{v}}, \qquad K_e^A = Z_{A_e} Z^a_{\partial_1^a[e]}. \tag{117}$$

This compact expression has a concrete meaning. If $d_0^a \mathbf{v} = \sum_e c_e \mathbf{e}$, then $X^A_{d_0^a \mathbf{v}} = \prod_e (X_{A_e})^{c_e}$: it acts on all gauge-field qubits incident on v. If $\partial_1^a[e] = \sum_v r_v[v]$, then $Z^a_{\partial_1^a[e]} = \prod_v (Z_{a_v})^{r_v}$: it acts on all temporary ancillas incident on e. These are exactly the ordinary graph-cluster stabilizers, expressed through the incidence maps of $A_\bullet$.

2.14.2.1 Reading Eq. (117) one edge at a time It is worth slowing down here, because the superscripts and the maps in Eq. (117) encode three different ideas at once. The superscript a means "temporary vertex-ancilla qubit," whereas the superscript A means "surviving gauge-field qubit." Thus X_{a_v} flips the qubit placed at the vertex label v, while X_{A_e} flips the qubit placed at the edge label e. The subscript $d_0^a \mathbf{v}$ does *not* name one qubit. It names a binary 1-cochain, and X^A turns that binary list into a product of X's on the selected A qubits.

For an ordinary graph, suppose that the edges incident on v are $e_1, \dots, e_r$. Then the coboundary of the indicator cochain of v is

$$d_0^a \mathbf{v} = \mathbf{e}_1 + \cdots + \mathbf{e}_r \quad \text{in } A^1,$$

where addition is in $\mathbb{F}_2$. Consequently Eq. (117)'s first operator is nothing more mysterious than

$$K_v^a = X_{a_v} X_{A_{e_1}} \cdots X_{A_{e_r}} = X_{a_v} \prod_{e \ni v} X_{A_e}.$$

It is one stabilizer for vertex v: it flips that vertex ancilla and every gauge-field qubit on an edge touching it.

Now take one edge e with endpoints u and w. Its boundary is

$$\partial_1^a[e] = [u] + [w] \quad \text{in } A_0.$$

The second operator of Eq. (117) therefore reads

$$K_e^A = Z_{A_e} Z_{a_u} Z_{a_w}.$$

It is one stabilizer for edge e: it compares the Z-parity of the gauge-field qubit on the edge with the Z-parity of the two temporary ancillas at its endpoints. In particular, for the one-edge graph

```
u  --- e ---  w
```

the complete set of cluster stabilizers is

$$K_u^a = X_{a_u} X_{A_e}, \qquad K_w^a = X_{a_w} X_{A_e}, \qquad K_e^A = Z_{A_e} Z_{a_u} Z_{a_w}.$$

This tiny example is also a useful check that Eq. (117) is consistent. For example, K_u^a and K_e^A overlap on two qubits: on a_u, where they contain X and Z, and on A_e, where they again contain X and Z. Each overlap supplies a minus sign on swapping the operators, and the two signs cancel. Hence the stabilizers commute.

They commute. For an incident pair (v, e), K_v^a and K_e^A have one X/Z overlap on a_v and a second one on A_e; the two minus signs cancel. For a nonincident pair they have no overlap. No spatial lattice is needed for this argument—only the common incidence coefficient $(\partial_1^a)_{ve}$.

2.14.3 The two ancilla symmetries, and why cocycles matter

The cluster state has two relevant families of symmetries.

First, if $\eta \in \ker \partial_1^a \subseteq A_1$ is an ancilla 1-cycle, then

$$G_A(\eta) = Z_\eta^A \tag{118}$$

is a symmetry. Indeed, its possible sign when commuting past K_v^a is the mod-two overlap $(d_0^a \mathbf{v})(\eta) = \mathbf{v}(\partial_1^a \eta) = 0$. These are the ancilla's 1-cycle symmetries and become the familiar magnetic-type structure of the gauge field.

Second, if $\mu \in \ker d_0^a \subseteq A^0$ is a 0-cocycle, then

$$G_a(\mu) = X_\mu^a = \prod_{v \in V_A} (X_{a_v})^{\mu([v])} \tag{119}$$

is a symmetry acting only on the temporary a qubits. To see it, commute this operator through K_e^A. The sign is $\mu(\partial_1^a[e]) = (d_0^a \mu)([e]) = 0$. It clearly commutes with all K_v^a, which are X-type on the a qubits. These are called **0-cocycle subsystem symmetries** because a nonzero μ can select only part of the set of a qubits.

Why is the kernel condition more than a formal label? After measuring every a_v in the X basis and correcting to the $+1$ branch, the first family of cluster stabilizers becomes the X-check family $X^A_{d_0^a \mathbf{v}}$ on the remaining A qubits. Multiplying these checks according to a cocycle gives

$$\prod_{v \in V_A} \left(X^A_{d_0^a \mathbf{v}}\right)^{\mu([v])} = X^A_{d_0^a \mu} = I \qquad (\mu \in \ker d_0^a). \tag{120}$$

Thus μ labels a relation, or redundancy, among the remaining X-checks. On a single connected square lattice, the familiar example is the product of all star checks, which is the identity. In the layer-resolved toy model, each independent layer supplied one such relation. This is exactly why the number and shape of 0-cocycles determine how many symmetry actions the ancilla can keep track of.

2.14.4 Symmetry enrichment: attaching an input-code gate to each cocycle

Let $\Omega_{a,BC}$ denote the finite-depth circuit that entangles the ancilla with the two input codes. The central condition of the subsection is

$$\Omega_{a,BC}\, X_\mu^a\, \Omega_{a,BC}^\dagger = X_\mu^a\, U_\mu \qquad \text{for every } \mu \in \ker d_0^a. \tag{121}$$

The left side asks what happens to an exact ancilla symmetry under the entangling circuit. The right side says that it remains a symmetry, but is now **decorated** by a transversal Clifford operation U_μ on the B, C codes. This is the abstract qLDPC version of the partial controlled-Z patch in Section II: a transformation in the ancilla/gauge sector is tied to a type-changing symmetry action in the matter sector.

Equation (121) must hold for every cocycle, not merely for one chosen generator. It is allowed that several cocycles produce the same gate, or that some produce $U_\mu = I$. At least one must produce a nontrivial encoded action, otherwise the gauging is physically trivial. The linearity of the left side also forces compatible gates to compose consistently: because $X^a_{\mu+\nu} = X^a_\mu X^a_\nu$, the encoded actions must satisfy the corresponding product rule on the code space (up to stabilizers, which act trivially there).

Once Eq. (121) is satisfied, the temporary a qubits are projected into $|+\rangle$ states. As in Section II.C, actual X-measurements can give -1 outcomes; a classically chosen Pauli correction returns the desired stabilizer sector. The paper later proves this deterministic feedforward step in the general setting. What remains after removing the a qubits is a gauge-field code on A, coupled to the input-code symmetry that has been gauged.

2.14.5 What to retain before the two constructions

1. The qLDPC construction gauges transversal Clifford **emergent symmetries**, not the logical Pauli operators labelled by H_1 and H^1.
2. The ancilla complex $A_\bullet$ supplies temporary vertex qubits a, surviving gauge-field qubits A, and a cluster state specified entirely by ∂^a_1 and d^a_0.
3. Every ancilla 0-cocycle $\mu \in \ker d^a_0$ is both an X-symmetry of the temporary ancillas and a relation among the final A-sublattice X-checks.
4. Symmetry enrichment attaches a transversal gate U_μ to each such cocycle through Eq. (121).
5. The next subsection compares two ways to choose $A_\bullet$: a homological choice that mirrors $D_\bullet$, and a graph choice that needs only the symmetry required for one global gate.

2.15 Section III.D: two approaches to gauging qLDPC codes

This short subsection is a decision point, not yet the full construction of either resulting code. Section III.C established the common requirement: every ancilla 0-cocycle $\mu \in \ker d^a_0$ must be decorated by an input-code transversal gate through

$$\Omega_{a,BC} X^a_\mu \Omega^\dagger_{a,BC} = X^a_\mu U_\mu. \tag{122}$$

The choice of ancilla complex $A_\bullet$ determines the available 0-cocycles, hence the set of gates that Eq. (122) must encode. The paper develops two natural choices. They are not competing approximations; they are designed for different symmetry resources supplied by the input code.

	homological gauging	graph gauging
ancilla complex	$A_\bullet \simeq D_\bullet$	$A_\bullet$ is a connected graph complex
0-cocycle symmetries	a structured family	one global generator
what is gauged	a compatible collection of gates	one transversal gate
extra input structure	local Poincare duality	none beyond the chosen gate
canonical example later	addressable transversal CZ's	transversal SWAP

(123)

The word "structured" in the second row is deliberately cautious: the relevant number is not simply the number of physical ancilla qubits, but the dimension and support structure of $\ker d^a_0$. These cocycles label the independent relations among final ancilla X-checks, and therefore the independent symmetry sectors that can be enriched.

2.15.1 Homological gauging: preserve the input's higher-dimensional structure

The homological route applies when the input complex $D_\bullet$ has a suitable algebraic version of **Poincare duality**. The paper requires a *local isomorphism* between

$$H^2(D_\bullet) \quad \text{and} \quad H_0(D_\bullet). \tag{124}$$

At this stage, the most useful reading is operational. An isomorphism says that these two spaces carry the same independent information; *local* says that the identification can be implemented using bounded-support incidence data, rather than by a dense matrix reaching across the whole code. It is much stronger than the bare numerical statement $\dim H^2 = \dim H_0$. The later construction uses this local dictionary to turn the check-redundancy data of one type into the local support data needed to define transversal controlled gates.

For intuition, a two-dimensional toric-code lattice has a dual lattice: vertices of one lattice correspond locally to plaquettes of the other. Poincare duality abstracts precisely the part of this picture that survives without drawing a surface. A generic qLDPC code need not possess such a local primal–dual dictionary, even though its chain complex still obeys $\partial_1\partial_2 = 0$.

When the dictionary does exist, the authors choose an ancilla that mirrors the input complex,

$$A_\bullet \simeq D_\bullet. \tag{125}$$

This does not mean that the ancilla qubits are the input qubits. It means that their check–qubit incidence pattern is an isomorphic copy of the one used by B and C. Consequently its 0-cocycles have the same structured family of relations that the duality can connect to collections of transversal gates on B, C. Eq. (122) can then enrich different μ's by different, compatible gates. This is the sense in which the resulting symmetries are **addressable**: one is not limited to gauging only the product of all available gates at once.

The later Poincare-duality section makes this statement concrete using transversal controlled-Z gates. For now, retain the logical order:

$$\text{local duality of } D_\bullet \implies \text{structured 0-cocycles of } A_\bullet \simeq D_\bullet \implies \text{a compatible family of enrichable gates.} \tag{126}$$

2.15.2 Graph gauging: retain only one global symmetry

Suppose instead that the input code offers one useful transversal gate, but does not come with a local Poincare-duality structure or a decomposition into independently addressable pieces. The authors then choose $A_\bullet$ to be the chain complex of a **connected graph**:

$$0 \longrightarrow A_1 \xrightarrow{\partial_1^a} A_0 \longrightarrow 0. \tag{127}$$

There is no A_2 in an ordinary graph. Put a temporary a qubit on each graph vertex and a surviving A gauge qubit on each graph edge, as before. Over $\mathbb{F}_2$, a connected graph has one independent 0-cocycle, represented by the uniform vertex cochain

$$\mu_{\text{all}} = \sum_{v \in V_A} \mathbf{v}, \qquad d_0^a \mu_{\text{all}} = 0. \tag{128}$$

The second equation is easy to verify. Each edge has two endpoints. When d_0^a tests the cochain on that edge, it adds one contribution from each endpoint, obtaining $1 + 1 = 0$ in $\mathbb{F}_2$. Conversely, on a connected graph the condition $d_0^a \mu = 0$ forces neighbouring vertices to have equal values; connectivity then forces every vertex to have the same value. Thus the kernel is spanned by μ_{all}.

The enrichment condition now contains only one nontrivial instruction:

$$\Omega_{a,BC} X^a_{\mu_{\text{all}}} \Omega^\dagger_{a,BC} = X^a_{\mu_{\text{all}}} U. \tag{129}$$

This construction can gauge that one gate U, but it cannot separately gauge a family $U_1, U_2, ...$: the necessary independent 0-cocycle symmetries are absent. This is exactly the distinction already visible in the stacked-toric-code toy model, now expressed without layers.

There is an important universal example. Any two identical quantum codes have a physical transversal swap:

$$U_{\mathsf{SWAP}} = \prod_{q \in Q} \mathsf{SWAP}(B_q, C_q). \tag{130}$$

It exchanges every physical qubit of copy B with its counterpart in copy C, so it maps each B stabilizer to the corresponding C stabilizer and vice versa. It is therefore an emergent symmetry of the two- copy code space. Graph gauging is available whenever this one global gate can be matched by a suitable symmetry-enrichment circuit; it requires no Poincare duality of $D_\bullet$.

"Choose a graph" does not mean choose one arbitrarily. Its incidence matrix must still let a finite-depth $\Omega_{a,BC}$ satisfy Eq. (129), and it must retain bounded-weight checks if the final code is to remain qLDPC. Hence the appropriate graph depends on the input complex and on the gate being gauged.

2.15.3 The practical decision rule

Before reading the two detailed constructions, use the following rule.

1. If the input code has a local Poincare-duality structure and the goal is to gauge a structured, addressable family of transversal gates, follow **homological gauging**.
2. If the goal is to gauge one global transversal gate—especially the inevitable swap between two identical copies—and no usable local duality is available, follow **graph gauging**.

Both routes use the same physical mechanism: an ancilla cluster state, symmetry enrichment, X-measurement of temporary ancillas, and feedforward. They differ only in the symmetry algebra deliberately built into the ancilla.

2.16 Section III.E: non-Abelian codes from gauging with graph states

The graph-state construction is the paper's general-purpose route when the input qLDPC code has a useful **single global** transversal symmetry but does not have the local Poincare-duality structure required by homological gauging. Its central idea is simple:

replace the input's missing geometric duality by a deliberately chosen connected ancilla graph. (131)

The graph is not meant to represent the geometry of the qLDPC code. It is an auxiliary device whose one global 0-cocycle symmetry supplies exactly one symmetry generator to gauge. The input code contributes the transversal gate; the graph contributes the gauge-field degrees of freedom and the symmetry that is tied to that gate.

2.16.1 What this section does, and does not, assume

We again begin with two identical CSS qLDPC codes B and C, whose common check–qubit incidence data are described by $D_\bullet$. Suppose there is a transversal operation U on the pair that has $\mathbb{Z}_2$ character: applying it twice acts trivially on the code space. The canonical case is

$$U_{\mathsf{SWAP}} = \prod_{q \in Q} \mathsf{SWAP}(B_q, C_q). \tag{132}$$

No Poincare duality, cup product, or layer decomposition of $D_\bullet$ is assumed. This is why the graph route is more broadly applicable than the homological one. But "more general" does not mean automatic: we must still choose a graph and an entangling circuit that meet three precise conditions.

The distinction from the earlier homological route is useful to keep in view:

$$\begin{aligned} \text{homological route: input local primal–dual structure and a gate family} \\ \iff \text{ancilla: a matching copy of that structure;} \\ \text{graph route: input one uniform transversal } \mathbb{Z}_2 \text{ gate} \\ \iff \text{ancilla: one global symmetry of a chosen graph.} \end{aligned} \tag{133}$$

2.16.2 The graph ancilla, written without a picture

Choose a finite connected graph $G = (V, E)$. Its chain complex is just

$$0 \longrightarrow A_1 = \mathbb{F}_2[E] \xrightarrow{\partial_1^a} A_0 = \mathbb{F}_2[V] \longrightarrow 0. \tag{134}$$

For an edge $e = \langle v, w \rangle$, the boundary map is

$$\partial_1^a[e] = [v] + [w]. \tag{135}$$

The ancilla has two kinds of qubit:

- a temporary qubit a_v at every graph vertex $v \in V$;
- a gauge-field qubit A_e at every graph edge $e \in E$.

The notation is the same as before, but now it has an ordinary graph interpretation. Let $\mathbf{v} \in A^0$ be the indicator cochain at vertex v. Since $d_0^a = (\partial_1^a)^\mathsf{T}$, $d_0^a \mathbf{v}$ selects exactly the graph edges incident on v. The graph cluster state is stabilized by

$$K_v^a = X_{a_v} \prod_{e \ni v} X_{A_e}, \qquad K^A_{\langle v,w\rangle} = Z_{A_{\langle v,w\rangle}} Z_{a_v} Z_{a_w}. \tag{136}$$

This is Eq. (117) specialized to an ordinary graph. The first stabilizer ties a vertex qubit to the incident gauge-field qubits. The second ties an edge gauge qubit to the two endpoint vertex qubits. For an endpoint–edge pair, the two stabilizers have two anticommuting overlaps, one on the vertex qubit and one on the edge qubit, so they commute overall.

2.16.3 Why connectedness produces exactly one symmetry to gauge

The temporary vertex qubits have a global X-symmetry

$$X^a_{\mu_{\text{all}}} = \prod_{v \in V} X_{a_v}, \qquad \mu_{\text{all}} = \sum_{v \in V} \mathbf{v}. \tag{137}$$

It is a 0-cocycle because each edge sees both of its endpoints:

$$(d_0^a \mu_{\text{all}})([\langle v,w\rangle]) = 1 + 1 = 0 \qquad \text{in } \mathbb{F}_2. \tag{138}$$

Conversely, let $\mu \in \ker d_0^a$. Equation (138), with 1's replaced by the values of μ, says that the values of μ at the endpoints of every edge are equal. If the graph is connected, that equality propagates along paths, so all vertices have the same value. Therefore

$$H^0(A_\bullet) = \ker d_0^a = \text{span}_{\mathbb{F}_2}\{\mu_{\text{all}}\} \cong \mathbb{Z}_2. \tag{139}$$

This proof explains the role of connectedness more deeply than the slogan "one connected graph gives one symmetry." A graph with r connected components has r independent uniform cocycles, one per component. The authors deliberately take $r = 1$, because the graph construction is intended to gauge one global gate rather than an independently addressable family.

2.16.4 Symmetry enrichment becomes a group-commutator test

The graph's sole nontrivial cocycle must be linked to the desired input-code gate. In the notation of Section III.C, the condition is

$$\Omega_{a,BC} X^a_{\mu_{\text{all}}} \Omega^\dagger_{a,BC} = X^a_{\mu_{\text{all}}} U. \tag{140}$$

Because $(X^a_{\mu_{\text{all}}})^2 = I$, multiply Eq. (140) on the left by $X^a_{\mu_{\text{all}}}$ to obtain the equivalent form used in the paper:

$$X^a_{\mu_{\text{all}}}\, \Omega_{a,BC}\, X^a_{\mu_{\text{all}}}\, \Omega^\dagger_{a,BC} = U. \tag{141}$$

The left-hand side is a group commutator: it measures the failure of $\Omega_{a,BC}$ to commute with the global ancilla symmetry. The required failure is not an error. It is exactly the desired physical statement: flipping every temporary ancilla vertex performs the transversal gate U on the two matter codes. After the a qubits are measured and removed, this relation is what makes U a gauged, rather than externally imposed, symmetry.

2.16.5 The three graph requirements

The paper's requirements can now be stated sharply.

requirement	reason
connected graph	$\ker d_0^a \cong \mathbb{Z}_2$, so the ancilla supplies exactly the one global symmetry intended for gauging.
enrichment circuit Ω	It must satisfy Eq. (140), or equivalently Eq. (141), so the global cocycle symmetry implements the chosen transversal gate U.
LDPC preservation	Every final stabilizer must have bounded weight, and every qubit must occur in only a bounded number of final stabilizers. This constrains both graph degrees and the locality/sparsity of Ω.

(142)

The final row is essential. A connected graph always supplies the desired one-dimensional symmetry, but a dense graph can give a high-weight vertex stabilizer K_v^a, and a nonlocal enrichment circuit can spread one input check across too many ancilla qubits. Either failure would destroy the LDPC property, even though the abstract symmetry condition was correct.

The next subsection gives the promised worked example: gauge U_{SWAP}. It will introduce a sparse assignment from input code data to graph vertices, construct $\Omega_{a,BC}$ explicitly, and derive the resulting non-Abelian stabilizers.

2.17 Section III.E.1: example - gauging the transversal SWAP gate

We now work through the paper's first graph-state example. Its target symmetry is the transversal interchange of the two input copies:

$$U_{\mathsf{SWAP}} = \prod_{e \in Q} \mathsf{SWAP}_{B_e, C_e}. \tag{143}$$

Here Q is the set of input qubits. Each $e \in Q$ labels one qubit in copy B and its counterpart in copy C. The two-qubit operator SWAP_{B_e,C_e} exchanges their states:

$$\mathsf{SWAP}_{B_e,C_e}(|b\rangle_{B_e}|c\rangle_{C_e}) = |c\rangle_{B_e}|b\rangle_{C_e}. \tag{144}$$

The product in Eq. (143) is unambiguous because its factors act on disjoint pairs. The word *transversal* means precisely this: the gate is a product of identical, pairwise local gates, one per input-qubit label. If the two copies use the same CSS code, this gate exchanges the entire two code blocks without spreading a single physical error into many qubits within either block.

The aim is not merely to implement Eq. (143). It is to make it the global $\mathbb{Z}_2$ symmetry which is subsequently gauged by the graph-state ancilla. The result is the qLDPC analogue of the layer-exchange gauging that led to non-Abelian excitations in Section II.

2.17.1 Reading the enrichment circuit in practice

The formula for Ω looks more formidable than the operation it describes. Start by keeping three label sets apart:

- Q labels the input qubits. For each $e \in Q$, there is one qubit B_e and one qubit C_e, which are the two qubits to be swapped.
- V labels vertices of the new auxiliary graph. For each $v \in V$, there is a temporary ancilla qubit a_v. This qubit is used as a control and is measured away later.
- E labels edges of that graph. For each graph edge $f \in E$, there is a surviving gauge qubit A_f. These edge qubits are not the controls in the SWAP circuit; they become important after the measurement.

The paper sometimes uses e for both an input-code edge/qubit and a graph edge. In this note, e always means an input-qubit label in this example, and f means a graph-edge label.

Now choose, for each input qubit e, which temporary ancilla should control its swap. Call that vertex v_e. This choice is what the paper calls ϕ:

$$\phi(e) = v_e.$$

Read this sentence literally:

> The temporary qubit a_{v_e} controls the swap of B_e and C_e.

The occupation number

$$n^a_{v_e} = \frac{I - Z^a_{v_e}}{2}$$

has value 0 when a_{v_e} is $|0\rangle$, and value 1 when it is $|1\rangle$. Therefore the enrichment circuit is simply

$$\Omega_{a,BC} = \prod_{e \in Q} \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n^a_{v_e}}.$$

In words: for every input-qubit pair (B_e, C_e), do a controlled SWAP, with a_{v_e} as its control. This is the whole practical definition of symmetry enrichment in this example.

So where does the paper's integral $\int_{\phi(\mathbf{e})} v$ come from? It is only a compact way to write the preceding choice while keeping the product over *all* vertices $v \in V$. Its value is

$$\int_{\phi(\mathbf{e})} v = \begin{cases} 1, & v = v_e, \\ 0, & v \neq v_e. \end{cases}$$

Thus, in the paper's double product, every vertex gives the identity except the one chosen by $\phi(e)$. The integral is not a spatial integral. It is a binary yes/no test: "is this particular vertex v the chosen controller for input qubit e?" Substituting the two possible answers into the paper's expression immediately recovers the simple controlled-SWAP circuit above.

The paper allows a more general choice in which $\phi(e)$ names several vertices. Then several temporary qubits control the same SWAP. Its condition that $\int_{\phi(\mathbf{e})} \mu = 1$ says that their number must be odd. The authors then choose the simplest case: exactly one controller v_e for every e.

One piece of notation is implicit in the paper. It says that ϕ maps into D_0, even though the controls live at vertices of the distinct ancilla graph. To read the formula, identify the relevant basis labels of D_0 with the graph vertices V, or equivalently supply a map from those labels to V. The paper does not

display that identification, so we will always state it in words as "$\phi(e)$ chooses the controlling ancilla vertex.'' Likewise, its use of both D^1 and D_1 is a way of saying that the same qubit-label assignment is used whether e is encountered in an X-check or a Z-check.

2.17.2 A warning about labels: e, e, and ϕ

This example uses the chain-complex notation introduced above, but it is easy to lose track of what is a qubit and what is a binary vector. The following dictionary is useful.

paper notation	meaning in this example
$e \in Q$	one physical input-qubit label (equivalently, a basis 1-cell)
$[e] \in D_1$	the corresponding basis vector, regarded as a chain
$\mathbf{e} \in D^1$	the coordinate cochain which reads whether $[e]$ is present
$v \in V$	one vertex of the temporary-ancilla graph
$v \in D^0$	the associated coordinate 0-cochain
ϕ	the sparse rule which assigns the qubit e to an ancilla vertex.

(149)

The paper writes

$$\phi : D^1 \oplus D_1 \longrightarrow D_0. \tag{150}$$

At first sight this may look strange: why can ϕ accept both a cochain and a chain? The construction encounters an input-qubit label in two complementary ways. An X-type CSS check is naturally described by a 1-*cochain* $\mathbf{e}$, while a Z-type check is naturally described by a 1-*chain* $[e]$. Once bases have been chosen, both carry the same finite list of qubit labels. Equation (150) records that the same sparse incidence rule is used in both appearances. Operationally, you may remember just the basis-level statement

$$e \longmapsto \phi(e) = v_e \in V : \qquad \text{"qubit } e \text{ is controlled by temporary ancilla } a_{v_e}\text{."} \tag{151}$$

As explained in the preceding practical guide, the paper tacitly identifies the output D_0 of ϕ with labels of ancilla-graph vertices. More formally, one must choose a labelled copy of the basis of D_0 as V, or supply a map into A_0. This is why a symbol such as $\phi(e)$ may be used both as a 0-cell label and as a graph vertex. It is a label, not an additional quantum degree of freedom.

The map is required to be sparse and local. Sparsity says that its binary matrix has only $O(1)$ nonzero entries in each relevant row and column. The stronger geometric condition mentioned by the paper is

$$\phi(\mathbf{e}) \in \partial[e]. \tag{152}$$

When the input code has a geometric cellulation, this says that the vertex chosen for qubit e is one of the endpoints of that edge. For a general qLDPC complex, it should be read as an incidence condition: $\phi(e)$ must be a 0-cell participating in the boundary data of e. This is not needed to define the symmetry, but it is the natural condition for keeping the later stabilizers local in the code's incidence graph.

2.17.3 The enrichment circuit is a collection of controlled swaps

For a temporary ancilla vertex v, define its computational-basis occupation operator by

$$n_v^a := \frac{I - Z_v^a}{2}. \tag{153}$$

It has eigenvalue 0 on $|0\rangle_a$ and 1 on $|1\rangle_a$. Thus $W^{n_v^a}$, for an involution $W^2 = I$, means "apply W if the a_v qubit is $|1\rangle$, and otherwise do nothing." This compact exponent notation is used throughout the paper.

The paper's enrichment circuit is

$$\Omega_{a,BC} = \prod_{v,e} \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n_v^a \int_{\phi(\mathbf{e})} v}. \tag{154}$$

The integral is not a continuum integral. It is the $\mathbb{F}_2$ pairing between a 0-chain and a 0-cochain:

$$\int_{\phi(\mathbf{e})} v = v(\phi(\mathbf{e})) = \begin{cases} 1, & v = \phi(e), \\ 0, & v \neq \phi(e). \end{cases} \tag{155}$$

Consequently, only one term in the product over v is active for a given qubit e. With the basis-level assignment $v_e = \phi(e)$, Eq. (154) is simply

$$\boxed{\Omega_{a,BC} = \prod_{e \in Q} \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n_{v_e}^a}.} \tag{156}$$

It is a product of Fredkin gates: temporary qubit a_{v_e} controls a swap of the two matter qubits B_e, C_e. Controlled swaps for different e's commute, even when they share their control, because their target pairs are disjoint. This makes the circuit particularly transparent.

2.17.4 Why this circuit enriches the global cocycle symmetry

The unique nontrivial graph cocycle is the operation $X_{\mu_{\text{all}}}^a = \prod_{v \in V} X_v^a$. Conjugating the occupation projector by its control flip gives

$$X_v^a n_v^a X_v^a = I - n_v^a. \tag{157}$$

Use this in one controlled-swap factor. Since $\mathsf{SWAP}^2 = I$,

$$X_{v_e}^a \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n_{v_e}^a} X_{v_e}^a \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n_{v_e}^a} = \mathsf{SWAP}_{B_e,C_e}. \tag{158}$$

Multiplying this result over all e gives exactly the required commutator condition from Eq. (141):

$$X_{\mu_{\text{all}}}^a \, \Omega_{a,BC} \, X_{\mu_{\text{all}}}^a \, \Omega_{a,BC}^\dagger = \prod_e \mathsf{SWAP}_{B_e,C_e} = U_{\mathsf{SWAP}}. \tag{159}$$

This is the central calculation of the example. Globally flipping every temporary ancilla reverses every control bit. Each controlled swap is then applied exactly once in the commutator, so the global ancilla symmetry acts on the matter as layer exchange. If a qubit were assigned to an even number of controlling vertices, its swaps would cancel pairwise; this is the paper's condition that $\int_{\phi(\mathbf{e})} \mu = 1$ for each e. Choosing one vertex $\phi(e)$ per qubit is the simplest way to satisfy it.

2.17.5 Conjugating the input stabilizers before measurement

It is helpful to pause at the intermediate, premeasurement stabilizers. For one input qubit, controlled swap changes a Pauli on the B copy as

$$\begin{aligned} \Omega X_e^B \Omega^\dagger &= X_e^B \left(X_e^B X_e^C\right)^{n^a_{\phi(e)}}, \\ \Omega Z_e^B \Omega^\dagger &= Z_e^B \left(Z_e^B Z_e^C\right)^{n^a_{\phi(e)}}. \end{aligned} \tag{160}$$

Check the two possible control values. If $n^a_{\phi(e)} = 0$, the right side is the original B Pauli. If it is one, the extra product turns X_e^B into X_e^C, or Z_e^B into Z_e^C, exactly as a swap should. The corresponding formulas for a C Pauli have the same correction factor.

Therefore an X-check labelled by the 0-cochain v becomes

$$\widetilde{S}^X_{B,v} = X^B_{d_0 v} \prod_{\mathbf{e}\in d_0 v} \left(X^B_{\mathbf{e}} X^C_{\mathbf{e}}\right)^{n^a_{\phi(\mathbf{e})}}, \tag{161}$$

This equation is simply the original B-check with the one-qubit rule in Eq. (160) applied to every qubit in that check. To read it, start with the 0-cochain $v \in D^0$. Its coboundary

$$d_0 v \in D^1$$

is a binary list of input qubits: its value on the basis chain $[e]$ is either zero or one. The shorthand

$$X^B_{d_0 v} = \prod_{e\in Q} \left(X^B_e\right)^{(d_0 v)([e])}$$

therefore means "put X_e^B on precisely those qubits for which $(d_0 v)([e]) = 1$." In the displayed formula, $\mathbf{e} \in d_0 v$ is only a compact shorthand for the same condition. A cochain is not literally a set: the product is over the *support* of the binary cochain $d_0 v$.

For a small graph-like example, suppose v selects a vertex whose check contains three input qubits e_1, e_2, e_3. Then

$$d_0 v = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \qquad X^B_{d_0 v} = X^B_{e_1} X^B_{e_2} X^B_{e_3}.$$

Equation (161) says that this becomes

$$\begin{aligned} \widetilde{S}^X_{B,v} = {} & X^B_{e_1} X^B_{e_2} X^B_{e_3} \\ & \times \left(X^B_{e_1} X^C_{e_1}\right)^{n^a_{\phi(e_1)}} \left(X^B_{e_2} X^C_{e_2}\right)^{n^a_{\phi(e_2)}} \left(X^B_{e_3} X^C_{e_3}\right)^{n^a_{\phi(e_3)}}. \end{aligned}$$

For any one qubit e_i, the corresponding temporary-control value decides what happens. If $n^a_{\phi(e_i)} = 0$, its extra factor is the identity and the check still contains $X^B_{e_i}$. If $n^a_{\phi(e_i)} = 1$, then

$$X^B_{e_i}\left(X^B_{e_i}X^C_{e_i}\right) = X^C_{e_i}.$$

That local part of the stabilizer has been swapped from copy B to copy C. The compact formula in Eq. (161) says this independently for every qubit in the original check. A general v can label a product of check generators rather than one independent generator; if $d_0 v = 0$, it labels the identity. This is why the paper uses all 0-cochains rather than choosing an independent check basis at this stage.

The corresponding C-check has exactly the same correction factors, with B and C exchanged. Similarly, a Z-check labelled by a 2-chain $[p]$ becomes

$$\widetilde{S}^Z_{B,[p]} = Z^B_{\partial_2[p]} \prod_{e\in\partial_2[p]} \left(Z^B_e Z^C_e\right)^{n^a_{\phi(e)}}. \tag{162}$$

The temporary-vertex graph stabilizer has an equally important change. A control flip X^a_v toggles every controlled swap assigned to v, so

$$\widetilde{K}^a_v = X^a_v X^A_{d^a_0 v}\mathsf{SWAP}_{\phi^T(v)}, \qquad \mathsf{SWAP}_{\phi^T(v)} := \prod_{e:\,\phi(e)=v} \mathsf{SWAP}_{B_e,C_e}. \tag{163}$$

Equation (163) is the original vertex stabilizer of the graph cluster state, now conjugated by the enrichment circuit. Before enrichment it was only

$$K^a_v = X^a_v X^A_{d^a_0 v}.$$

Here d^a_0 is the coboundary map of the *ancilla graph*, not the coboundary map of the input code. If the graph vertex v is incident on edges f_1, f_2, f_3, then

$$X^A_{d^a_0 v} = X^A_{f_1}X^A_{f_2}X^A_{f_3}.$$

Thus the original cluster constraint says: flip the temporary qubit a_v, and simultaneously flip every surviving gauge-field qubit on a graph edge incident on v.

The final factor in Eq. (163) records which matter-qubit swaps were controlled by a_v:

$$\mathsf{SWAP}_{\phi^{\mathsf{T}}(v)} = \prod_{e:\,\phi(e)=v} \mathsf{SWAP}_{B_e,C_e}.$$

The superscript T is the paper's matrix-transpose notation. In the chosen bases, $\phi^{\mathsf{T}}(v)$ means the list of all input-qubit labels e whose assigned controlling vertex is v. For example, if

$$\phi(e_1) = v, \qquad \phi(e_3) = v,$$

then

$$\mathsf{SWAP}_{\phi^{\mathsf{T}}(v)} = \mathsf{SWAP}_{B_{e_1},C_{e_1}} \mathsf{SWAP}_{B_{e_3},C_{e_3}}.$$

If, in addition, v has the two incident graph edges f_1, f_2, the full stabilizer is

$$\widetilde{K}_v^a = X_v^a X_{f_1}^A X_{f_2}^A \mathsf{SWAP}_{B_{e_1},C_{e_1}} \mathsf{SWAP}_{B_{e_3},C_{e_3}}.$$

Why does this new SWAP factor appear? The enrichment circuit attached controlled swaps to a_v. Conjugating X_v^a through them flips their control bit. For every input pair controlled by a_v, this leaves one ordinary swap:

$$\Omega_{a,BC} X_v^a \Omega_{a,BC}^\dagger = X_v^a \prod_{e:\, \phi(e)=v} \mathsf{SWAP}_{B_e,C_e}.$$

The $X_{d_0^a v}^A$ factor is unaffected, which gives Eq. (163). Physically, the equation says that a local ancilla/gauge transformation at vertex v is tied to swapping the two matter-code layers on precisely the qubits assigned to v.

Finally, the graph-edge stabilizers $Z_f^A Z_{\partial f}^a$ are unchanged because $\Omega_{a,BC}$ is diagonal in every control Z_v^a.

2.17.6 Measuring the temporary qubits: how strings of Z^A appear

Vertex stabilizers. First consider only the graph-cluster vertex stabilizer $\widetilde{K}_v^a$ in Eq. (163). Its X-measurement has a particularly direct effect: it removes X_v^a and leaves a constraint on the surviving A, B, C qubits.

The protocol now measures every a_v in the X basis and, after the standard Pauli-frame correction, takes the outcomes to be $+1$. Thus the factor X_v^a can be removed from Eq. (163), leaving the gauged vertex constraint

$$G_v^A = X_{d_0^a v}^A \mathsf{SWAP}_{\phi^T(v)}. \tag{164}$$

Thus G_v^A is precisely what the graph cluster state's vertex stabilizer becomes after symmetry enrichment and the X-measurement (with postselection or Pauli-frame correction) removes the temporary qubit a_v.

It is called a constraint because physical code states must satisfy

$$G_v^A |\Psi_{\text{gauged}}\rangle = |\Psi_{\text{gauged}}\rangle.$$

And it is called gauged because it is now a local relation involving only surviving degrees of freedom:

- an X transformation on gauge-field edges adjacent to v;
- the associated swaps of B, C matter qubits.

Without enrichment, the surviving relation would have been only

$$X_{d_0^a v}^A = 1,$$

which is an ordinary pure-$\mathbb{Z}_2$ gauge-type vertex constraint. Enrichment dresses it by layer exchange.

A useful global check is to multiply all these local constraints. Every A-edge appears at both of its endpoints, so its two X^A's cancel. If every input qubit is assigned to exactly one vertex, the SWAP factors multiply to the global transversal swap:

$$\prod_{v\in V} G_v^A = U_{\mathsf{SWAP}}.$$

This is the essential gauging idea: the former global $B \leftrightarrow C$ symmetry has been distributed across local vertex constraints of the enlarged A, B, C system.

B- and C-code stabilizers. We now return to a different family of stabilizers: the dressed input-code checks in Eqs. (161) and (162). Their measurement problem is not solved merely by setting an X_v^a factor to $+1$, because they contain Z^a-dependent occupation projectors. They must first be rewritten using the graph-edge stabilizers.

There is one remaining issue. Equations (161) and (162) still contain operators on measured a qubits. More precisely, the problematic operators are the occupation projectors in their exponents,

$$n^a_{\phi(e)} = \frac{I - Z^a_{\phi(e)}}{2}.$$

For example, the factor

$$\left(X_e^B X_e^C\right)^{n^a_{\phi(e)}}$$

is a controlled operator: it acts as I or as $X_e^B X_e^C$, depending on the Z-basis value of the temporary qubit $a_{\phi(e)}$. Thus the n^a's are not harmless numerical exponents; before measurement they are genuine operators acting on the temporary ancilla and the matter system. The same point applies to the $n^a_{\phi(e)}$'s in Eq. (162).

We measure every a qubit in the X basis, whereas n^a is made from Z^a. They do not commute:

$$[X_v^a, n_v^a] \neq 0.$$

Consequently, after projecting a qubit onto $|+\rangle$, one cannot replace n_v^a by either 0 or 1, nor can one leave it in a stabilizer of the surviving system. In contrast, a Z-basis measurement would supply a definite value for n_v^a, but that is not the measurement used by the gauging protocol.

The paper therefore first changes to an equivalent set of stabilizer generators in which the unwanted temporary-ancilla dependence occurs in pairs such as $Z_u^a Z_w^a$. The graph cluster-state stabilizers then replace each such pair by a string of Z^A's on the surviving edge qubits. Only after this rewriting is it safe to measure and discard the a qubits. Here "equivalent'' means *equal on the prepared stabilizer subspace*, after multiplication by already satisfied input and cluster stabilizers; it does not mean that the displayed operators are unconstrained identities on the whole Hilbert space. This distinction is important.

The key cluster-state relation on any graph path $\gamma_{u,w}$ from vertices u to w is

$$\prod_{f\in\gamma_{u,w}} \left(Z_f^A Z_{\partial f}^a\right) = Z^A_{\gamma_{u,w}} Z_u^a Z_w^a \doteq I. \tag{165}$$

Read Eq. (165) directly as a product of stabilizers. Every factor $Z_f^A Z_{\partial f}^a$ in the product is a graph-edge stabilizer, so it acts as $+1$ on the premeasurement cluster state. Their product must therefore also act as $+1$ on that state.

Now take a simple path from u to w. Every temporary vertex in the interior of the path belongs to exactly two successive graph edges. Its Z^a operator consequently occurs twice in the product and cancels:

$$(Z^a_{\text{interior vertex}})^2 = I.$$

The two endpoint vertices u, w occur only once, so their Z^a's do *not* cancel. Each edge of the path also occurs once, leaving one Z^A on each surviving edge qubit. Thus it is the combined operator

$$Z^A_{\gamma_{u,w}} Z^a_u Z^a_w$$

that is equivalent to the identity on the premeasurement stabilizer subspace. Hence,

$$Z^a_u Z^a_w \doteq Z^A_{\gamma_{u,w}}. \tag{166}$$

The symbol $\doteq$ means equality on that stabilized subspace, not an operator identity on every state. Equation (166) is the precise reason that a pair of unwanted temporary-ancilla Z's can be replaced by a string of Z's on the surviving A edge qubits. The notation $\langle u, w\rangle$ in the paper denotes a chosen such path; it need not be a single graph edge.

After this generator rewriting, the paper gives the surviving matter checks in the following form. For an X-check v, choose the graph path $\langle \phi(\mathbf{e}), v\rangle$ for every qubit in its support:

$$\begin{aligned} S^{X,\text{gauged}}_{B,v} &= X^B_{d_0 v} \prod_{\mathbf{e}\in d_0 v} \left(Z^A_{\langle \phi(\mathbf{e}),v\rangle}\right)^{m^B_{\mathbf{e}}+m^C_{\mathbf{e}}}, \\ S^{X,\text{gauged}}_{C,v} &= X^C_{d_0 v} \prod_{\mathbf{e}\in d_0 v} \left(Z^A_{\langle \phi(\mathbf{e}),v\rangle}\right)^{m^B_{\mathbf{e}}+m^C_{\mathbf{e}}}, \end{aligned} \tag{167}$$

Here $m^B_{\mathbf{e}} = (I - X^B_{\mathbf{e}})/2$, and similarly for C. The sum in each exponent is modulo two.

Deriving Eq. (167), including the origin of the path. Temporarily use the simpler symbol s for one input X-check. If

$$E_s := \text{supp}(s),$$

then its checks in the two copies are

$$S^X_{B,s} = \prod_{e\in E_s} X^B_e, \qquad S^X_{C,s} = \prod_{e\in E_s} X^C_e.$$

In the paper's notation, s is the basis label v, its coordinate cochain is v, and $E_s = \text{supp}(d_0 v)$. The paper also uses the same label s for a vertex of the ancilla graph. This identification is implicit in the paper; formally one could instead introduce a map sending each check label to a chosen graph vertex.

For each input qubit e, the map ϕ chooses one ancilla vertex $\phi(e)$ whose temporary qubit controls the SWAP of B_e and C_e. This controller need not equal s. Indeed, one physical qubit can belong to several X-checks, whereas the simplifying assumption in the paper assigns it only one controller $\phi(e)$. If e belongs to two checks s and t, one cannot in general have both $\phi(e) = s$ and $\phi(e) = t$.

Let

$$P_e := X_e^B X_e^C.$$

Equation (161) says that after symmetry enrichment the dressed B-check is

$$\widetilde{S}_{B,s}^X = S_{B,s}^X \prod_{e \in E_s} P_e^{n^a_{\phi(e)}}. \tag{167a}$$

Only the controller occupations $n^a_{\phi(e)}$ occur here; no operator on a_s has appeared yet. We now introduce one in a way that does not change the enriched state.

First define

$$Q_s := S_{B,s}^X S_{C,s}^X = \prod_{e \in E_s} P_e. \tag{167b}$$

The two original checks give $Q_s = +1$ before enrichment. More importantly, this remains true afterwards. A SWAP on B_e, C_e leaves $P_e = X_e^B X_e^C$ invariant, so every controlled-SWAP in the enrichment circuit commutes with Q_s. If $|\Psi_{\text{enr}}\rangle = \Omega|\Psi_0\rangle$, then

$$Q_s|\Psi_{\text{enr}}\rangle = \Omega Q_s|\Psi_0\rangle = |\Psi_{\text{enr}}\rangle. \tag{167c}$$

Since n_s^a acts on an ancilla and commutes with Q_s, the controlled operator

$$Q_s^{n_s^a} = (I - n_s^a) + n_s^a Q_s$$

also acts as the identity on the enriched state. We may therefore multiply Eq. (167a) by it. Using Eq. (167b), this gives the equivalent representative

$$\widetilde{S}_{B,s}^X \doteq S_{B,s}^X \prod_{e \in E_s} P_e^{n^a_{\phi(e)} + n_s^a}. \tag{167d}$$

This multiplication is the precise step that introduces the previously absent operator on a_s.

For each e, the following is an operator identity:

$$P_e^{n^a_{\phi(e)} + n_s^a} = \left(Z^a_{\phi(e)} Z_s^a\right)^{m_e^B + m_e^C}, \qquad m_e^B = \frac{I - X_e^B}{2}, \tag{167e}$$

with the exponent understood modulo two. To verify it, work in a joint X_e^B, X_e^C eigenstate. If their eigenvalues agree, then $P_e = +1$ and both sides are the identity. If they disagree, then $P_e = -1$, and the left side is $(-1)^{n^a_{\phi(e)} + n_s^a} = Z^a_{\phi(e)} Z_s^a$, exactly the right side.

We now have two temporary-ancilla Z's. The second one is essential: a finite graph path has two endpoints, so the cluster-state path relation cannot replace a single $Z^a_{\phi(e)}$ by a finite A-string. For each $e \in E_s$, choose a path in the ancilla graph

$$\gamma_{e,s} : \ \phi(e) \longrightarrow s.$$

The paper denotes it by $\langle \phi(e), s \rangle$. Equation (166) gives

$$Z^a_{\phi(e)} Z^a_s \doteq Z^A_{\gamma_{e,s}} := \prod_{f \in \gamma_{e,s}} Z^A_f. \tag{167f}$$

Substituting Eq. (167f) into Eq. (167d) removes every measured a qubit and yields Eq. (167). Thus the path for the term associated with e is specifically the ancilla-graph path from the SWAP controller $\phi(e)$ to the vertex carrying the label of the whole check s. If $\phi(e) = s$, this is the empty path and its Z^A product is the identity.

The Z-type checks have the parallel form

$$\begin{aligned} S^{Z,\text{gauged}}_{B,[p]} &= Z^B_{\partial_2[p]} \prod_{e \in \partial_2[p]} \left(Z^A_{\langle \phi(e), v(p) \rangle} \right)^{n^B_e + n^C_e}, \\ S^{Z,\text{gauged}}_{C,[p]} &= Z^C_{\partial_2[p]} \prod_{e \in \partial_2[p]} \left(Z^A_{\langle \phi(e), v(p) \rangle} \right)^{n^B_e + n^C_e}, \end{aligned} \tag{168}$$

where $n^B_e = (I - Z^B_e)/2$, and likewise for C. Let us unpack the choice of $v(p)$ without using geometrical language. A basis element $[p] \in D_2$ labels one Z-check, and

$$\partial_2[p] = \sum_{e \text{ acted on by } p} [e]$$

lists the physical qubits on which that check acts. For each such qubit, $\partial_1[e] \in D_0$ lists the X-check labels that also act on e. The paper's "closure'' of p is therefore the set

$$\mathrm{cl}(p) := \bigcup_{e \in \mathrm{supp}(\partial_2[p])} \mathrm{supp}(\partial_1[e]). \tag{168a}$$

In words,

$$\boxed{\mathrm{cl}(p) = \{X\text{-check labels touching at least one qubit of the } Z\text{-check } p\}.} \tag{168b}$$

To choose $v(p)$, pick one physical qubit e on which the Z-check p acts, then pick an X-check v that also acts on that qubit. Call the selected X-check label $v(p)$. Because the ancilla-graph vertices are indexed by X-check labels, $v(p)$ also names an ancilla vertex. In the Tanner-incidence structure, this is simply the two-step choice

$$p \text{ (a } Z\text{-check)} \longrightarrow e \text{ (a data qubit)} \longrightarrow v(p) \text{ (an } X\text{-check/ancilla vertex).} \tag{168c}$$

One must not replace the union in Eq. (168a) by $\mathrm{supp}(\partial_1\partial_2[p])$. The chain-complex identity $\partial_1\partial_2 = 0$ makes that support empty: the incident X-check labels cancel modulo two after all qubits of the Z-check are summed. The closure instead collects the supports of the individual $\partial_1[e]$'s *before* those cancellations. Geometrically, for a square toric-code plaquette, $\partial_2[p]$ gives its four boundary edges, $\mathrm{cl}(p)$ gives its four corner vertices, and $v(p)$ is one chosen corner. No literal face or corner is required in a general qLDPC code.

The notation in Eqs. (167)–(168) deserves one last reading. For example, if $m_e^B + m_e^C = 0$, the associated path contributes nothing. If it is one, multiply Z^A over every graph edge on the chosen path from $\phi(e)$ to v for an X-check, or to $v(p)$ for a Z-check. These are finite products of *controlled-Pauli* factors; as whole operators they need not be bare Pauli products. In a fixed matter eigenvalue sector each factor reduces to either I or an ordinary Z^A string. Together with Eq. (164), they are the stabilizers of the gauged model written in the paper's occupation-number convention.

2.17.7 Where the qLDPC difficulty has moved

The symmetry calculation was easy; code locality is the nontrivial part. Equations (167) and (168) contain graph paths. A long path makes one stabilizer high weight, while too many selected paths through one graph edge make that qubit participate in too many checks. Either outcome violates the qLDPC conditions.

The paper therefore proposes designing the graph so that the needed pairs are near each other: include short connections between $\phi(e)$ and v whenever $e \in d_0 v$, and between $\phi(e)$ and $v(p)$ whenever $e \in \partial_2[p]$. A bounded-degree implementation of those connections is the remaining graph-design problem. It must also retain the single connected component required by Eq. (139). This is a good point at which to separate the two achievements of the example:

1. Eqs. (154)–(159) prove that the graph ancilla gauges transversal SWAP.
2. Eqs. (164)–(168) show what the resulting constraints are and expose the precise locality conditions a qLDPC graph must satisfy.

The construction is consequently a template, not yet a universal qLDPC theorem. It tells us exactly which sparse incidence data and short graph paths would turn the elegant gauging idea into a family of good non-Abelian codes.

2.18 Section IV.A: non-Abelian fractons from gauging transversal CZ gates

The SWAP example established the graph-gauging mechanism in its most universal form. This example is more directly connected to quantum computation: it gauges a transversal CZ, the logical Clifford operator whose measurement will later prepare a magic state. It is also the paper's first fully spatial qLDPC example.

There are two things to keep distinct from the outset. The **ALP code** is an Abelian CSS code with fracton/subsystem structure. The construction then takes two copies of that code and gauges a symmetry between the copies. The gauged code is the non-Abelian object. In this first pass we will follow the graph construction through its local circuit and its resulting constraints. The later subsections on addressable gates and cup products are deliberately postponed: they answer how to gauge a family of CZ gates, not what the basic CZ-gauging mechanism is.

2.18.1 The unusual qubit placement in the ALP code

The anisotropic lineon-planeon (ALP) model lives on a periodic cubic lattice. Its CSS chain complex is

$$0 \longrightarrow D_2 \xrightarrow{\partial_2} D_1 \xrightarrow{\partial_1} D_0 \longrightarrow 0. \tag{165}$$

The word *anisotropic* is literal: the three spatial directions are not treated alike. The bases have the following geometric interpretation:

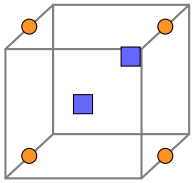
space	basis object	role
D_2	cubes c	Z-check labels
D_1	plaquettes in xy planes and edges parallel to z	physical qubits
D_0	vertices v	X-check labels.

(166)

The middle line is the surprise. A horizontal *xy plaquette* carries a qubit, whereas a vertical *z edge* carries a qubit. There are no qubits on horizontal edges. This is why the ALP stabilizers do not look like the usual cubic-lattice toric-code stabilizers.

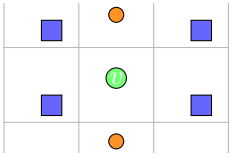
The following deliberately flattens the geometry into two readable 2D schematics. Blue squares stand for horizontal-plaquette qubits and orange circles for vertical-edge qubits; they are not physical projections of every part of the 3D lattice.

(a) Cube Z check



Six qubits: two xy-face qubits and four vertical-edge qubits.

(b) Vertex X check



Six qubits: four adjacent xy-plaquette qubits and two vertical-edge qubits.

In the paper's compact notation, the two stabilizer families are therefore

$$A_v = X_{d_0 v}, \qquad B_c = Z_{\partial_2[c]}. \tag{167}$$

The chain-complex condition $\partial_1 \partial_2 = 0$, or dually $d_1 d_0 = 0$, guarantees that each A_v and B_c overlap on an even number of qubits and hence commute. This is the only fact needed for the gauging construction; one does not need the detailed fracton mobility theory yet. The name "lineon-planeon" signals that some excitations in this model have restricted motion, but the present section uses the model principally as a concrete CSS qLDPC code.

For an $L_x \times L_y \times L_z$ periodic lattice, the paper reports $2(L_x + L_y - 1)$ logical qubits and distance scaling as $\min(L_x, L_y, L_z)$. This is a useful reminder that it is a subsystem-like 3D code rather than a conventional topological code with a constant number of logical qubits.

2.18.2 The transversal CZ symmetry between two copies

Make two identical ALP blocks, named B and C. The paper specifies a periodic, local pairing $\mathcal{P}$ of a B-qubit with a C-qubit in each cubic unit cell. Its drawing uses black line segments between the paired qubits. Write the resulting gate without committing to one misleading 2D projection as

$$U_{\mathsf{CZ}} = \prod_{(e,e')\in\mathcal{P}} \mathsf{CZ}_{B_e, C_{e'}}. \tag{168}$$

"Transversal" means that each physical qubit occurs in only a bounded, fixed number of these elementary gates (in this pattern, one). The paired qubits need not be drawn directly on top of one another: the pairing is a local matching determined by the cubic unit cell. This is a point at which the paper's 3D perspective matters more than an attempted planar picture.

The elementary conjugation rules are

$$\mathsf{CZ}_{B,C} X_B \mathsf{CZ}_{B,C} = X_B Z_C, \qquad \mathsf{CZ}_{B,C} X_C \mathsf{CZ}_{B,C} = Z_B X_C, \tag{169}$$

while both Z_B and Z_C are unchanged. The chosen pairing makes the extra Z factors fit the ALP check structure. This sentence hides an important qualification: a *single physical* X_e, Z_e, or $\mathsf{CZ}_{B_e,C_{e'}}$ is generally **not** a symmetry of the code. For example, a lone Z_e anticommutes with the X-checks incident on that qubit and creates syndrome defects. Equation (169) is only a local operator identity; it does not say that its individual factors preserve check eigenvalues.

The simple lattice-level picture is this. Pick **one whole** B-block X-check A_v^B, not one physical X_e. Every CZ touching that six-qubit check contributes a Z on a qubit of the C block. When all those contributions are multiplied, the special pairing drawn in the paper makes them close up into one or a few complete C-block Z-checks. Symbolically,

$$\begin{aligned} U_{\mathsf{CZ}} A_v^B U_{\mathsf{CZ}}^\dagger &= A_v^B \prod_{c \in R_C(v)} B_c^C, \\ U_{\mathsf{CZ}} A_v^C U_{\mathsf{CZ}}^\dagger &= \left(\prod_{c \in R_B(v)} B_c^B \right) A_v^C. \end{aligned} \tag{170}$$

Here $R_C(v)$ and $R_B(v)$ are small, local sets of cubes determined by the CZ-line pattern. The important point is not their exact shape yet; it is that the added Z's form **complete stabilizers**, not an open string with endpoints. This is the cancellation you had in mind: when the gates are applied throughout the lattice, intermediate Z's cancel modulo two and only closed Z-check products remain.

Thus it is fair to say, informally, "an X-check on one copy acquires Z-checks on the other copy." It is not quite right to say "all the X-stabilizers and all the Z-stabilizers are added." The statement is local and check-by-check: a particular A_v^B acquires a particular nearby product $\prod_{c \in R_C(v)} B_c^C$. On the codespace every factor in Eq. (170) has eigenvalue $+1$, so the check value remains $+1$. The generator is merely replaced by another generator of the same stabilizer group.

The Z-checks themselves do not acquire anything, because CZ is diagonal:

$$U_{\mathsf{CZ}} B_c^B U_{\mathsf{CZ}}^\dagger = B_c^B, \qquad U_{\mathsf{CZ}} B_c^C U_{\mathsf{CZ}}^\dagger = B_c^C. \tag{171}$$

So the correct conclusion is not "single X's and Z's commute with the stabilizers." They generally do not. Rather, the *whole* transversal CZ maps every stabilizer to a product of stabilizers, and so preserves the codespace while acting as a logical Clifford operation. If you later want the compact binary test for whether a proposed pairing has this property, it is exactly the condition that its added Z support be in $\operatorname{im} \partial_2$ for every X-check support in $\operatorname{im} d_0$.

2.18.3 The cubic ancilla graph and the controlled-CZ enrichment

The graph method now supplies a third set of qubits. Put temporary qubits a_v on every cubic-lattice vertex and gauge-field qubits A_f on every cubic-lattice edge $f = \langle v, w \rangle$. After making the graph cluster state (the paper applies a Hadamard to the A sublattice), its convenient stabilizer generators are

$$K_v^a = X_v^a \prod_{f \ni v} X_f^A, \qquad K_f^A = Z_f^A Z_v^a Z_w^a. \tag{172}$$

Because the cubic graph is connected, the temporary vertices possess exactly one global X symmetry,

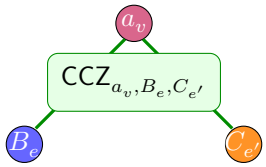
$$X_{\text{all}}^a = \prod_v X_v^a. \tag{173}$$

The enrichment must tie this single symmetry to Eq. (168). It uses CCZ, the three-qubit controlled-controlled-Z gate. Its action is diagonal in the computational basis:

$$\mathsf{CCZ}_{a,B,C} |r, b, c\rangle = (-1)^{rbc} |r, b, c\rangle. \tag{174}$$

Thus when the temporary control is $|0\rangle_a$, it does nothing; when the control is $|1\rangle_a$, it applies CZ to the selected B, C pair. The essential local motif is

A 2D schematic of the local 3D enrichment rule



The actual cubic construction assigns a bounded nearby collection of B-C pairs to each vertex v; the schematic shows one such pair.

The paper denotes this local assignment by $\phi^{\mathsf{T}}(v)$. In this example $\phi^{\mathsf{T}}(v)$ is a set of **pairs** (e, e'), not merely a set of single qubits as in the SWAP example. In a basis-independent form, the enrichment circuit is

$$\Omega = \prod_v \prod_{(e,e') \in \phi^{\mathsf{T}}(v)} \mathsf{CCZ}_{a_v, B_e, C_{e'}}. \tag{175}$$

The map must be local and surjective in the paper's sense: every required physical CZ in U_{CZ} is assigned to an odd number of controls, preferably exactly one, and each control reaches only $O(1)$ nearby pairs. This is the CZ version of the sparse assignment $e \mapsto \phi(e)$ in the SWAP construction.

To see the symmetry enrichment explicitly, use the one-gate identity

$$X_a \, \mathsf{CCZ}_{a,B,C} \, X_a \, \mathsf{CCZ}_{a,B,C} = \mathsf{CZ}_{B,C}. \tag{176}$$

It follows just by checking $a = 0, 1$: flipping the control interchanges the "apply CZ" and "do nothing" branches. Multiply Eq. (176) over all temporary vertices and their assigned pairs. The odd-assignment condition then gives

$$X^a_{\text{all}}\Omega X^a_{\text{all}}\Omega^\dagger = U_{\mathsf{CZ}}. \tag{177}$$

This is the same group-commutator test as Eq. (155), with controlled swaps replaced by controlled CZ gates. It is the one equation that shows the cubic ancilla is gauging the desired logical symmetry rather than merely adding extra qubits.

2.18.4 What survives measurement, and why the final formulas look difficult

Measure every temporary a_v qubit in the X basis. For the moment, pretend every outcome is $+1$; the next paper subsection explains how a decoder and feedforward correct the actual random signs. From Eq. (172), the temporary-vertex stabilizer becomes the surviving gauge constraint

$$\widetilde{K}^A_v = \left(\prod_{f\ni v} X^A_f\right)\left(\prod_{(e,e')\in\phi^{\mathsf{T}}(v)} \mathsf{CZ}_{B_e,C_{e'}}\right). \tag{178}$$

Read this slowly. The first factor is an ordinary X-type gauge transformation on the six edges incident to a cubic vertex. The second factor simultaneously performs the local piece of the original B-C symmetry. This is the concrete form of the slogan "a gauge transformation carries the old symmetry."

The matter checks become dressed as well. The useful one-qubit identity is

$$\mathsf{CCZ}_{a,B,C} X_B \mathsf{CCZ}_{a,B,C} = X_B \mathsf{CZ}_{a,C}, \tag{179}$$

and the analogous equation holds with B and C exchanged. Therefore each X-type ALP check picks up controlled-Z factors involving temporary ancillas. Before measurement, those factors can be traded, using products of the edge relations K^A_f, for operators on paths of surviving A-edge qubits - precisely the path-replacement mechanism explained in Eqs. (161)–(162). The Z-type checks are simpler at the first step because CCZ is diagonal, but the complete final commuting projectors still include the corresponding gauge dressing.

This explains why the paper presents the final stabilizers mainly as 3D figures rather than as a short one-line formula. Three distinct incidence structures must be superimposed: the six-qubit ALP checks, the cubic ancilla graph, and the local B-C pair assignment. The final operators are local, but "local" now means local in that combined three-layer incidence structure, not local in a single planar projection.

There is a second subtlety. The raw dressed generators can commute only **up to** the gauged U_{CZ} symmetry. To define a Hamiltonian with mutually commuting projectors, the paper symmetrizes them with respect to that $\mathbb{Z}_2$ action. This is not a failure of the construction; it is the non-Abelian feature in algebraic form. A bare matter-check label can be changed by the gauge transformation, so the gauge-invariant term must include both members of its symmetry orbit.

2.18.5 Why this example deserves priority for the magic-state goal

The SWAP example taught the general graph procedure. This ALP example adds the specific ingredient needed later for computation:

$$\text{temporary control} \quad + \quad \mathsf{CCZ} \quad \Longrightarrow \quad \text{gauged transversal logical } \mathsf{CZ}. \tag{180}$$

The later ungauging protocol reverses the ancilla construction and measures the gauge qubits. Equation (177) then reappears as a projector onto an eigenvalue of the logical CZ acting on the original B, C blocks. On appropriate logical input states, that Clifford measurement prepares a non-stabilizer state. Thus the CCZ gates used here are not being claimed as directly applied logical non-Clifford gates; rather, they are the local measurement gadget from which a non-Clifford *resource state* emerges after gauging and ungauging.

For the next reading step, retain the following four statements:

1. The ALP code has qubits on xy plaquettes and z edges, giving six-qubit cube and vertex checks.
2. Two copies possess a local transversal CZ matching.
3. A cubic graph cluster state plus local CCZ enrichment satisfies the global commutator identity (177).
4. Measuring the temporary ancillas turns this into a non-Abelian gauge constraint (178); ungauging later converts the same construction into a logical CZ measurement.

2.19 Section III.E: measurement preparation of non-Abelian qLDPC codes

Up to now, the gauging discussion has used an idealized instruction: "project every temporary ancilla a_v onto $|+\rangle$." A physical X-basis measurement cannot promise that outcome. It returns a random number

$$x_v \in \{+1, -1\}, \qquad X_v^a \longmapsto x_v. \tag{181}$$

This section explains why that randomness does not turn gauging into a postselected protocol. Assuming that the Abelian code on the ancilla complex has a decoder, the random signs can be removed deterministically by a Pauli correction on the surviving A qubits. The word *deterministic* means that every possible measurement record is accepted and corrected; it does not mean that individual measurement outcomes cease to be random.

2.19.1 First ignore the matter codes: measuring a graph cluster state

It is useful to strip the problem to its simplest form. Before the measurement, the temporary-vertex graph-cluster stabilizer is

$$K_v^a = X_v^a X_{d_0^a v}^A. \tag{182}$$

For an ordinary graph, $X_{d_0^a v}^A$ is the product of X on all A-edge qubits incident to vertex v. Since $K_v^a = +1$ on the cluster state, measuring $X_v^a = x_v$ leaves the surviving A qubits with the signed check

$$x_v X_{d_0^a v}^A = +1. \tag{183}$$

The usual Z-type ancilla checks, labelled schematically by $Z_{\partial_2[p]}^A$, are unaffected. Thus the only difference between the desired Abelian ancilla code and the actually prepared one is the list of signs $\{x_v\}$ in Eq. (183).

Turn these signs into a binary syndrome vector,

$$s = \sum_v s_v[v] \in A_0, \qquad s_v = \begin{cases} 0, & x_v = +1, \\ 1, & x_v = -1. \end{cases} \tag{184}$$

Why are there an even number of -1 outcomes in the clean target sector? Multiply every premeasurement vertex stabilizer in Eq. (182). Each A-edge occurs at the two ends of that edge, so its two factors cancel:

$$\prod_v K_v^a = \left(\prod_v X_v^a\right)\left(\prod_{f\in A_1}(X_f^A)^2\right) = \prod_v X_v^a = X_{\text{all}}^a. \tag{185}$$

All the K_v^a's have eigenvalue $+1$, hence the cluster state has $X_{\text{all}}^a = +1$. The individual X_v^a measurements commute, so their outcomes must obey

$$\prod_v x_v = +1. \tag{186}$$

A product of signs is $+1$ precisely when an even number of its factors are negative. For three temporary qubits, for example, $(+,+,+)$ and $(+,-,-)$ are allowed, whereas $(-,+,+)$ is not. After symmetry enrichment the corresponding global relation is $X_{\text{all}}^a U = +1$. In the clean target-preparation sector, the input matter state has $U = +1$, so the same even-parity conclusion follows. An odd parity would instead record the opposite global symmetry sector, not an ordinary local syndrome that an A-edge correction can erase.

This is exactly the familiar CSS situation: a -1 outcome says that the corresponding X-check has the wrong sign. On a 2D surface-code graph, one would draw the -1 vertices and connect them pairwise by a path of edges. In a general qLDPC complex there may be no literal picture, but the same binary linear-algebra question remains: find a 1-chain

$$r \in A_1 \quad \text{such that} \quad \partial_1^a r = s. \tag{187}$$

The decoder is the classical algorithm that finds a suitably likely such chain r from the syndrome. It is not necessary for this discussion to know how that algorithm works internally.

The corresponding quantum correction is the Pauli product

$$Z_r^A := \prod_{f\in A_1}(Z_f^A)^{r(f)}. \tag{188}$$

Why does it fix the signs? A factor Z_f^A anticommutes with an X-check precisely when edge f occurs in that check. Therefore

$$Z_r^A X_{d_0^a v}^A Z_r^A = (-1)^{v(\partial_1^a r)} X_{d_0^a v}^A = (-1)^{s_v} X_{d_0^a v}^A. \tag{189}$$

Because $x_v = (-1)^{s_v}$, conjugating Eq. (183) by Z_r^A changes it to the desired $+X_{d_0^a v}^A = +1$ check. The correction commutes with all Z-type ancilla checks, since it consists only of Z's. This is the full deterministic-preparation argument for the Abelian ancilla code.

Worked nearest-neighbour example: two -1 outcomes.

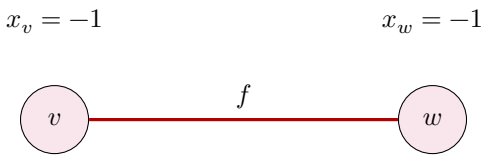
Take two neighbouring vertices v and w of the cubic ancilla graph, joined by one surviving A-edge $f = \langle v, w\rangle$. Suppose that the two temporary measurements give $x_v = x_w = -1$, while every other

temporary measurement gives +1. This is the smallest nontrivial syndrome compatible with a closed, periodic graph:

$$s = [v] + [w] = \partial_1^a [f]. \tag{190}$$

The decoder can therefore choose the shortest possible correction chain, $r = [f]$. Its quantum correction contains just **one** single-qubit Pauli:

$$Z_r^A = Z_f^A. \tag{191}$$



wrong sign at v wrong sign at w
apply Z_f^A: the one selected edge has
exactly the two defective vertices as endpoints

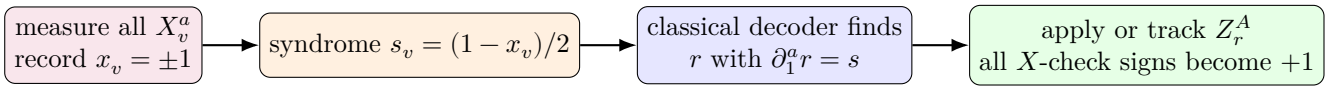
The two signed checks before correction are

$$-X^A_{d_0^a v} = +1, \qquad -X^A_{d_0^a w} = +1. \tag{192}$$

Edge f occurs once in each of the two vertex checks. Hence

$$Z_f^A X^A_{d_0^a v} Z_f^A = -X^A_{d_0^a v}, \qquad Z_f^A X^A_{d_0^a w} Z_f^A = -X^A_{d_0^a w}. \tag{193}$$

The correction flips both minus signs to plus signs at once. At every other vertex u, the edge f does not occur in $d_0^a u$, so the u-check is unchanged. In the gauged code, exactly the same calculation applies to $-X^A_{d_0^a v}\mathcal{U}_v$ and $-X^A_{d_0^a w}\mathcal{U}_w$, because Z_f^A commutes with the matter dressing $\mathcal{U}$. This is the most concrete example of the general rule $\partial_1^a r = s$.



2.19.2 Why exactly the same decoder works after gauging

Now restore the matter blocks B, C and the enrichment circuit. After the X_a measurements, the gauged vertex constraint has the general form

$$\mathcal{G}_v[x_v] = x_v X^A_{d_0^a v}\mathcal{U}_v. \tag{194}$$

The factor $\mathcal{U}_v$ is the local piece of the old transversal symmetry. In the SWAP example it is a product of SWAP_{B_e,C_e}'s; in the ALP CZ example it is a product of $\mathsf{CZ}_{B_e,C_{e'}}$'s. Crucially, the random outcome x_v occurs in exactly the same place as it did in Eq. (183): it merely multiplies the X^A factor by a sign.

The correction Z_r^A has two equally important properties:

1. By Eq. (189), it flips precisely the unwanted signs in $\mathcal{G}_v[x_v]$.
2. It commutes with $\mathcal{U}_v$, because Z_r^A acts on the surviving gauge qubits whereas $\mathcal{U}_v$ acts on B, C. It also commutes with the other Z-type and matter constraints.

Hence the same correction selected by the *Abelian* ancilla decoder turns

$$\mathcal{G}_v[x_v] \quad \longmapsto \quad \mathcal{G}_v[+1] \tag{195}$$

without changing the non-Abelian matter dressing. This is the central observation of the paper's subsection. Measurement randomness creates only an Abelian syndrome, even though the final code has non-Abelian structure.

2.19.3 What "feedforward" means here

The decoder may be a nonlocal classical computation: it reads the full list of outcomes before deciding on r. But once r is known, every factor of Z_r^A is a one-qubit Pauli. They can all be performed in one quantum layer. Often one does not physically apply them at all. One instead stores the binary vector r as a **Pauli frame** and flips the interpretation of later measurements accordingly.

This is why the paper calls the preparation a finite-depth *adaptive* circuit: the quantum circuit has bounded depth, while a classical decoder and a classically chosen Pauli frame connect the measurement step to later operations. This is a deterministic state-preparation statement assuming an appropriate decoder exists. It is not yet a complete proof of fault tolerance against noisy gates and noisy measurements; the paper identifies that stronger question as future work.

2.19.4 What to retain for the computation section

The ideal $|+\rangle_a$ projection used in a formal gauging derivation is therefore operationally replaced by

$$\text{measure } X_v^a \quad \longrightarrow \quad \text{decode the sign syndrome} \quad \longrightarrow \quad \text{update by } Z_r^A. \tag{196}$$

After this correction, we may use the same clean $+1$-outcome gauged state in subsequent reasoning. In particular, the next section can safely ungauge it and interpret the overall protocol as a logical Clifford measurement rather than as a postselected experiment.

2.20 Section IV: logical Clifford measurement from gauging and ungauging

We now reach the computational claim of the paper. Gauging was introduced as a way to make a transversal symmetry local and dynamical. **Ungauging** reverses that enlargement: undo the ancilla cluster entangler and measure away the surviving A gauge qubits. The surprising result is that the combined operation is a measurement of the logical symmetry that was gauged.

The paper writes the derivation for a transversal CZ in the homological construction, where there can be several cocycle symmetries. The conclusion is broader. In the connected-graph construction there is only one nontrivial cocycle, so the same derivation reduces to a measurement of its one chosen transversal Clifford gate.

2.20.1 The claim in one line

Let U_μ be the transversal logical Clifford symmetry associated with an ancilla 0-cocycle $\mu \in \ker d_0^a$. Gauging and then ungauging applies the projector

$$P_{x_\mu} = \frac{I + x_\mu U_\mu}{2} \tag{191}$$

to the two input code blocks. Here $x_\mu = \pm 1$ is obtained from the temporary-ancilla measurement outcomes. Equation (191) is precisely the projective measurement of the Hermitian involution U_μ: it selects its $+1$ or -1 eigenspace.

For the connected graph used in the SWAP and ALP examples, $\ker d_0^a$ has just the uniform cocycle μ_{all}. The formula is then simply

$$\frac{I + x\,U}{2}. \tag{192}$$

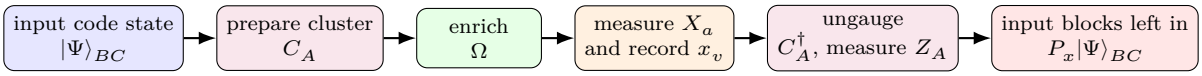
The rest of this section derives this statement. The logical information never has to be decoded into individual physical qubits; all operations are performed on the qLDPC blocks and the ancillary qubits.

2.20.2 Ingredients and notation

Let $|\Psi\rangle_{BC}$ be an arbitrary state in the two input code spaces. Let C_A be the Clifford circuit which prepares the ancilla cluster state from

$$|0\rangle_A^{\otimes} |+\rangle_a^{\otimes}. \tag{193}$$

The gauging protocol applies the enrichment circuit Ω, measures every temporary qubit a_v in the X basis, and leaves the A qubits in place. The ungauging protocol does the reverse ancilla operation: apply $C_A^\dagger$, then measure every A qubit in the Z basis. The full conceptual circuit is



We now unpack only the two algebraic moves which make this work.

2.20.3 Reset: chains and cochains on the ancilla graph

Before using the calculation, it is worth resetting the notation on the smallest ordinary graph. Temporarily ignore the word "dual." Imagine the ancilla graph

```
v1 ----- e12 ----- v2 ----- e23 ----- v3
 a1                 a2                 a3

          A12                 A23
```

Each vertex v carries a temporary qubit a_v, and each edge e carries a surviving gauge qubit A_e. This is the same vertex-and-edge picture used in ordinary lattice gauging. The qLDPC construction permits the graph to be an abstract incidence structure rather than a planar drawing, but the following bookkeeping is identical.

The symbols A_0 and A_1 mean, respectively, the binary vector spaces of collections of vertices and collections of edges:

mathematical object	plain-English meaning	physical qubits
A_0	collections of vertices	a_v live at vertices
A_1	collections of edges	A_e live on edges.

Because the coefficients are in $\mathbb{F}_2$, a collection means "mark some objects," and marking an object twice cancels: $1 + 1 = 0 \pmod 2$. For example,

$$r = e_{12} + e_{23} \in A_1$$

means "mark both edges," while

$$s = v_1 + v_3 \in A_0$$

means "mark vertices v_1 and v_3." These are **chains**.

The boundary map turns a marked set of edges into the marked endpoints left over:

$$\partial_1^a(e_{12}) = v_1 + v_2, \qquad \partial_1^a(e_{23}) = v_2 + v_3,$$

and therefore

$$\partial_1^a(e_{12} + e_{23}) = (v_1 + v_2) + (v_2 + v_3) = v_1 + v_3.$$

The interior vertex v_2 cancels because the two selected edges meet there. This is the familiar fact that a path has endpoints but its interior has no boundary. In measurement decoding, the equation

$$\partial_1^a r = s$$

means: choose an edge string r whose endpoints are exactly the vertices with unwanted measurement signs s.

Now introduce a 0-cochain $\phi \in A^0$. For intuition, draw it as another assignment of bits to vertices:

```
vertex:        v1   v2   v3
phi(v):         1    0    1
```

The coboundary $d_0^a\phi$ assigns bits to edges. On an edge, it asks whether the endpoint bits differ:

$$(d_0^a\phi)(e_{12}) = \phi(v_1) + \phi(v_2) = 1 + 0 = 1,$$

$$(d_0^a\phi)(e_{23}) = \phi(v_2) + \phi(v_3) = 0 + 1 = 1.$$

Thus $d_0^a\phi$ marks the edges crossing from a vertex with bit 0 to one with bit 1:

```
vertex bits phi:       1 ---- 0 ---- 1
edge bits d phi:           1       1
```

The distinction between the two maps is therefore

$$\partial_1^a : \underbrace{\text{marked edges}}_{\text{chain}} \longrightarrow \underbrace{\text{their endpoints}}_{\text{chain}},$$

$$d_0^a : \underbrace{\text{vertex-bit assignment}}_{\text{cochain}} \longrightarrow \underbrace{\text{edge-bit assignment}}_{\text{cochain}}.$$

In the measurement-and-ungauging calculation, ϕ is used only to choose a term in an expansion. Its coboundary specifies the surviving edge qubits on which to apply X:

$$X^A_{d_0^a\phi} = \prod_{e:\,(d_0^a\phi)(e)=1} X_{A_e}.$$

For the displayed example, this gives

$$X^A_{d_0^a\phi} = X_{A_{12}} X_{A_{23}}, \qquad X^A_{d_0^a\phi}|0\rangle_A = |1\rangle_{A_{12}}|1\rangle_{A_{23}}.$$

Thus a subsequent Z_A measurement records two minus signs. Its edge-outcome pattern is written as the 1-cochain $z = d_0^a\phi$.

Only after this concrete picture is solid do we restore the formal statement

$$A^0 = \mathrm{Hom}(A_0, \mathbb{F}_2), \qquad A^1 = \mathrm{Hom}(A_1, \mathbb{F}_2).$$

Formally, cochains are linear functions. But because there is a preferred basis of vertices and edges, such a function is completely specified by one bit per vertex or edge. That is why a cochain can be drawn just like a marked collection. For a first reading, retain the following rule:

```
chain:    "Which vertices or edges have I selected?"
cochain:  "What bit do I assign to each vertex or edge?"
```

They look alike in coordinates, but they participate in different maps:

$$\partial : \text{chains} \longrightarrow \text{chains}, \qquad d : \text{cochains} \longrightarrow \text{cochains}.$$

2.20.4 Step 1: the X_a measurements create a sum over ancilla cochains

For a record $x_v = \pm 1$, the one-qubit measurement projector is

$$\Pi_v(x_v) = \frac{I + x_v X_v^a}{2}. \tag{194}$$

The cluster relation lets us move this projector through the preparation circuit. In the notation used throughout the note,

$$\Pi_x\, \Omega C_A(|\Psi\rangle_{BC}|0\rangle_A|+\rangle_a) \;\doteq\; C_A \prod_v \frac{I + x_v X_v^a X^A_{d_0^a v}}{2} \Omega(|\Psi\rangle_{BC}|0\rangle_A|+\rangle_a). \tag{195}$$

This equation says: "instead of viewing the X_a measurements as acting after cluster preparation, we can pull their projectors through the cluster-preparation circuit." In that new viewpoint, each measured temporary qubit is accompanied by X's on all adjacent surviving A-qubits.

Write the initial state as

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|0\rangle_A|+\rangle_a.$$

For a particular set of recorded measurement outcomes $x = \{x_v\}$, with $x_v = \pm 1$,

$$\Pi_x = \prod_v \Pi_v(x_v), \qquad \Pi_v(x_v) = \frac{I + x_v X_v^a}{2}.$$

This is just the projector onto the stated outcomes: if $x_v = +1$, it projects a_v onto $|+\rangle$; if $x_v = -1$, onto $|-\rangle$.

The cluster-preparation circuit C_A is a product of CNOTs from a temporary vertex qubit a_v to each neighbouring gauge-field qubit A_e. Its basic conjugation rule is

$$C_A^\dagger X_v^a C_A = X_v^a X_{d_0^a \mathbf{v}}^A.$$

Here $d_0^a \mathbf{v}$ is the binary list of edges incident on v. Thus, if v has neighbouring edges $e_1, \ldots, e_r$,

$$X_{d_0^a \mathbf{v}}^A = X_{A_{e_1}} \cdots X_{A_{e_r}}.$$

For example, for a degree-three vertex,

$$C_A^\dagger X_v^a C_A = X_{a_v} X_{A_{e_1}} X_{A_{e_2}} X_{A_{e_3}}.$$

Now insert $I = C_A C_A^\dagger$ around each measurement projector:

$$\Pi_v(x_v) C_A = C_A (C_A^\dagger \Pi_v(x_v) C_A) = C_A \frac{I + x_v X_v^a X_{d_0^a \mathbf{v}}^A}{2}.$$

Doing this for every vertex gives the displayed equation:

$$\Pi_x \Omega C_A |\Phi_0\rangle \doteq C_A \prod_v \frac{I + x_v X_v^a X_{d_0^a \mathbf{v}}^A}{2} \Omega |\Phi_0\rangle.$$

The enrichment circuit Ω commutes with C_A: both are controlled by the a-qubits, but they act on different targets (Ω on B, C, C_A on A). That is why their order can be exchanged here.

So the important content is not a new operation. It is a change of description:

measuring X_{a_v} after cluster preparation	$\longleftrightarrow$	measuring the cluster stabilizer $X_{a_v} X_{d_0^a \mathbf{v}}^A$ before it.

The sign $\doteq$ suppresses normalization and already-known Pauli-frame byproducts. The crucial feature is the factor $X_v^a X_{d_0^a v}^A$: measuring an a qubit is tied to an X operation on the adjacent A-gauge qubits.

Expanding the product in Eq. (195) means choosing either the I term or the $X_v^a X_{d_0^a v}^A$ term at every vertex. A choice is a binary 0-cochain $\phi \in A^0$. Concretely, this means that ϕ is a binary label assigned to every temporary-ancilla vertex:

$$\phi : V_A \longrightarrow \mathbb{F}_2 = \{0, 1\}.$$

If there are N temporary a-qubits, ϕ is simply an N-bit string. In this equation it is not a new quantum state, a measurement outcome, or a physical excitation. It is an accounting device for one term in the algebraic expansion. At each vertex v, it records which of the two terms was selected:

$\phi(v)$	selected term
0	I
1	$x_v X_v^a X_{d_0^a v}^A$.

For example, for three vertices, $\phi = (1, 0, 1)$ means: select the non-identity term at vertices 1 and 3, and select I at vertex 2. The corresponding contribution is $x_1 x_3 K_1 K_3$, where $K_v = X_v^a X_{d_0^a v}^A$. Therefore

$$\prod_v \left(I + x_v X_v^a X_{d_0^a v}^A\right) = \sum_{\phi \in A^0} x_\phi \, X_\phi^a X_{d_0^a \phi}^A, \qquad x_\phi := \prod_v x_v^{\phi(v)}. \tag{196}$$

There is one important typography point. The definition is

$$x_\phi := \prod_v x_v^{\phi(v)},$$

not $\prod_v x_{\phi(v)}$. The exponent is essential. Since $\phi(v)$ is either 0 or 1,

$$x_v^{\phi(v)} = \begin{cases} 1, & \phi(v) = 0, \\ x_v, & \phi(v) = 1. \end{cases}$$

Thus x_ϕ is simply the product of the recorded signs x_v over those vertices selected by ϕ.

The equation itself is just the familiar expansion of a product of binomials. Define

$$K_v := X_v^a X_{d_0^a \mathbf{v}}^A.$$

Then Eq. (196) says

$$\prod_v (I + x_v K_v) = \sum_{\phi \in A^0} x_\phi \, X_\phi^a X_{d_0^a \phi}^A.$$

Here

$$X_\phi^a := \prod_v (X_v^a)^{\phi(v)}.$$

So ϕ tells us, vertex by vertex, which term we selected while expanding each factor:

$\phi(v)$	term selected from $I + x_v K_v$
0	I
1	$x_v K_v$.

For two vertices v_1, v_2, the left-hand side is

$$(I + x_1 K_1)(I + x_2 K_2) = I + x_1 K_1 + x_2 K_2 + x_1 x_2 K_1 K_2.$$

The four terms correspond exactly to the four binary cochains:

$\phi = (\phi(v_1), \phi(v_2))$	x_ϕ	Pauli operator
$(0,0)$	1	I
$(1,0)$	x_1	K_1
$(0,1)$	x_2	K_2
$(1,1)$	$x_1 x_2$	$K_1 K_2$.

Finally, the compact form of the Pauli operator follows because d_0^a is linear over $\mathbb{F}_2$:

$$d_0^a \phi = \sum_v \phi(v)\, d_0^a \mathbf{v}.$$

Therefore, selecting K_v at every vertex with $\phi(v) = 1$ gives

$$\prod_{v:\phi(v)=1} K_v = \left[\prod_v (X_v^a)^{\phi(v)}\right] \left[\prod_v \left(X^A_{d_0^a \mathbf{v}}\right)^{\phi(v)}\right] = X_\phi^a X^A_{d_0^a \phi}.$$

The missing factors of $1/2$ are intentional: Eq. (196) expands the numerators of the measurement projectors. The full projector has an overall factor $2^{-|V_A|}$, which is irrelevant once the state is normalized.

This is not a new physical superposition of measurement records. It is an algebraic expansion of the fixed-record projector. It is useful because the co-boundary $d_0^a \phi$ will be tested by the later Z_A measurements.

2.20.5 Step 2: measuring Z_A keeps only cocycles

Ungauging first applies $C_A^\dagger$, which turns the A subsystem back into qubits initially prepared in $|0\rangle_A$. Let the subsequent Z_A measurement record be a binary 1-cochain z: $z(f) = 1$ means the outcome on edge f is -1.

Here the cleanest way to read the notation is: choose the natural bases of vertices and edges, then represent each cochain by its list of values on those basis objects. Thus $\phi \in A^0$ is formally a linear function on vertex chains, but we draw it as vertex bits; likewise, $z \in A^1$ is formally a linear function on edge chains, but we draw it as edge bits. The important point is that z and $d_0^a \phi$ belong to the same space, A^1. Equation (197) is an equality of two edge cochains, meaning

$$z(e) = (d_0^a \phi_0)(e) \qquad \text{for every edge } e.$$

Take the concrete vertex assignment

```
vertex:       v1   v2   v3
phi(v):        1    0    1
```

Its coboundary is

$$(d_0^a\phi)(e_{12}) = 1 + 0 = 1, \qquad (d_0^a\phi)(e_{23}) = 0 + 1 = 1.$$

Thus its edge values are

```
edge:              e12  e23
(d phi)(e):          1    1
```

The operator $X^A_{d_0^a\phi}$ applies X precisely on those edge qubits with assigned bit 1. Therefore

$$X^A_{d_0^a\phi}|0\rangle_A = X_{A_{12}}X_{A_{23}}(|0\rangle_{A_{12}}|0\rangle_{A_{23}}) = |1\rangle_{A_{12}}|1\rangle_{A_{23}}.$$

The later Z_A measurement reads this computational-basis string. With the convention $z(e) = 0$ for a $+1$ Z_{A_e} outcome and $z(e) = 1$ for a -1 outcome, the displayed state has only one possible record:

```
edge:       e12  e23
z(e):         1    1
```

Hence $z = d_0^a\phi$. If instead the measurement reported $z = (0, 1)$, its overlap with this branch would vanish:

$$\langle 0, 1|1, 1\rangle = 0.$$

This is the meaning of "its overlap with the observed record is nonzero only if" there is some 0-cochain ϕ_0 with

$$z = d_0^a\phi_0 \tag{197}$$

The subscript 0 says only: choose one cochain whose coboundary matches the actual record. It need not be unique. This is an outcome constraint, not a postselection condition: the actual record automatically has this form because it came from the state in Eq. (196).

Indeed, if $\mu \in \ker d_0^a$, then

$$d_0^a(\phi_0 + \mu) = d_0^a\phi_0 = z.$$

These alternative ϕ's are precisely the branches that remain after the Z_A measurement, and they differ by ancilla symmetries.

Now compare any other surviving summand ϕ with ϕ_0. It must obey

$$d_0^a\phi = d_0^a\phi_0 \quad \Longleftrightarrow \quad d_0^a(\phi + \phi_0) = 0. \tag{198}$$

Thus every surviving label has the form

$$\phi = \phi_0 + \mu, \qquad \mu \in \ker d_0^a. \tag{199}$$

This is the mathematical heart of ungauging. The final Z_A measurement eliminates all non-symmetry components of the temporary-ancilla expansion; only the 0-cocycles, i.e. the ancilla symmetries, remain.

The offset ϕ_0 is an ordinary measurement byproduct. Applying or tracking $X^a_{\phi_0}$ removes it. After this feedforward, the part that acts nontrivially on the input blocks is

$$\sum_{\mu \in \ker d_0^a} x_\mu X_\mu^a \, \Omega. \tag{200}$$

Notice the contrast with the previous measurement-preparation section. Local byproducts such as ϕ_0 are correctable. The signs x_μ attached to genuine cocycles are not local syndrome data to erase: they are the logical measurement outcomes we want to retain.

2.20.6 Step 3: remove the ancilla and reveal the logical projector

Undo the enrichment circuit and use its defining symmetry-enrichment identity,

$$\Omega^\dagger X_\mu^a \Omega = X_\mu^a U_\mu. \tag{201}$$

The remaining a qubits are in $|+\rangle_a$, so X_μ^a acts as $+1$ and can be discarded. Choose a basis $\{\mu_i\}$ of the cocycle space. The output on the input codes is, up to normalization,

$$|\Psi\rangle_{BC} \longmapsto \prod_i \frac{I + x_{\mu_i} U_{\mu_i}}{2} |\Psi\rangle_{BC}. \tag{202}$$

2.20.6.1 What $\mu \in \ker d_0^a$ means here The expression does not mean "the kernel of μ." Rather, μ is one 0-cochain belonging to the kernel of the coboundary map:

$$\mu \in \ker d_0^a \qquad \Longleftrightarrow \qquad d_0^a \mu = 0.$$

For an ordinary ancilla graph, on an edge $e = \langle v, w \rangle$, this condition says

$$(d_0^a \mu)(e) = \mu(v) + \mu(w) = 0 \pmod 2.$$

Thus neighbouring vertices must have equal μ-bits. On a connected graph, μ must be constant everywhere, leaving just two possibilities:

```
mu = 0:       0 ---- 0 ---- 0 ---- ...
mu = mu_all:  1 ---- 1 ---- 1 ---- ...
```

For the three-vertex line used above,

$$d_0^a(m_1, m_2, m_3) = (m_1 + m_2,\ m_2 + m_3),$$

so $d_0^a \mu = 0$ requires $m_1 = m_2 = m_3$. Hence

$$\ker d_0^a = \{(0,0,0),(1,1,1)\}.$$

Why do precisely these cochains survive Step 2? Suppose one vertex cochain ϕ_0 explains the final edge record:

$$z = d_0^a \phi_0.$$

For any $\mu \in \ker d_0^a$,

$$d_0^a(\phi_0 + \mu) = d_0^a \phi_0 + d_0^a \mu = z.$$

Therefore ϕ_0 and $\phi_0 + \mu$ create exactly the same bit pattern on every surviving A-edge qubit. The final Z_A measurements cannot distinguish them. In a connected graph, adding μ_{all} simply flips every vertex bit; each edge still has the same endpoint *difference*. The kernel is therefore the residual global ambiguity left after the edge measurements have removed all local information.

Step 3 translates that ambiguity into a logical operation. The enrichment circuit was chosen so that every such ancilla symmetry obeys

$$\Omega^\dagger X_\mu^a \Omega = X_\mu^a U_\mu.$$

Moving X_μ^a backward through enrichment produces the same ancilla operator times the logical Clifford symmetry U_μ on the B, C code blocks. The a qubits are in $|+\rangle_a$, so X_μ^a acts as $+1$ and disappears. What remains is

$$\sum_{\mu \in \ker d_0^a} x_\mu U_\mu.$$

For a connected graph, the kernel has only 0 and μ_{all}. Thus, after normalization, this is simply

$$\frac{I + x_{\text{all}} U_{\text{all}}}{2}, \qquad x_{\text{all}} = x_{\mu_{\text{all}}} = \prod_v x_v.$$

Although every temporary qubit was measured separately, only this global parity of their outcomes remains after ungauging. It is exactly the outcome of measuring the logical symmetry U_{all}.

This is Eq. (191), now derived. In the homological construction, each independent cocycle can measure a related, potentially addressable logical CZ. In the connected-graph construction, there is one cocycle and Eq. (202) is one logical-gate measurement.

2.20.7 Final summary: the full symbolic procedure

Now that Steps 1–3 have each been derived, we can summarize the complete physical protocol in one equation. The labels "Step 1" and "Step 2" below refer to parts of the derivation, not to extra physical operations.

Let

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|0\rangle_A|+\rangle_a, \qquad P_z^A = \prod_{e\in E_A} \frac{I+(-1)^{z(e)}Z_{A_e}}{2},$$

where P_z^A is the projector for the final, observed Z_A-measurement record. Choose one ϕ_0 satisfying $d_0^a\phi_0 = z$. Then

$$\begin{aligned}
|\Phi_0\rangle &\xrightarrow{C_A,\,\Omega,\,\Pi_x} \Pi_x \Omega C_A|\Phi_0\rangle \\
&\doteq C_A\, 2^{-|V_A|} \sum_{\phi\in A^0} x_\phi\, X_\phi^a X^A_{d_0^a\phi}\, \Omega|\Phi_0\rangle && \text{(Step 1)} \\
&\xrightarrow{C_A^\dagger,\,P_z^A} \sum_{\phi:\, d_0^a\phi = z} x_\phi\, X_\phi^a\, \Omega|\Phi_0\rangle && \text{(Step 2)} \\
&= x_{\phi_0} X^a_{\phi_0} \sum_{\mu\in\ker d_0^a} x_\mu X_\mu^a\, \Omega|\Phi_0\rangle \\
&\xrightarrow{\text{correct/track }\phi_0,\,\Omega^\dagger} \left[\prod_i \frac{I + x_{\mu_i}U_{\mu_i}}{2}\right] |\Psi\rangle_{BC}\otimes|+\rangle_a.
\end{aligned}$$

In the last line, $\{\mu_i\}$ is a basis of $\ker d_0^a$. The A qubits have been measured and discarded; the temporary a qubits can also be discarded because they are again in $|+\rangle_a$. The only nontrivial action left is the product of logical projectors on the original B, C code blocks.

The five lines have a simple interpretation:

1. Prepare the cluster ancilla, enrich it, then measure the temporary a-qubits: this is physical **gauging**.
2. **Step 1** rewrites that fixed record as a sum over vertex cochains ϕ. Each term leaves the edge pattern $d_0^a\phi$ on the surviving A qubits.
3. Undo C_A and measure the A qubits: this begins **ungauging**.
4. **Step 2** retains only terms with $d_0^a\phi = z$, namely $\phi = \phi_0 + \mu$ with μ a cocycle.
5. **Step 3** removes the correctable local piece ϕ_0 and turns each surviving cocycle into the logical projector $(I + x_{\mu_i}U_{\mu_i})/2$.

Thus gauging creates a local ancilla record of the symmetry, while ungauging filters away all local information and leaves only the global symmetry eigenvalue on the original code blocks.

2.20.8 The magic-state example

Take the connected-graph case and choose the input logical state $|\overline{+}\rangle_B|\overline{+}\rangle_C$. A $+1$ logical CZ outcome gives

$$\begin{aligned}
\frac{I+\overline{\mathsf{CZ}}}{2}|\overline{+}\rangle_B|\overline{+}\rangle_C &\propto |\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle, \\
|\overline{M}_{\mathsf{CZ}}\rangle &:= \frac{|\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle}{\sqrt{3}}.
\end{aligned} \tag{203}$$

The $|\overline{11}\rangle$ component is absent because it is the unique computational-basis component with CZ eigenvalue -1. The state in Eq. (203) is non-stabilizer: a stabilizer state has a computational-basis support

of size a power of two when its nonzero amplitudes have equal magnitude, whereas this state has three equal nonzero amplitudes. It is therefore a magic-state resource.

For this particular input, the $+1$ outcome occurs with probability $3/4$; the -1 outcome gives $|\overline{11}\rangle$. Thus this simple demonstration is heralded rather than a deterministic magic-state factory: retain a $+1$ trial or repeat the preparation. The paper's central achievement is that the required logical CZ measurement has been compiled into gauging, local measurements, decoding, and feedforward on qLDPC code blocks.

2.20.9 The essential takeaway

The construction is not using a transversal CCZ as a direct logical non-Clifford gate. It uses local controlled-symmetry enrichment, then removes the ancillary gauge system, to perform a **logical Clifford measurement**. On a suitable logical input, that measurement supplies the non-Clifford resource needed for universal computation.

2.21 Sources and scope

- Target text: arXiv-2602.12228v1/main.tex, Sections II.A–III.C, "Two Approaches to Gauging qLDPC Codes," and the introduction, "Graph requirements," and "Example: Gauging transversal SWAP gate" of "Non-Abelian Codes from Gauging with Graph States," plus the opening graph-gauged ALP fracton example and "Measurement Preparation of Non-Abelian qLDPC Codes," and "Logical Clifford Measurement from Gauging and Ungauging" (from about line 421).
- Context for gauging and $\mathcal{D}(D_4)$: arXiv-2209.06202v2/main.tex, especially its discussion of Abelian Kramers-Wannier gauging and D_4 quantum doubles.
- Experimental and twisted-$\mathbb{Z}_2^3$ presentation: arXiv-2305.03766v2/main.tex, especially "The model," "Ground State Preparation," and its table equating $\mathcal{D}(D_4)$ with $\mathcal{D}^{\alpha}(\mathbb{Z}_2^3)$.